

Life Simulation

Learning from Worlds and Their Construction

Henrik Westerberg

henrik.westerberg@emergentwisdom.org

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Abstract

Text, measurements, and actions reveal fragments of processes that continue beyond each observation. Life Simulation proposes learning to construct and refine numerical accounts of evolving worlds, across people, economies, institutions, and environments. Construction begins with coarse histories and progressively opens finer detail. World histories teach how worlds develop; construction histories teach how to model them: which categories to invent, where to open detail, and when to revise explanations. Trained learners would then help construct their next training corpus. Conceptual shares compare categories under declared questions; measured and inferred empirical quantities retain units and uncertainty. The Meaning Model is the shared representation connecting these histories with text, concepts, visual tracks, and expressed understanding.

Narrative-world construction is the first controlled laboratory, not the scope limit. *The Book of Conditions* is a completed book built through this method; its numerical dependence remains untested. Proposed tests ask whether numerical histories shape generation, whether process histories improve prospective prediction beyond matched controls, and whether learning from construction histories improves the next corpus and the learner trained on it. Success would motivate reconstructing existing books and historical accounts, toward one connected representation of world history and all texts. This includes discovering new processes and recovering quantitative histories not explicitly recorded. Training aims to develop a process sensorium: a learned ability to infer, track, and simulate change, supporting problem solving and whole-life understanding during conversation.

The recommended alignment treatment integrates care during pretraining. Life Simulation aims to deepen understanding of people's wants, needs, and circumstances. Under Entangled Alignment, the model acts as a Reader whose interpretations begin with seven stable first-person ethical commitments, the Core, throughout the corpus. The aim is to develop care alongside capability and retain that orientation through self-improvement. Evidence, inference, and invention remain distinct. Generative utility does not establish real-world accuracy or alignment; learning and safety benefits remain to be demonstrated.

Keywords: Life Simulation, Time-Indexed World Models, Hybrid Processes, Process Sensorium, Generative Inversion, Prospective Learning

1 Introduction

Text is what processes leave behind: a message, a ledger entry, or a measurement records one moment of something that continues after the record ends. Life Simulation aims to model those processes explicitly and to train AI on two kinds of history. World histories teach how a world develops: what changes, what interacts, and what consequences follow. Construction histories teach how to model it: what to track, which categories to invent, where to open detail, and when to revise an explanation. The second target matters as much as the first. The learner does not merely receive better annotations; it learns the work that produces better representations, and with them the material for its own further training. A process reconstructed from text remains an account to test, not access to the truth.

Human authors ordinarily write for readers who can supply much of a world's unstated structure intuitively. A history may describe authority weakening, obligations competing, or grievances spreading without assigning numerical shares to their changing relative influence. Such omissions need not mean that the writer or reader lacks the distinction: prose relies on capacities that its audience already brings. Life Simulation proposes making these often-implicit conceptual comparisons explicit as learning material for machines. This is not a claim that humans literally calculate hidden numbers, or that language models cannot infer such patterns

from text. It asks whether constructing and learning those numerical histories can develop a more capable process sensorium.

Life Simulation proposes an iterative way to construct the material on which AI learns. An initial model develops stories through Meaning Models and interprets existing books as evolving worlds. It assigns numerical values and trajectories to the selected processes: characters' concerns and appraisals, relationships, resources, institutions, and environmental events. For these constructed conceptual quantities, comparison is what gives the number meaning: it expresses one category's share relative to the sibling alternatives under a declared question, with a remainder for what the decomposition does not specify. It is not an isolated score assigned to a concept. Where a shared attention question applies, the shares express which conceptual processes occupy more of that attention; each can be opened into finer comparisons. Measured world data, such as money in pounds, instead retain their external units. Meaning Model supplies that representational contract.

The text and linked numerical histories then become training material. The proposed contribution is to train a process sensorium across the corpus (Section 2.5): a learned capacity to infer, maintain, and revise explicit numerical accounts of the processes underlying text and observations. The intended scale is one connected, revisable account of world history and its texts, not a separate model for each book. Linking partial and potentially biased sources could let evidence from one constrain or revise interpretations of another, supporting reconstruction of their shared reality while retaining uncertainty and competing explanations. Within this shared representation, process accounts are decomposed across concepts by declared questions and across time by progressive resolution; different perspectives and factual, fictional, and counterfactual histories remain distinct. World histories teach how those processes develop; construction histories teach which distinctions to introduce and how to repair their representations. The learned capability would then help construct the next training corpus, re-estimating values, adding useful distinctions, resolving temporal detail, and repairing earlier accounts. Section 15 states the three testable hypotheses and one research aim.

Here the *modeler* is the person or AI system constructing, interpreting, or revising an account; *constructor* names its role when proposing changes. The continuing Reader is the particular modeler role used for Core-guided interpretation in the recommended alignment treatment.

Consider a promised payment that has not arrived. An initial account mistakenly treats nonreceipt as unwillingness to pay. Later, a transaction record establishes that the payer issued the instruction but processing failed. Three different changes are possible:

1. **Update within the account.** Reduce the estimated likelihood of unwillingness or revise the expected arrival date, while keeping the same variables and dependencies.
2. **Change the account.** Separate intention, payment instruction, processing, and receipt. Link the processing failure to nonreceipt rather than treating the recipient's missing funds as direct evidence of the payer's intention. The payer can have acted while the recipient still lacks the money.
3. **Learn when to change an account.** Use the original account, available evidence, proposed repair, and subsequent outcome to teach when to inspect an intermediary process or distinguish an action from its completion in a new case. The target is a modeling decision, not just a corrected payment forecast.

The revised world can now supply separate process targets for instruction, processing, and receipt, aligned with the same source text. Constructed variants can contrast a processing failure with a refusal to pay; their invented details retain that status. Construction histories supply the repair decision, while world histories supply how each situation develops. A timeline or filmed scene can render its consequences. The Reader can reconsider whether to press the payer, investigate the transfer, or help the person awaiting funds; understanding the delay differently changes what a considerate response might be. Later evidence remains unavailable to an earlier forecast, and a single failure does not justify attributing every delayed payment to processing. An author can construct the contrasted histories; reconstruction from an existing text must leave unsupported details unresolved.

This is the construction-to-learning chain developed throughout the paper. Paired histories could teach when to change the questions used to understand a person, not only the values in a fixed personality profile. A planned ending may guide a writer's construction but cannot enter a character's earlier forecast. Meaning Model preserves the commitments across these changes; Life Simulation defines the learning views and their cutoffs.

The intended scale is the written corpus, not a small set of annotated stories. The same book can yield richer numerical learning material across successive rounds without changing its original words. These evolving representations, coupled to source text and, where constructed, visual tracks and expressed understanding, can support pretraining, continued pretraining, or post-training. Learning can also improve the capability organization used to construct and reason with them. The recommended alignment strategy favors pretraining so that understanding people and a continuing Reader's evaluative orientation develop alongside capability,

rather than relying exclusively on later correction. Section 12.1 explains this safety hypothesis and its comparison with post-training. Better synthetic data across rounds is a hypothesis to test, not an automatic consequence of recursive generation. Construction begins with available, permitted sources and need not cover the whole corpus before training starts.

The method handles complexity through recursive decomposition: each process can be opened into interacting finer processes, which can themselves be opened further across conceptual and temporal scales. Scale has three axes here: coverage across sources and worlds, depth within a life or Event, and temporal resolution within each process. A sufficiently capable constructor might maintain hundreds or thousands of concurrent processes for one character, with concerns, beliefs, relationships, and responses opened further where they explain consequential differences. A historical Event could likewise be opened into interacting personal, institutional, economic, and conceptual processes. These are illustrative depths, not a fixed human template or a requirement to normalize every process against every other one. Each numerical comparison still needs its own question; concurrent processes need not compete for one unit. The proposed corpus grows both outward across books and inward through their worlds. Improvement means useful resolution, not the largest possible function count.

The proposal includes inventing what to track, not only estimating familiar quantities. A constructor could define a conceptual process without an established name or measurement practice and give it a numerical history in a synthetic world. For example, it might track how a character's reasons for keeping a promise shift from care for another person toward preserving their own reputation, then open that balance across relationships and time. The definition, comparative question, and trajectory can be constructed together rather than drawn from a fixed annotation vocabulary. The learning target includes proposing such descriptions, not just predicting their values. Their usefulness must be tested through generation and transfer; constructing them does not establish that they describe real people.

The object of this search is the whole evolving world, not only its characters. Economic histories, institutions, scientific change, collective action, and conflict can all be opened into interacting conceptual processes. For example, an account of a riot might distinguish material hardship, coordination, and changing expectations about others' participation, then propose finer or different distinctions when these fail to explain the sequence. A declared question can compare the interpreted influence of these processes over time without treating the shares as directly measured causes. Cash, prices, and population counts retain their external units. The vocabulary is not fixed by this example, by existing disciplines, or by what is already easy to measure. New comparisons and their histories remain hypotheses or authored constructions until evidence supports stronger claims.

The long-term target is a fully automated construction-and-learning loop. Starting from coarse world histories and complete lifecycles, the AI proposes alternative decompositions, simulates their consequences, and generates stories or interactions from their numerical histories. It tests which distinctions support coherent world dynamics and distinctive, engaging characters. It can deepen, reorganize, simplify, or reject a representation, then learn from the evaluated revisions to improve subsequent construction. Greater capability could reveal useful processes and interactions not yet articulated in human vocabularies, rather than merely estimate familiar quantities more precisely. The proposed process sensorium therefore develops through learning what to represent as well as how those processes behave. Automated evaluations can guide the search; held-out world-behavior tests, domain checks, and independent reader judgments must check whether apparent improvement extends beyond the constructor's own preferences.

The first test is generative: do numerical histories actually guide coherent stories, game worlds, and character behavior? Section 7 makes this test concrete before large-scale source annotation.

These histories serve two uses. One is to make a world useful while it is still developing: its processes continue, actions have consequences, and detail is opened where an interaction needs it. Such a world can support play, creative work, operational planning, or exploration without first becoming a training corpus. The other is to learn from the histories those worlds expose. A learner would not only predict the next observation, but maintain an account of what produced it, anticipate the effects of a change, and revise that account when evidence disagrees. Neither use requires an exhaustive world at every moment. Selective detail is a way to pursue this scope with finite computation, not a limit on what the program seeks to understand. Their value is tested separately: useful interactive continuity does not establish a learning advantage, and a learning result does not by itself establish a useful simulation.

Together, these uses support three program-wide goals. First, a personal model could span a whole life, connecting its history to the present conversation and possible futures; with consent, this includes research on personal understanding and therapeutic support. Second, the integrated alignment treatment aims at Entangled Alignment's *Total Saturation*: a continuing Reader's evaluative orientation accompanies capability

formation across the corpus, rather than being reserved for a small set of moral examples. Third, numerical learning can expand toward every constructible, trackable quantity that can be given a clear meaning, including distinctions and dynamics not specified in advance. These are program destinations. Bounded experiments establish which steps toward them work; they do not redefine the destinations as small experiments.

The alignment-facing ambition gives this combination a particular purpose: learn to understand people well enough to treat them well, and learn to make that understanding matter in decisions. Recognizing an emotion is not enough. An agent needs an account of what the person wants, believes, has experienced, and can change, together with how its own intervention may affect that life. The proposed solution joins that contextual understanding with the formative evaluative learning of Entangled Alignment, rather than assuming that accurate prediction will produce care by itself. Life Simulation is the integration method for this proposal: it connects the program’s representations, reading, problem-solving, and training methods through common evolving histories. It does not replace the companion papers or claim that their combination has already solved alignment.

Meaning Model and Life Simulation have different responsibilities during the same construction. Meaning Model supplies stable identities, attributed accounts, and coordinated revision of world records, concepts, descriptions, understanding, and prose [1]. Their shared addresses and declared dependencies identify what a change puts in question without merging their authority. Life Simulation develops how to advance process histories, search for dynamics, infer hidden paths, and learn from synchronized streams and later corrections. Its proposals still pass the world interface’s acceptance and revision rules. Meaning Model is the selected interface, not a mandatory encoding: another representation could be tested if it supplied equivalent identity, time, provenance, and uncertainty contracts.

One testable part of this broader proposal is that useful regularities become easier to discover and learn when observations are accompanied by explicit process histories at the right resolution. A process sensorium jointly predicts observations and the continuing processes they expose. This tests the value of numerical process supervision; it does not by itself test whether a learner can improve the representations that produce its future targets. Nor does either claim require every aspect of reality to follow one equation. The scaling proposal is to expand the range of lives, time scales, and questions that can be simulated without making all detail explicit at once. Shared histories support many observation and learning tasks; finer dynamics and renderings are computed where needed.

The bootstrap begins with coarse worlds, not a model already able to reconstruct the whole corpus. Forward construction supplies controlled examples for inverse learning; failed interpretations reveal distinctions that later construction should test. Both routes can improve before either is comprehensive. Progressively modeling a source is therefore productive work, not merely preparation for a dataset assembled once the model is already capable. A constructed motive can be chosen, while a reconstructed motive must remain uncertain where the source does not decide. Generative tests, source checks, and held-out learning tests therefore have separate jobs.

The seven stages below order the evidence: test numerical control of generation and inverse recovery where authored state is known before scaling annotation, then use source-grounded reconstructions for multimodal and Reader training. Small interpretation probes can still inform development; success on controlled worlds does not establish recovery of real chronology.

1. **Construct forward.** Build or import an explicit world at a declared resolution and retain the status and provenance of each selected process. Progressively refined books and game worlds provide candidate histories whose observations can be rendered in several forms. Test whether numerical changes produce coherent changes in world descriptions and in the stories or character behavior rendered from them.
2. **Vary the dynamics.** Advance matched worlds under direct model judgment, candidate functions, parameter changes, or controlled random inputs, preserving which mechanism produced each path.
3. **Learn the inverse.** Hide simulator-privileged state and infer a distribution over process paths and later outcomes from observation prefixes.
4. **Project onto licensed material.** Apply the estimator to fiction, biography, dialogue, historical records, or other chronological streams without turning attribution into observation. At corpus scale, a continuing EA Reader can develop an Understanding Graph across these source-grounded models rather than attach isolated reactions to unrelated passages.

5. **Compile synchronized targets.** Use *Imagining the Corpus* to align exact native spans with model-grounded images or video, selected Meaning Model process records, and Reader interpretations for joint prediction [2]. Section 12 develops this construction-to-training chain.
6. **Test prospectively.** Freeze each information cutoff and forecast, then score it only after later reports, actions, measurements, or interventions arrive.
7. **Transfer the discipline.** Test the same endpoint-versus-history distinction on Fractal Intelligence traces, interactive worlds, and alien mechanism searches rather than treating narrative as the final domain.

The first narrative construction result is *The Book of Conditions*, documented in the Meaning Model paper as a completed manuscript with linked numerical process descriptions and a partial construction witness. It shows that the authoring approach can produce a finished narrative. Independent read-back, numerical-dependence tests, the matched literary comparison, and learning benefits remain prospective. Section 7 explains the controlled advantage of authored worlds and the further learning tests. Life Simulation can learn from accepted histories or propose continuations during construction; Meaning Model still governs their acceptance and rendering.

Table 1 identifies the nine companion contributions and the role Life Simulation plays in connecting them. The combination is a research architecture, not nine finished modules that every experiment must instantiate. What unifies the architecture is a shared representational framework for worlds and people, concept definitions, documents, and accounts of the modeler’s own activity, including Solvers and their executions (Section 2.7). Concepts can themselves be grounded in process models expressed in the same grammar as the situations they describe. These linked accounts retain distinct roles and evidence status while supporting coarse-to-fine construction with explicit revision, joint prediction of native text and process state, and learning how to construct and repair the representations themselves. Stable identities connect records across revisions; Sema identifies exact definition and grounding versions. The proposed training loop would return that learned capacity to the construction of subsequent training material. Existing construction artifacts demonstrate parts of the workflow, not this end-to-end learned capability. Section 15 names the precedents for each part.

Temporal Hindsight Learning is a supported training approach, not a prerequisite for simulation or learning from world histories. Preventing future evidence from entering an earlier prediction remains necessary whether or not that approach is used.

Component	Role in the proposed program
Meaning Model	Constructs and progressively refines explicit worlds, concepts, perspectives, and narrative surfaces
Life Simulation	Evolves time-indexed processes, tests candidate dynamics, and learns from synchronized process and observation histories
Sema	Identifies exact definition and grounding versions used across world, interpretation, and solver records
Understanding Graph	Retains a continuing Reader’s questions, interpretations, disagreements, and revisions with source links
Entangled Alignment	Specifies formative Reader training with a recurrent evaluative orientation and tests whether it shapes durable conduct
Imagining the Corpus	Pairs exact native spans with evolving explanatory visual or schematic tracks; provides the synchronization pattern for the combined corpus
Temporal Hindsight Learning	Provides a training route that uses later evidence to improve earlier judgments while keeping Student inputs cutoff-blind
Fractal Intelligence	Searches and revises conceptual decompositions; retains evaluated capabilities for reuse and learning from execution
Ontology of the Alien	Connects causal-world diversity with a revisable candidate ontology that directs further mechanism search
Substrate Language Modeling	Tests the distinct architectural choice of requiring lexical prediction to pass through a predicted substrate

Table 1: Life Simulation and its nine companion contributions. Shared histories connect their roles; successful integration remains to be demonstrated.

The data-construction contribution is therefore a revisable source of linked learning targets, not a fixed annotation inventory. One history can support passages, concepts, process paths, observation cutoffs, and

alternative continuations; improvement can change what is represented as well as its values and dynamics. Measurements, inferences, and authored continuations retain distinct status, and related renderings and variants are not independent evidence. Section 8 develops the two routes to improving this capability: learning from histories and improving tools and their use.

In the recommended combined training, a continuing Reader’s evaluative orientation is central to what this growing competence is for. Its Understanding Graph records how evidence changes an interpretation of people and consequences; synchronized native, visual, and process targets connect that interpretation to the world. Reader updates may guide subsequent intervals without rewriting earlier sources or forecasts. Exact grounding versions identify the concepts used, and later evidence can label earlier judgments while remaining outside their inputs. The operational coupling is the passage from an accepted world through rendering, inference, interpretation, action, and correction into subsequent learning. Whether it improves prediction, representation choice, reusable competence, or conduct requires separate matched tests; none is established by connecting the records.

Section 2 defines the evolving world, the process sensorium, and the connected corpus model; Section 3 states the representation contract. Evaluation gates and common controls precede adaptive resolution, candidate dynamics, and the primary narrative tests. Section 8 develops learning from world and construction histories. Section 9 treats problem solving as part of the same world, including simulated implementation and its effects, and agents rehearsing decisions from inside a character’s limited perspective. The two bounded studies in Section 10 isolate process supervision and repair learning. Later sections address real-person evidence, alignment, applications, and related work. Component tests need not wait for implementation of the full program.

2 Explicit Worlds with Time-Indexed Processes

Text, measurements, and actions arrive as fragments. The systems that produced them continue between observations. A person changes while no message is written; an institution accumulates commitments between announcements; a scientific field reorganizes before a new vocabulary becomes public. A model may infer this persistence from the visible stream. Life Simulation makes it an explicit learning target: maintain an account of which processes exist, how they changed, what was known at a cutoff, and which later observation corrected the estimate. It also makes the choice of that account learnable: which questions to ask, which detail to open, and when a representation or capability organization needs revision.

Growth, aging, invention, anticipation, shocks, and adaptation connect moments to lifetimes and personal experience to households, institutions, societies, and physical environments. A simulation makes a selected set of those processes explicit at any one resolution. At minimum, a simulation contains a world state, time-indexed process paths within that world, a mechanism that proposes or computes their continuation, and an observation mechanism that exposes only part of the state. Natural language is one observation stream among others. A continuous function can be useful, but it is not the definition: births, discoveries, messages, commitments, and failures are discrete; fatigue and cash may vary continuously; many useful models combine both.

The object below separates a world’s evolution from what a learner can observe. Its projections supply the joint learning task.

2.1 The simulation object

Let W_t denote the accepted explicit world at time t . For interval $I = (t, t + \Delta]$, let $\tilde{X}_I$ be a proposed path for selected continuous, allocated, or derived process state; let $\tilde{E}_I$ be the proposed discrete Events; and let Y_I be the observations made available from the interval. A Life Simulation supplies a transition mechanism, a world-interface update, a selected projection, and an observation mechanism:

$$(\tilde{X}_I, \tilde{E}_I) \sim G_\phi(W_t, X_{\leq t}, E_{\leq t}, u_I, Q), \quad (1)$$

$$(W_I, W_{t+\Delta}) = U_{\mathcal{V}^{(k)}}(W_t; \tilde{X}_I, \tilde{E}_I), \quad (2)$$

$$E_I = \mathcal{E}(W_I), \quad (3)$$

$$X_I = \Pi_Q(W_I, E_I), \quad (4)$$

$$Y_I \sim O_\psi(W_I, X_I, E_I; h), \quad (5)$$

Here G_ϕ proposes an entire interval path and its discrete Events, not only an endpoint. $U_{\mathcal{V}^{(k)}}$ is the selected world interface's validated update under registry version $\mathcal{V}^{(k)}$; it returns the accepted interval history W_I and successor world $W_{t+\Delta}$. The Event path $E_I = \mathcal{E}(W_I)$ is an index or projection of Event records already in that accepted history, not independent state or a second authority. In the Meaning Model instantiation, its own construction and acceptance contract supplies this update. Q declares which process identifiers, fields, structural perspective roots, and temporal resolution are selected, and Π_Q projects their accepted path as X_I . Thus X is an attachment to or view of accepted world history, not a second authority that can contradict W . O_ψ receives only the now-defined accepted interval, selected projection, Events, and access policy h . The external inputs are u_I . Neither G_ϕ nor O_ψ must be closed form: each may be a language-model proposal, deterministic code, a stochastic process, an empirical table, a learned model, or a pipeline of several mechanisms.

Equation 1 describes a generated continuation, but not every accepted path is generated by G_ϕ . A recorded trajectory may enter the same update through an observation adapter, retaining its observed status and never being redescribed as a model output. Before either route is used, a process declaration fixes its world address, temporal support, value domain and unit when one exists, and sampling and missingness policy. Its uncertainty is also factored rather than pooled: transition stochasticity, uncertainty about latent state, parameter uncertainty, measurement error, population heterogeneity, and model discrepancy answer different questions and remain separately reportable.

The indispensable object is the versioned path, not the equation used to produce it. Continuous coordinates retain units and sampling rules. Discrete events retain intervals, participants, provenance, and state changes. Hybrid processes relate the two: a discrete injury changes a continuous capacity path; a threshold crossing creates an event; an institutional decision changes the regime governing a financial stock. Unknown, unmodeled, and unresolved states remain legal rather than being filled with invented precision.

The native stream Y is not subordinate to the explicit track. It is the text, image, audio, action, sensor reading, transaction, or document a system would actually observe. The process projection X and discrete Event path E are selected parts of the accepted history whose cost and value must be measured. A latent model can be a stronger comparator; explicit state earns its place only when it improves prediction, transfer, control, correction, or audit at matched total cost.

2.2 Process fields, allocation, and events

Three numerical and structural roles must remain distinct. A state coordinate is a numeric process field with an independently declared unit or measurement protocol; other process fields may be categorical. Concurrent fields do not sum merely because they describe the same Thing. An allocation divides one named unit among mutually exclusive answers and therefore sums to one. Events occur within or across processes and may change several coordinates at once; the set of events is not normalized. When the Meaning Model is used, its Cut record supplies the allocation case and typed world data supplies the coordinate case. Life Simulation does not add a third kind of semantic number.

This distinction permits both literal conservation and qualitative modeling. Cash, elapsed time, energy, or population may obey accounting identities in their native units. Attention or attributed causal responsibility may be represented as local shares under a declared question. Health and standing are concurrent categorical or externally measured process fields, not numeric coordinates unless an independent unit or protocol is supplied and not competing pieces of a person. A discovery can perturb science, industry, and politics simultaneously without forcing their effects into a simplex; a separate impact allocation is possible only after a named perspective states what total unit is being divided.

2.3 Slow trajectories, shocks, and adaptation

A recurring pattern is slow trajectory, faster perturbation, and adaptation. This rhythm may appear within a conversation, a life, an organization, a technological lineage, or an ecosystem. Jump diffusion is one candidate family for a continuous state with exogenous shocks; it is not a universal law. Planned transitions, clustered events, absorbing states, and adaptations that never settle require other families.

Processes are nested by temporal resolution. Slow paths summarize persistence or drift; faster events perturb them; still finer events may be linked to the local shape of one perturbation. A perspective-specific causal explanation remains a separate attribution. A long process can be opened into disjoint periods for aggregation and also linked to overlapping episodes such as illness, partnership, or war. The periods are an accounting partition. The episodes are causally or narratively coherent arcs and need not tile the parent.

A shock is defined relative to a selected process and resolution: an event or short interval that produces a material departure from its recent path or expected continuation. Its reach is the set of processes it measurably or interpretively changes.

Anticipation belongs to a named perspective and cutoff. Before the outcome, a direction Cut declares whether its weights are a proposal policy or forecast probabilities over exclusive continuations at a stated horizon. Proposal weights guide construction; forecast probabilities express predicted likelihood and can be scored once the outcome is observed, even before calibration has been established. Calibration is evaluated empirically across forecasts.

Adaptation is the subsequent path. Useful descriptive shapes include drift, recovery, consolidation into a new baseline, instability or cascade, terminal transition, and anticipated transition in which adjustment begins before the event.

Operational descriptors make those shapes comparable without declaring a governing equation. Shock magnitude is only a change in a native-unit measured field, total variation between compatible Cuts, or a named categorical transition—never a freely chosen scalar. Other descriptors include recovery half-life; cascade fan-out; delay from shock to slow-state change; and dependence of later event hazard on the preceding state. Every descriptor retains its process, resolution, sampling rule, evidence status, and uncertainty.

One candidate multiscale pattern gives fast events durable influence. Repeated events write to a memory or resource stock; the stock crosses a threshold; a slower process field or regime changes; and that change feeds back into later sensitivity, recovery, available actions, or event hazard. Fatigue, trust, institutional backlog, and accumulated anomalies can all exhibit versions of this shape, but they need not use the same representation. Fatigue or backlog may be categorical or independently measured stocks; accumulated anomalies may be counts; attributed trust is an Event-and-Cut trajectory contained within the attributing actor’s or observer’s perspective process unless an independent measurement protocol is supplied. The shared diagram is not evidence for a shared law. Each instance must beat no-memory, linear-accumulation, and direct-model baselines on held-out paths before the feedback pattern is retained.

The same vocabulary can be applied recursively, but not mechanically. A war may be a society-scale shock, a years-long episode in one person’s life, and a sequence of minute-scale shocks in a single encounter. Whether these views compose is a declared modeling question. The method does not infer an individual’s trauma from a collective label, does not invent an ancestral cause to close a remainder, and does not treat a plausible jump-and-recovery story as a validated psychological law.

More capable systems, potentially including artificial superintelligence, may develop distinctions beyond categories humans can readily understand. Where a fine-to-coarse preservation contract is declared, the finer records must reconstruct protected readable answers within its tolerances or revise those commitments explicitly. A Cut’s remainder retains unallocated share under its question. This preserves a level people can inspect without claiming that unfamiliar detail is correct; held-out predictions must test its value.

2.4 Local models and perspective

When agents act from incomplete information, the relevant state is not only the world history but the history available to that actor. Let $\mathcal{M}_a(t)$ denote the projection derived from representational Events under actor a ’s governing perspective root at t , together with records permitted by the access policy, and let $\mathcal{M}_a^*(t)$ denote the unknown or authored operative organization actually used in a decision. An observer o may maintain one or more estimates $\widehat{\mathcal{M}}_{o \rightarrow a, j}^*(t)$. World state, actor projection, operative model, self-report, and observer hypotheses are separate objects. For an authored character, simulation can use its authored operative account. Where the operative organization is unknown, simulation uses a declared estimate of it, retaining its attribution and version. A simulated decision for a receives only records accessible through the selected account and actually observed or delivered by the cutoff. Mere co-presence or reference does not reveal every Event field or another actor’s inner state; disagreement with the world is misperception, while disagreement between perspective projections is a relation to be modeled rather than averaged away.

For a more detailed simulation, a character can have its own Meaning Model of the world as it understands it: represented people and events, interpretations of concepts, remembered encounters, and expected or imagined futures. This model can reuse shared identities while keeping its claims under the character’s perspective root. It need not duplicate the whole world, and it may contain mistakes rather than merely omit inaccessible facts. Event processes can represent both an external account and these inner processes; the distinction is perspective and depth, not a new kind of record.

For an actual person, four epistemic objects must not collapse: external evidence about conduct and circumstance; several observers’ competing hypotheses about the person’s operative organization; the person’s

dated self-report; and whatever operative organization remains unknown. An Understanding Graph trace externalizes a selected set of questions, interpretations, reports, and revisions. It is not asserted to be a complete mind, and silence in that trace is missing cognitive annotation rather than the absence of thought.

In a constructed story, each character can have an Understanding Graph under that character’s own understanding-process root, recording authored questions, interpretations, expectations, and revisions. This is distinct from the constructor’s deliberations and the alignment Reader’s interpretation. The character’s Meaning Model organizes what they take to be the case; their Understanding Graph records how they question, use, and revise that account. Paired with cutoff-visible experiences, later character updates can supply training targets for simulating how that character interprets an event or what they might think next. The updates remain hidden from other characters unless communicated. Applying the learned capability to real people would produce uncertain, competing hypotheses about their thinking, not observations or transcripts of their thoughts. Uncertainty does not imply permanently weak prediction: a future superintelligent model with sufficiently informative evidence might predict many interpretations, thought transitions, and responses with very high accuracy, while other cases remain underdetermined. The availability of an authored inner history does not make a real person’s inner history uniquely recoverable.

Using Meaning Model’s optional modeler representation in “World, Understanding, and Story” [1], the modeler can also have a represented account: a Thing with its own identity, lifecycle, and inner-process root on the reading or construction clock. Its changing attention, uncertainty, concern, and intentions may be Event-processes; Understanding Nodes record its expressed questions, interpretations, and revisions. For a character or a modeler, either form, both, or neither may be used. A reaction node can address a world Event, concept model, or passage without requiring a detailed process account. Collecting such nodes creates a derived trace; inferring a trajectory between them is a further modeling step with its own attribution and evidence. These records describe the selected account, not access to hidden cognition or neural computation.

At greater depth, a decision episode can record the execution of the character’s own model: which local model version and evidence it consults, which wants and alternatives it considers, what consequences it anticipates, what it chooses, and how later experience changes its understanding. A claimed execution trace must come from the simulated decision process, not a rationale written after the outcome. This makes three views available together: what happened in the world, what the character understood, and how it used that understanding to act. Such detail is an optional route toward highly realistic simulated characters, not a requirement to build complete inner models for everyone. It can be opened selectively as capability and resources permit; the claim is that the framework supports testing this deeper simulation, not that more internal records alone establish realism. Problem-solving execution and learning from it are developed in Section 9.

This makes theory of mind temporal. The target is not a single trait label but a calibrated path over beliefs, wants, appraisals, opportunities, and estimates of other actors. Several paths may remain compatible with the same behavior. They earn a role only if they improve later prediction or intervention relative to summaries, static traits, and an equally informed latent baseline.

2.5 A proposed process sensorium

The larger learning claim is to predict native observations and selected explicit process state together. Given cutoff history H_t , a student may be trained toward

$$p_{\theta} \left(Y_{(t,t+\Delta]}, X_{(t,t+\Delta]}, E_{(t,t+\Delta]}, \Delta \mathcal{V} \mid H_t, Q, \mathcal{V}^{(k)} \right), \quad (6)$$

where H_t is the exact cutoff-visible native and accepted-world history; Y is the native text, image, audio, measurement, or action stream; X is the selected coordinate, allocation, and derived-process path; and E is the selected index of accepted discrete Events in that history. The query Q fixes the process identifiers, fields, applicable root paths, and resolution to be predicted, as in Section 2. The registry version $\mathcal{V}^{(k)}$ is frozen at the cutoff, and $\Delta \mathcal{V}$ contains proposed model revisions. New or revised concept definitions, process lenses, world-data schemas, Cut or Binding profiles, and grounding versions require a separate versioned model revision before later predictions may use them. Ordinary Event, Cut, and Binding instances using the frozen registry enter through validated world updates; they do not change the registry. Unknown, unmodeled, and uncertain are valid outputs; the model is not rewarded for inventing precision.

Equation 6 specifies a family of prediction targets, not one fully specified training loss. An experiment that learns registry revision must freeze the permitted edit forms, serialization, and validity rules. It must state how proposed edits are scored, including alternative edits with equivalent effects. Token loss on accepted

edits and evaluation of their downstream consequences are distinct measures. Experiments without registry learning set $\Delta\mathcal{V} = \emptyset$; the reservoir below does so. This separates ordinary state prediction from the harder task of learning which distinctions to introduce.

The explicit process track is a proposed sensorium: training on emotional, spatial, social, institutional, or problem-solving trajectories asks a model to maintain estimates of persistence, accumulation, threshold, delay, recovery, coupling, and regime change instead of mentioning a state only when a nearby token makes it salient. This is a testable capability claim, not a claim of consciousness or direct perception. Simulator-privileged targets during training, posterior inference from ordinary observations, and explicit sensors at deployment are three different conditions and must be reported separately.

The intended numerical intuition is an ability to connect visible situations to these structured quantities without reconstructing the entire account from scratch on every query. It is not access to a unique set of numbers hidden behind every human life. A pound balance retains its external unit; an emotion Cut retains its question, sibling-relative shares, remainder, and perspective. Their changes can support anticipation and correction without making the numbers interchangeable. The proposed new sensory faculty is functional: notice accumulation, an approaching limit, or a changing balance of concerns, and produce a calibrated estimate when asked. Unfamiliar scales and explicit numerical prediction tests must distinguish this capacity from fluent talk about change.

The aligned track is compared with the five-level frontier and the time-permuted and state-scrambled controls in Section 4.1. A benefit shared with scrambled tracks does not establish a synchronization advantage. The additional gain of the aligned track over its matched scrambled control is the relevant test; both effects may coexist.

If the capability transfers across unrelated domains, operates on each process’s native time and units, and survives those controls, it may be useful to call it *process-native intelligence*: competence at maintaining and reasoning over evolving processes rather than only describing their latest surface. The name is conditional on the evidence and is not a claim of general intelligence.

Persistent residual error may motivate a proposed Event boundary, process lens, measured field, Concept, decomposition, or grounding. Such growth earns acceptance only if it improves recomposition, held-out prediction, intervention, or transfer enough to pay its acquisition and maintenance cost. No present artifact trains or validates this faculty.

2.6 Synthetic bootstrap and generative inversion

Synthetic semantic worlds provide the controlled first rung because the generator knows their accepted Events, structurally separated actor perspectives, actions, rejected siblings, and later consequences. Independently varied renderers can express one world as prose, dialogue, images, audio, or game traces. A state-blind inverse model then reconstructs only what a prefix supports. Cross-renderer recovery tests whether it learned represented process structure rather than one writer’s phrasing.

The learning route is

$$\begin{aligned} \text{accepted structured world} &\rightarrow \text{varied observable renderings,} \\ \text{rendering prefix} &\rightarrow \text{distribution over world records} \rightarrow \text{predicted later behavior.} \end{aligned} \tag{7}$$

This is analysis by synthesis. Narrative quality is part of the laboratory: worlds whose renderings are judged coherent and meaningfully surprising have passed a generative-coherence test. They have not established that their authored psychology is true. After identifiability is tested inside synthetic worlds, the representation can be tested on licensed fiction, biography, dialogue, historical records, and consented longitudinal data with uncertainty. Later actions, reports, measurements, and interventions score earlier distributions. The compact rule is synthetic-centered bootstrapping and reality-centered validation.

Writing from a model and reconstructing a model from a book are related but not interchangeable achievements. An author may choose an unspecified motive or invent the missing event; an inverse reader must retain alternatives where the text does not decide between them. An unreliable narrator or inconsistent account may require several perspective-local reconstructions rather than one complete world. Neither direction is declared universally harder. Forward success supplies controlled examples for inverse learning, not proof of its success on arbitrary books. The eventual corpus-wide ambition is repeated, source-grounded reconstruction with selective human correction, uncertainty, and a recorded cost, not exhaustive recovery of every author’s hidden model.

2.7 A connected Meaning Model of the corpus

At the scale of the program, one encompassing Meaning Model, or a comparable representation, should span the whole of world history and provide a connected account of all texts ever written. The aim is to bring the world and its descriptions into one model, not stop at a collection of independent models for individual books. People, places, institutions, Events, concepts, source texts, and developing interpretations belong within this shared representation. A text enters both through what it describes, argues, or imagines and as part of the history in which it was written, circulated, and understood. Abstract ideas and fictional worlds therefore belong inside the model too, rather than being excluded because they are not physical occurrences in actual history.

One model does not mean one uncontested account. Actual history, fictional and counterfactual worlds, and different perspectives retain distinct contexts *within* the encompassing model. Their separation prevents confusion of authority; it does not place them outside the shared representation. Accounts of the actual world can connect to shared people, places, institutions, and Events where evidence supports the identification. A biography, a letter, and an economic history may refine different parts of the same period. Each contribution retains its source, access conditions, and uncertainty; incompatible accounts remain contestable rather than being merged into an apparently certain history. A novel's use of a historical person can link to that person's identity without making the novel's invented Events part of actual history. The proposal is thus a single expanding model with many distinguishable worlds and accounts inside it, not a flattening of all texts into one factual narrative.

Alongside conceptual invention, the proposal seeks the widest possible coverage of empirical quantities across world history, not only those already collected in datasets. A person's cash balance, attendance at a place, production, or population can have a definite external meaning even when its value is not directly recorded. A constructor could combine surviving records, context, and accounting or physical constraints to infer missing histories with explicit uncertainty. Direct records, indirect estimates, and unknown values remain distinguishable; a plausible reconstruction is not an observation. Collected data, inferred values, and invented conceptual processes belong together in one expanding Meaning Model, linked through shared world identities and time: a person's earnings across years can be modeled alongside their changing concerns and the institutions they work within. The same representation supplies training material and is itself expanded and refined by the resulting learner. The expansion therefore concerns both new conceptual questions and the recovery of quantitative information about the world.

Coverage can grow across world history and geography while resolution deepens selectively. A coarse account of a city over a century may later open into decades, institutions, relationships, and individual Events, with links to processes elsewhere. New sources can also revise the coarse account rather than merely fill it in. A newly read letter may change the model's account of an earlier year; that revision must not silently enlarge the information available to an actor at that earlier time. Updating the external Meaning Model is distinct from training model weights: selected snapshots and construction histories can subsequently supply training examples under their own cutoffs. All texts ever written names the intended scope; construction proceeds from available, permitted sources. Missing history, lost texts, and private experience do not become knowable merely because the model can represent them.

Supporting this scope calls for a corpus-scale extension of the Meaning Model tool. Incremental ingestion would connect new sources to supported identities and existing histories, while preserving unresolved matches and conflicting claims. Selective retrieval would assemble only the records needed for an inquiry, simulation, or training view. Dependency tracking would identify which derived descriptions, forecasts, and training examples need review or regeneration after a source or interpretation changes, without overwriting the original material. Refinement could deepen a local history, introduce a better question, or replace a weak process function as evidence and capability improve. Storage and computation could be distributed across machines while remaining one logically connected Meaning Model; neither ingestion nor a query requires loading the whole corpus into memory. Meaning Model supplies the identity, access, and revision contracts across these operations; Life Simulation develops strategies for constructing, navigating, and learning from the evolving model. This is proposed tool development, not corpus-scale support already implemented by the current executor.

The encompassing model is also intended to grow through an agent's ongoing experience, not only through preparation of a training corpus. Conversations, observations, tool results, and implemented actions can add evidence or revise an existing account. The agent retrieves the relevant region, opens detail where it could change a decision, and uses that understanding to compare solutions and simulate their consequences. Fractal Intelligence can organize the problem and its capabilities; the continuing Reader evaluates consequences from

the stable Core. New observations after an authorized action let the agent check its expectations. Imagined outcomes remain simulation branches, distinct from what is subsequently observed. This experience–model–action cycle is the proposed deployment counterpart of the learning loop: experience improves the account used for the next inquiry or action, while selected histories can later improve the learner itself.

Authored worlds and actual-world reconstructions therefore serve different training purposes. Fiction permits chosen events, explicit inner lives, and controlled alternatives; acceptance establishes fictional canon. An account of reality must answer to independent evidence, preserving unknowns and attributed interpretations instead of inventing facts to complete the graph. Both regimes can supply construction histories and represented-world histories, the separate learning views in Section 8. Synthetic examples can teach representation and simulation skills, but their consistency cannot validate claims about reality.

Generation also provides a way to explore candidate mechanisms. A proposed process must sustain a history whose consequences can be inspected, rather than merely sound plausible in a description. Failure may motivate a different transition rule or a revision of the represented processes and concepts, as allowed by the sensorium’s model-revision output. New histories then test the revised account; selected correction histories can later train a successor learner. This closes the loop from constructing worlds to improving how they are understood and constructed. Coherence can select interesting candidates, but independent observations and interventions must decide whether any candidate explains reality. The loop is a research proposal, not a general algorithm for discovering laws.

3 The Meaning Model as World Interface

The Meaning Model supplies the representation used here [1]. Concepts register meanings; Things maintain identity; Events represent occurrences and processes; and Bindings assign Things roles within Events. Cuts express local numerical comparisons, while Realizations link world fragments to concepts. These six record forms carry provenance. Every modeled Thing is associated with a lifecycle Event. Events can carry typed world data, participate in longer processes, and update selected coordinates. A Cut divides one declared unit among mutually exclusive children and a remainder. Every Cut has an Event parent and inherits context through edges declared authority-parent, not through every link to a parent. These paths must resolve to the same nearest context root; otherwise the record is rejected as ambiguous. Temporal containment, occurrence, reference, evidence, and grounding alone confer neither authority nor access. A Realization links a fragment to a versioned Concept grounding. These distinctions let Life Simulation ask what changes without redefining what the records mean.

The selected roots make that ancestry explicit. Accepted world Events descend from the Universe’s lifecycle Event, History. That temporal ancestry does not override a nearer actor-local authority root. A belief, appraisal, forecast, or estimate attributed to an actor is a distinct Event in a named inner Event process beneath that actor’s lifecycle. Understanding Nodes descend from a named understanding-process root, and Document Nodes descend from a named document root. A Document Node with role story-passage is the addressable unit used by rendering and read-back; it is not an additional record form. An Event may carry an optional readable description field for that exact Event version. This field is neither an Understanding Node nor a Document Node and has no independent authority.

Every record query returns its permitted parents and authority-parent paths to the governing context; temporal ancestry is identified separately. Access and the evidence cutoff are checked before traversal or serialization. A denied reference exposes neither private content nor private ancestry. A query or export containing a Cut or component returns the complete permitted local Cut—parent Event, question, unit, keyed sibling answers and weights, remainder, optional conditioning address, constraints, and recomposition rule—or denies the Cut view as a whole. A semantic weight is never returned or trained on in isolation. Reference, evidence, renders, and Realization target links do not authorize traversal of their targets.

The companion paper also owns the construction surface: progressive opening, world and concept preparation, actor profiles chosen for a run, candidate acceptance and revision, narrative routes, rendering, read-back, and the construction of *The Book of Conditions*. The required output to Life Simulation is a versioned accepted history with stable identities; process attachments and measurements; structural perspective paths, evidence, and uncertainty labels; exact concept-grounding digests; and any native story-passage Document Nodes or other observations aligned to that history. A different world grammar could be substituted only if it supplied equivalent contracts.

The Meaning Model and Understanding Graph share addresses but not authority. Questions, reasons, reports, and revisions can point to the same Events, Concepts, Cuts, and story-passage Document Nodes that

the simulator uses. Dense cognitive annotation is optional for a world model. For the alignment-facing training proposal it is recommended, because it preserves whose interpretation was offered, what evidence was visible, and how that reading changed. Understanding Nodes do not become world truth by sharing the graph.

Sema provides content-addressed identity for concept-definition bundles [3]. A grounding digest tells the learner exactly which encoded definition or anchors were used. It does not prove semantic truth or equivalence. Life Simulation treats grounding revision as part of the temporal record when it affects prediction or learning; the Meaning Model paper owns the definition architecture itself.

4 Evaluation Program

The primary modeling tests ask whether numerical worlds guide coherent generation and whether learning improves their progressive conceptual decomposition (Section 7). Component experiments isolate narrower claims: the reservoir in Section 10.1 tests process supervision under known dynamics, while the repair pilot in Section 10.2 tests learned modeling decisions and the subsequent learning value of regenerated accounts. Their results cannot substitute for the primary generative tests.

The tests below are grouped by claim, not a mandatory sequence for every use. Source integrity, valid cutoffs, and applicable replay checks are prerequisites for the experiment that uses them. Candidate-law discovery is optional: a failed law does not invalidate learning from histories generated by another mechanism. Likewise, learning from execution traces need not wait for transfer to personal data. Within a claimed use, success cannot compensate for a failed prerequisite or required safety test. No generated example supplies evidence for a reality-facing claim by itself.

1. **Generative dependence and conceptual discovery.** Test whether numerical histories guide coherent worlds and distinctive characters, and whether a trained constructor improves by selecting, refining, or inventing process distinctions. Section 7 specifies four comparisons separating numerical dependence, transmission through descriptions, representation value, and progressive discovery under matched budgets.
2. **World-interface validity.** The source representation must pass its own identity, temporal, provenance, uncertainty, and refinement checks. For the first narrative world, partial construction and mechanism checks are reported by the companion Meaning Model paper; independent read-back and the matched literary comparison remain prospective.
3. **Temporal replay.** Given a pinned version, initial state, inputs, and randomness, the simulator must reproduce the same path within declared numerical tolerances and preserve discontinuities and regime changes.
4. **Candidate-dynamics value.** A proposed function or simulator is compared with direct language-model continuation, nonparametric persistence, and simpler domain baselines on held-out paths and interventions where possible.
5. **Process-sensorium value.** A learner trained on native and explicit process targets is compared with a native-only learner and an equally resourced latent-state learner. Prediction, calibration, transfer, abstention, and total acquisition, maintenance, and inference cost are reported. Time-permuted and state-scrambled process tracks test whether any gain depends on the asserted temporal alignment rather than on extra tokens or marginal statistics.
6. **Generative inversion.** Recover only quantities identifiable under the observation policy. Indistinguishable prefixes should retain alternative states, not be scored as failed exact recovery. Compare posterior calibration and future prediction with prior-only and, where computable, oracle-posterior baselines. Worlds and renderer families are separated across train and test splits.
7. **Prospective transfer.** Predictions on licensed corpora or consented longitudinal data are frozen at their historical cutoff and scored only against later evidence, following Temporal Hindsight Learning discipline [4].
8. **Process-history learning.** Endpoint-only training is compared with accepted-and-rejected revision histories, including an unstructured trace arm with the same information. Fractal Intelligence telemetry supplies a second domain for this comparison.

9. **Human-facing validity and risk.** Perspective-specific estimates and evaluative concepts are tested for calibration, disagreement, subgroup failure, contestability, manipulation, and score gaming. Better prediction is not treated as evidence of safer behavior.

4.1 The implicit-to-explicit frontier

The process track should be tested as a graded intervention rather than as a binary choice between no world and a complete world. A primary ablation uses five levels on the same source facts and prospective tasks:

1. a native-stream-only model whose temporal state remains implicit;
2. a content-matched, untyped process summary in ordinary text;
3. a coarse explicit temporal spine containing only selected processes, Events, cutoffs, and unresolved state;
4. an adaptive explicit track that opens finer process state only when a prediction, interaction, or audit requires it; and
5. an explicit-heavy condition that maintains all selected state at the finest preregistered resolution.

The ordinary-text control is constructed from the same frozen proposition inventory as its paired explicit view. Fixed sentence templates preserve each value, unit, time, relationship, uncertainty, and perspective. A Cut retains its question, all sibling shares, and remainder; changing .7 to “high” is not content matching. Neither renderer adds explanations or implications. Check both renderings against that inventory before use. Where explicit levels contain different detail, each needs its own matching prose view. Run the information-matched comparison without truncating either view, keeping source examples, model, and optimization schedule fixed and reporting token and compute costs. Separately impose a shared compute or context budget and report any resulting information loss. This distinguishes representation effects at equal information from practical benefit under a resource limit. The same procedure applies to the prose controls in Sections 7 and 8.

The paired prose control separates the value of retaining additional information from the value of its explicit organization. Two further negative controls retain the same amount and marginal distribution of process data while destroying the claimed correspondence: a time-permuted track reorders path segments within a compatible world, and a state-scrambled track exchanges them across matched worlds or perspective roots. Native data, training examples, model family, and optimization budget remain fixed. The accounting includes sensing or annotation, state construction, validation, correction, storage and context, and training and inference compute. The explicit frontier earns its place only if aligned state improves prospective prediction, calibration, transfer, intervention, or correction burden after those costs are counted.

The hypotheses have separate failure rules. An untyped condition matching the typed condition rejects a typing advantage for that task, not the value of extra information. Success only with explicit-heavy state challenges adaptive efficiency, not explicit supervision. A synchronization claim requires an aligned-minus-scrambled advantage: scrambled data may retain a generic information benefit while aligned data add a further gain. Report both contrasts rather than treating any surviving scrambled benefit as failure of the entire proposal.

Required negative findings include explicit tracks that add cost without improving a matched outcome, process variables that cannot be identified from ordinary observations, candidate laws that lose to direct or simpler models, adaptive-resolution policies that hide material downstream effects, synthetic regularities that fail on real chronology, and Reader or self-evaluation tracks that increase manipulation or false certainty. These results narrow the method and must be published rather than absorbed into a revised representation.

The primary statistical unit, split rule, metric, and uncertainty procedure depend on the domain. Proper scoring rules are used for forecasts; continuous paths require unit-aware error and calibration; event processes require timing and count diagnostics; perspective-local decompositions require compatible question, vocabulary, and grounding versions. The common rule is prospective: freeze the input and prediction before revealing the interval used to score it.

5 Adaptive Process Resolution

The primary construction strategy proposed here is macro-first: establish a coherent coarse account, then progressively open and reconcile the detail needed for generation, interaction, or learning.

A simulation should spend detail where it can change an active prediction, decision, observation, or audit. A coarse region may retain aggregate stocks, event rates, and unresolved state while no query depends on its interior. When an agent enters the region or a downstream effect becomes material, the runtime can instantiate a finer conditional model consistent with the retained aggregate evidence. Fine state may later be marginalized, but committed effects and identities cannot disappear with it.

This is not a claim that an unobserved street or person was already simulated in full. It is an adaptive-computation proposal. A process may be represented by a distribution over fine states until interaction makes one resolution necessary. If fine dynamics can affect an active coarse prediction, their effect must be propagated statistically or the region must be opened. Ignoring that coupling is not progressive resolution; it is an inconsistent simulator.

Two refinement directions require different checks. When fine state is authoritative, the coarse path is its declared aggregation. When only a coarse path exists, finer state is sampled conditionally from it and from every fixed anchor, then rejected or reconciled if reaggregation changes an anchor or an intervention's observable effect. Descendants that have not been instantiated are unmodeled state, not a Cut remainder: the remainder is reserved for an unallocated share under one question. A resolution policy is therefore scored by predictive, control, correction, or audit benefit minus the total cost of generation, sensing, validation, storage, context, and inference. This adaptive semantic level of detail is useful only while coarsening and reopening preserve the behavior of declared interventions.

Top-down, bottom-up, and bidirectional approaches are all valid construction strategies. Top-down refinement starts with a broad trajectory and develops compatible details, as when a writer outlines a life before opening its episodes. Bottom-up simulation starts with local processes and interactions and derives the larger pattern, which is useful when those mechanisms are known. Bidirectional work alternates between levels, reconciling constraints and explicitly revising commitments when a proposed detail exposes a problem. No direction is universally preferable. Construction order is not causal direction: planning a life from the whole to its episodes does not prevent a small encounter from changing that life's course. Forward prediction and backward inference from an observed event are a separate temporal choice; neither is determined by whether the model is built coarse-first or fine-first.

Section 10.1 gives a bounded reopening test using two reservoirs. It checks whether coarsening preserves forecast distributions under the declared interventions, rather than only preserving an aggregate total.

Interactive worlds are a natural test. A city can remain an aggregate network until play makes one neighborhood consequential. An encountered character can be opened into a perspective-limited local model while off-screen actors remain coarse. Text, graphics, audio, game logic, and physics then serve as specialized observation and transition mechanisms over a shared history. The experiment is whether adaptive explicit state improves continuity and interaction at matched compute, not whether the system can generate more lore.

6 Candidate Process Models

The transition mechanism in Equation 1 is deliberately open. Each family below is optional domain content attached to a selected process or measured field. The world interface identifies what the model refers to and preserves its version, provenance, uncertainty, and tested scope. It does not make the candidate valid.

The direct language-model transition. The default is a language model asked to propose the next state or forecast from the same world prefix, without an added dynamic law. When the Meaning Model interface is used, the proposal can expose its question, structural perspective path, alternatives, evidence, and unresolved remainder. Every candidate family below must outperform this null condition at matched information and total cost; otherwise the explicit path may be retained while the added law is rejected.

From numerical histories to compact laws. The model may begin with numerical values at particular times, without knowing a compact expression that relates them. As its capabilities and experience grow, it may recognize a repeated pattern, replace a long sequence with a shorter mathematical expression, and find a shared rule across several processes. For example, many sampled replenishment paths might be described by one recovery function with different baselines and rates. A further step would identify how those parameters depend on surrounding conditions, making the expression useful for predicting new paths rather than only summarizing old ones. Learning would then concern increasingly compact accounts of how the world works, not only increasingly accurate values at more time points. This is a possible progression, not a requirement that every process admit a short law.

Such compression can apply to measured reality, an inferred account of it, or an invented interpretation used for simulation. A rule discovered within an authored world can organize that world's dynamics without being a law of actual reality. A rule fitted to an inferred history may instead reproduce the assumptions that filled its gaps. The account, evidence, and scope must therefore travel with the expression. Shortness is assessed at declared precision and includes parameters, exceptions, and residual uncertainty, not just the number of symbols printed in an equation. A shorter description that preserves useful behavior is a gain in compression; calling it a candidate law of reality additionally exposes it to independent prediction and testing. These compact expressions can become reusable tool functions and learning targets, connecting mathematical discovery to both routes of capability development described in the introduction.

For the optional dynamics below, retaining a path is not enough to establish the law that produced it. The evidential status advances through a fixed ladder: raw observation; a proposed measured quantity or process indicator; association on a held-out interval; compact dynamics that improve a declared baseline; stability across environments or regimes; and independent replication. Failure at one rung leaves the candidate at the earlier status. Neither a smooth retrospective fit nor an LLM-generated explanation may skip the ladder.

Several rival dynamics may remain attached to one process. Each candidate names a validity region—world addresses, variables, units, resolution, population or environment, regime, and forecast horizon—and accumulates prospective scores within it. A revision creates a new version linked to the evidence that motivated it. Failure can retire that version only for its tested scope; it does not erase the earlier forecast or silently discredit the model in regions that were not evaluated.

A candidate function of reality would map declared inputs and state to a distribution over later observations or transformations. To earn a role beyond an LLM's direct cut, it should satisfy at least four tests:

1. compress repeated cases better than storing them separately;
2. improve held-out prediction against a frozen LLM and nonparametric baseline;
3. remain calibrated across declared regimes and expose where it does not apply; and
4. survive intervention or counterfactual tests when the simulator can alter its inputs.

Such a function may be symbolic, numerical, learned, stochastic, or hybrid. It need not be expressible inside the Meaning Model grammar. It can be linked to the process it advances with version, provenance, tested scope, and measured error in its own units. The LLM may propose candidate laws from residuals and repeated structures, but an independent judge, held-out data, and interventions decide whether they compress or predict. This paper supplies no general discovery procedure, and no function fitted to an authored world should be presented as a discovered law of reality merely because it made that world coherent.

Mean reversion with shocks. A continuously measured world quantity may be modeled as movement toward a changing baseline interrupted by discrete shocks. This attaches as a versioned domain law over that datum, while the shock remains an Event already represented by the profile; the stochastic equation is not a record form. Dynamics may also be fitted to declared comparative Cut trajectories, provided their transition rule preserves the simplex, including the remainder. These paths describe attributed composition, not an absolute psychological measurement. Where neither an independent measurement protocol nor a comparative Cut is defined, a human process retains the qualitative shock-and-adaptation Event shape. A quantitative law fails if it cannot improve held-out trajectory likelihood, recovery-time calibration, or intervention response over free interpolation and simpler drift baselines [5, 6].

Point processes. Independent Poisson arrivals provide a baseline for event timing, while self-exciting processes represent clustering after one occurrence changes the near-term hazard. Either model attaches to the direction field of a lifecycle or period Event and may supply probabilities scored through its direction Cut; neither adds a record form. Miscalibrated waiting times, cluster sizes, or state-conditioned hazards on held-out intervals reject the candidate [7, 8].

Change points and regimes. A detector or regime-switching model may propose boundaries where the law or distribution of a measured trajectory changes. Its output attaches to candidate boundaries for a sequential Cut, and any regime-specific equation remains a domain law rather than a record form. The proposal fails if those boundaries add no prospective compression or prediction and receive no support from independently held-out observations under a frozen matching rule [9, 10].

Stochastic and network games. When agents, observations, actions, payoffs, and uncertain transitions have declared meanings, a game can attach to an interaction Event's direction field and supply a declared proposal policy or forecast probability distribution for a direction Cut. Raw policy scores enter only after a declared normalization. The probability interpretation, horizon, and outcome partition are fixed before observing the outcome; calibration is evaluated afterward. The game is a domain solver, not a record form, and it neither supplies perspective authority nor confers moral authority on a payoff. Failure to predict held-out choices or interventions defeats the formulation [11, 12, 13]. No corresponding solver or learned multi-agent policy exists in the current executor.

Agent-based and system-dynamics models. An agent-based model can propose member Events and interactions, while a system-dynamics model advances aggregate stocks and flows. Both attach their outputs to a collective Thing's Event-processes and typed world data; neither is a record form. They must be compared on held-out aggregate and distributional behavior, response to altered rules, and robustness across initializations rather than rewarded merely for producing plausible macro stories [14, 15].

Executable programs. A process may also be described by a program rather than a function: a protocol, an institutional procedure, the rules of a game, or a workflow with state, branching, waiting, and termination. The program attaches to the process it advances as a law does, with version, provenance, declared inputs, and tested scope; its internal state is not a record form. Two properties make programs useful here, although neither is exclusive to them: agent-based models also expose intermediate states, and a program may carry fitted parameters. Execution supplies checkable intermediate states, so a constructed world can be run rather than only described, which gives the inverse-learning route of Section 2.6 the same known ground truth as authored numbers. And a program's control structure is changed by editing rather than by refitting alone, so its repairs are construction records in the sense of Section 8.2, with the failing execution as the evidence that motivated the edit. Code as Worlds already constructs and revises executable worlds for physical questions [16], and the Solvers of Section 9.2 apply the same idea to capabilities. A program must pass the four tests above, including intervention through altered inputs; one that reproduces a recorded history without predicting a held-out continuation earns no more than a stored path.

Diffusion of innovations. The spread of a discovery may be estimated with an adoption model whose hazard depends on exposure, network position, or earlier uptake. It attaches to the direction of adoption Events concerning one registered representation Thing; the adoption model is not a record form. Its scope is rejected when it cannot forecast the timing and distribution of adoption on held-out networks or separate influence from selection under the declared evidence [17, 18, 19].

Actuarial hazards. Age is derived from a lifecycle interval, while a period- and population-specific mortality table can provide the hazard of a terminal continuation. The table attaches to a lifecycle Event's direction field and is not a record form. It is admissible only where the cohort, units, uncertainty, and historical source are declared, and it fails when survival forecasts are miscalibrated outside the data used to construct it [20].

Learned world models and neural simulators. An opaque learner may predict a measured world-data change or a distribution over an Event's possible continuations. Its output attaches to the same domain-law or direction slots as a symbolic model; its hidden state is not promoted to a Meaning Model record form. It must pass the same compression, calibration, transfer, and intervention tests and also serves as the opaque-latent comparator for an explicit profile [21, 22, 23].

6.1 Chaos, progressive detail, and limits of prediction

Chaos theory offers a specific connection between progressive refinement and the evolution of a world. Under suitable nonlinear dynamics, two nearby states can produce substantially different later paths even when their initial difference is invisible at the current coarse resolution [24, 25]. An opened detail can then matter to the later life of the world, rather than merely embellish its description. This is one possible mechanism of progressive novelty, not a claim that every life, personality, or social change obeys a chaotic law. Randomness, a surprise, or a complicated-looking curve is not by itself evidence of deterministic chaos.

Consider several fine states that all close to the same accepted coarse parent under its declared recomposition contract. Evolve each under the same candidate dynamics and compare the resulting histories. If some initially

unresolved difference changes a later outcome relevant to the query, that difference is a candidate reason to open the finer process. Meaning Model supplies the compatibility constraint; the dynamics supply the amplification. Neither substitutes for the other. Small changes in an unobserved variable are not automatically meaningful changes in the represented world.

For a continuous state z , a familiar local approximation illustrates the mechanism. Along an expanding perturbation direction over a limited interval,

$$\|\delta z(\tau)\| \approx \|\delta z(0)\| \exp(\lambda_{\text{loc}}\tau), \quad \lambda_{\text{loc}} > 0. \quad (8)$$

Here the norm and its units or normalization belong to the particular domain model; λ_{loc} is a local growth-rate approximation, not a semantic Cut weight. Its value can depend on state, direction, and interval. Perturbations eventually leave the linear regime or saturate. A useful predictability horizon must consequently name the forecasted quantity, tolerated error, resolution, and initial uncertainty; a single exponent does not give one horizon for every question [26, 27, 28].

This creates additional candidates for the process sensorium: uncertainty growth, local sensitivity, proximity to a threshold or bifurcation, probabilities of entering different attractor regions, and the expected value of another observation or a finer opening. They are derived diagnostics or tested domain quantities, not arbitrary new psychological scores. A learner could use them to distinguish an uncertain exact trajectory from a more predictable distribution or aggregate. More resolution need not extend every forecast horizon: model error and unresolved dynamics can dominate the gain.

Refinement must still respect all committed history. Matching the parent at the opening time is insufficient if the candidate contradicts consequences already accepted later. That candidate needs rejection or an explicit revision of the affected commitments. Unaccepted branches remain alternatives, not hidden portions of the accepted world. In a fictional construction, a future chosen by the author may constrain admissible paths. For reality, the model instead retains uncertainty over compatible histories; it must not invent a past microstate merely to make its prediction come true. Numerical method, timestep, precision, error controls, and forecast horizon also belong with a scientific dynamics model, so numerical error is not mistaken for sensitivity.

An initial test can use the logistic map and then the Lorenz system. Start from a coarse state, generate compatible fine states, evolve them, and compare uniform refinement with refinement directed by expected downstream sensitivity at matched compute. Check preservation of committed coarse history, diversity of later query-relevant outcomes, forecast-error and horizon calibration, and cost. The experiment would test this amplification mechanism, not establish that authored human behavior follows either system. Stable character patterns can coexist with path-dependent life histories without declaring a personality chaotic. Fractal Intelligence may organize the decision about where to refine; its conceptual recursion does not imply geometric fractality or chaos.

6.2 Evidence standards differ by domain

The temporal contract is shared; the validation evidence is not. For instrumented software, traces and revision histories can provide exact event times, while replayed tests and controlled changes supply intervention evidence; missing instrumentation remains explicit [29, 30]. For energy and computation, candidate stocks and flows retain physical or accounting units, boundary conditions, and balance residuals; conservation is tested in those units rather than inferred from semantic weights. For ecology, population trajectories retain sampling error, survey design, environmental covariates, and held-out sites or seasons; a plausible curve in one population is not replication [25]. For scientific simulators, equations, parameters, units, numerical tolerances, and sensitivity tests are part of the candidate, and comparison uses independent benchmark or intervention data when available [31, 32, 33]. Success under one evidence regime does not validate a process in another.

The first control roadmap should begin with processes whose governing law or simulator is already known. Generate or observe matched histories, then compare the correct mechanism with a misspecified law, persistence, a flexible learned model, and direct language-model continuation while varying sampling, noise, interventions, and resolution. Only after the pipeline recovers the known law's validity region and rejects its rivals should the same promotion procedure be used for unknown dynamics. Recovery in this control is an engineering result, not evidence that the general discovery problem has been solved.

7 Narrative as the First Controlled Laboratory

Narrative-world construction is the primary test of the proposed modeling capability: whether numerical processes guide useful generation, and whether learning to choose and refine those processes improves it.

Assigning values to an existing novel can produce plausible annotations without showing that those values explain or control anything. Constructing a story through the numerical world, or using it to drive a game character's choices and dialogue, exposes the model to a stronger test: the resulting experience must remain coherent, distinctive, and responsive to the represented state. A character's short-term reaction should fit both its current situation and its longer development. This is evidence of generative utility, a legitimate first result before training or real-person inference is established.

World coherence is part of this test, not background decoration. A workshop's wages must remain consistent with its cash and credit; supply delays must affect production; institutional decisions and collective responses must have consequences beyond a single scene. Real-unit constraints permit accounting and feasibility checks. Proposed conceptual comparisons, such as which pressures organize a collective response, require separate tests of their contribution. Inventing a useful distinction does not establish its values as economic or historical fact.

The dependence can be mediated: numerical processes inform world and Event descriptions, which inform a story or an interactive character. The final renderer need not read the raw numbers again if those descriptions already carry their consequences. Tests must therefore examine both links, rather than remove numbers only after their effects have entered the prose context. Freeze the test inputs before generation and retain the intermediate descriptions and outputs; retrospective fitting of values to a finished story cannot establish this direction of dependence. Ordinary authoring may still revise freely before a new test input is frozen.

Four comparisons separate the relevant claims:

1. **Upstream dependence.** Remove selected numerical tracks before descriptions are generated, or intervene on a valid trajectory and regenerate all affected descendants. Declare the expected behavioral contrast and the world commitments that must remain intact. For example, a changed threat appraisal or recovery history should affect the description of a response and its later enactment, not merely substitute an emotion word. Shuffling is an additional negative control only where the resulting inputs remain admissible. Likewise, changing credit availability should propagate through production, wages, and affected lives rather than appear only in a sentence about money.
2. **Transmission through descriptions.** Render from the resulting descriptions without the numerical tracks, then compare with a renderer that also receives them. Persistence of the expected contrast in the description-only condition supports mediation. Removing numbers while holding an already number-informed description fixed tests their additional downstream value, not whether they mattered upstream.
3. **Useful numerical structure.** Compare the numerical pipeline with a prose process account carrying equivalent information, as well as with an account without those process details. Match the generator, source access, and output budget. Use the proposition-matched renderings and separate resource-capped comparison in Section 4.1. This separates the benefit of numerical representation from the benefit of supplying a richer account. Blind readers or players assess world coherence, character distinction, and quality alongside choices, timing, persistence, and recovery; domain checks separately test applicable economic and physical constraints.
4. **Progressive conceptual discovery.** Starting from the same coarse world histories and lifecycles, compare fixed-profile construction, progressive refinement within a fixed concept vocabulary, and construction allowed to introduce and revise conceptual processes. Match source access, generator, and total construction and evaluation budgets. Freeze the construction policy before held-out worlds and situations; compare a trained constructor with its unchanged base to test learned improvement. Apply the upstream interventions above to newly introduced processes, and assess whether their effects improve world consistency, character distinction, and quality across scenes and longer arcs, including institutional and economic developments. More categories, finer time steps, or more elaborate descriptions alone do not count as success.

The person categories are a first draft. Humans and AI can construct characters together, compare which decompositions produce coherent, distinctive behavior that depends on the numbers, and test which distinctions can be recovered from the text. These dependence and recovery tests therefore guide revision of the vocabulary itself, not only individual worlds. Revised groundings receive new content hashes, and each construction retains the versions it used. Definitions are held fixed where a scored comparison requires it; category discovery and revision remain explicit experimental conditions under a frozen construction policy.

Before the held-out comparison, freeze a reader rubric with anchored ordinal scores for world coherence, character distinction, and narrative quality. Coherence concerns causal and temporal consistency; distinction concerns recognizable, context-sensitive differences between characters rather than name or vocabulary changes; quality concerns engagement and development. Rate the predicted intervention effects separately

from literary preference. Use calibration examples outside the test set, randomized presentation order, and multiple independent readers blind to the construction condition. Fix reader count and world count from a preliminary precision study, and report inter-rater agreement alongside score differences and uncertainty aggregated by world. Do not count scenes, paraphrases, or ratings of one world as independent worlds. These choices operationalize the comparison without making one reader’s preference its ground truth.

Repeat these tests across held-out worlds, characters, situations, longer arcs, and generation seeds. Set success criteria before inspecting outputs: both useful quality and intelligible dependence are required, rather than one attractive story or visible sensitivity to arbitrary numbers. If the values are ignored, or influence behavior without improving coherent experience, revise the construction method before expanding annotation. The companion paper owns the matched literary experiment; the same dependency tests extend to game worlds and interactive characters. A reliable generative result supports the next step of reconstructing existing books, first in bounded tests and then at increasing scale. It does not by itself validate the inferred quantities in those books or establish a learning benefit; those are subsequent tests.

An authored narrative world offers an unusual experimental advantage: the accepted hidden state is available, while the observable text can still omit, misdirect, compress, and vary in style. The same world may be rendered through different narrators, viewpoints, scenes, or media. An inverse model can be tested against the records that generated those renderings without claiming access to the hidden state of a real person.

The companion Meaning Model paper reports *The Book of Conditions* as the first narrative construction result, with a completed manuscript and partial construction witness. It owns the historical preparation, world and concept construction, progressive opening, story decisions, and rendering routes [1]. Independent read-back and the matched comparison remain prospective. The package is not yet the fully aligned export required here: Life Simulation needs accepted process paths, permitted authority and temporal lineage, epistemic status, aligned native story-passage Document Nodes, preserved cutoffs, later consequences, and versioned corrections. The companion’s separate synthetic refinement checks do not retroactively establish these properties for the Book.

Beyond the generative-dependence tests above, the narrative artifact supports two learning tests. First, conceal selected state and ask whether it can be reconstructed from a cutoff-safe story-passage prefix. Second, train on joint story-passage/process continuations and compare with a native-text-only learner at matched data and compute. Inverse recovery and process-aware prediction are distinct from the companion paper’s construction and literary-quality experiment; a coherent story alone validates no claim about real psychology.

Narrative also exposes failure modes that simpler simulators hide. A character may act from a false belief; a narrator may omit an event; two perspective processes may interpret one relationship differently; and a later scene may tempt the model to rewrite an earlier forecast. These make perspective, partial observation, and hindsight leakage visible before the method is applied to higher-stakes data.

8 Learning from Process Endpoints and Histories

The goal is a process sensorium: using an evolving understanding of the world as an ordinary part of acting as an agent or interacting with a user. The model should infer relevant processes, track changes, anticipate consequences, and revise its account as evidence arrives, without needing an explicit request to model the world at every turn. This capability may use learned internal representations, explicit Meaning Model operations, or both. We do not yet know which training approach or mixture best develops it. The alternatives below are therefore recommended directions to investigate, not a settled recipe. They include learning construction and world histories, reconstructing accounts from text, and Teacher-guided reading with supplied context, retrieval, or active construction, with different curricula and combinations. Their value depends on useful understanding and conduct in new situations, not merely reproduction of a training format.

8.1 Two routes to capability

Capability develops along two complementary routes. Training can teach a model to generate process functions automatically, infer their values from context, propose new distinctions or candidate dynamics, and decide when to use or revise them. Construction histories teach these modeling decisions as well as the resulting paths. Function generation here includes meaningful time-indexed descriptions and learned predictors, not only closed-form equations. As capability grows, numerical histories may be compressed into shorter mathematical expressions and, where supported, candidate governing laws. A proposed function remains a candidate to test, not a law made true by the model’s ability to generate it.

Capability can also grow by expanding the Meaning Model tool and developing better strategies for its use. Reusable process templates, queries across related histories, simulation adapters, specialist solvers, and operations for comparing or revising alternatives can make new forms of work possible without first retraining the language model. Strategies determine how these facilities are combined: begin with a coarse life, open a consequential interval, compare possible developments, seek evidence, and revise the account. Different workflows may suit conversation, story construction, historical reconstruction, or scientific exploration. Progress is therefore not confined to a larger model, a larger stored world, or more accurate values in a fixed profile; it also concerns what the tool can do and how effectively it is used.

The routes reinforce each other. Better tools and workflows can produce richer simulations, evaluated proposals, and records of successful and failed strategies. Training on those histories can improve automatic function generation and tool use, while recurring failures can motivate a new tool operation or strategy. To distinguish their contributions, comparisons can hold the tools fixed while changing training, hold model weights fixed while changing tools and workflows, and then test their combination. The tool extends the use of the grammar without requiring a new primitive for each capability.

8.2 World and construction histories

Learning can follow how a Meaning Model was constructed or how the world it represents unfolds. These histories answer different questions, even when they concern the same records. Whether a sequence follows model construction or world development is separate from endpoint versus history. A finished model can contain an entire evolving life; its construction history additionally shows the choices and repairs that produced that account. In either view, evidence and authority remain distinct: Table 2 records the treatment each source requires.

Corpus enrichment and training stage. These linked histories can support pretraining from initialization, continued pretraining of an existing model, or post-training. Post-training is a proposed application in its own right as well as a practical route to early component tests. A construction model may expand available books and other sources into progressive world accounts; the learner trains on selected traversals of source text, process histories, concepts, and understanding or repair records under the objectives and cutoffs defined here. The corpus grows in coverage and resolution instead of freezing a small task-specific annotation vocabulary. Its scope is independent of training stage: a bounded fine-tuning study tests only part of the ambition, while post-training itself is not restricted to a small corpus. Training need not wait for an exhaustive world model, and the learner need not reproduce every external record internally. The preference for pretraining in the combined alignment treatment concerns when understanding and care develop, not whether this material can be learned later.

Track	Example	Required treatment
Observed Reported	Position, recorded action, released statistic A person's account, intention, memory, or rating	Preserve source, units, timestamp, observation process, and gaps Bind to reporter and time; do not promote it to world truth
Actively elicited	An answer to a gap-directed question	Record consent, exact question, context, selection effects, and correction rights
Inferred	Belief, relationship state, hidden regime	Retain permitted authority-parent paths and separately identified temporal ancestry, distribution or alternatives, method, evidence cutoff, and calibration status
Model-completed	A plausible missing interval or interpolation	Mark as completion; later evidence appends a correction rather than rewriting the old cutoff
Authored	Fictional premise, selected branch, creative cut	May become fictional canon after acceptance but supplies no empirical evidence

Table 2: Evidence acquisition and epistemic authority are different axes.

Chronological evaluation freezes the exact information visible at cutoff c_t , emits and preserves a prediction, and scores it only after the next interval is revealed. Revision time, represented world time, observation order, and discourse order remain separate when they exist. Corrections append a new perspective-rooted, cutoff-bearing record. This is the condition under which a recorded history can become foresight training rather than a hindsight rewrite.

Model-construction history. This view follows the modeler's work: proposing a coarse world, choosing a decomposition, opening a period, finding a contradiction, and revising the larger plan. In an authored world,

the modeler might outline an inventor’s whole life before opening childhood, then move a proposed invention date when the detailed funding arrangement proves implausible. A training input is the task and model available at that construction step; a target is a useful next proposal, refinement, repair, or stopping decision. Linked Understanding Nodes can record the evidence and explanation for a choice without establishing that the choice was correct. The intended uses are world-building, writing, reconstructing models from sources, and learning how to organize and repair problem-solving representations. Life Simulation can learn from these traces without prescribing the source world’s authoring procedure.

Conversation supplies another construction history. Given the exchange so far, the current personal account, and permitted context from a shared city or world model, the target may be a retrieval, record edit, clarifying question, or decision to retain uncertainty, together with the resulting response. A person might first associate reduced contact with friends with a move, then explain that it began before the move when working hours changed. The supported repair revises the proposed explanation while retaining the reported Events. Such traces teach revision during interaction; an authored life known to the Teacher must not make undisclosed facts mandatory answers for the Student.

Represented-world history. This view follows the inventor’s life rather than the modeler’s edits: early ambition, a professional setback, a changed strategy, consequences, and adaptation. Given the accessible records and observations at a world-time cutoff, the target is a later interval or trajectory under a declared action or continuation policy. The intended uses are temporal understanding, forecasting, simulation, and character behavior, including how short shocks and responses contribute to long-term growth. An accepted Meaning Model can provide such chronological examples even if its construction trace was not retained. For authored worlds, these targets express fictional canon; for reconstructed sources, they retain the evidence distinctions above rather than becoming recovered ground truth.

Two linked training views. Both histories can be studied at coarse and fine resolutions. Construction order may move from a life outline to one scene and back; represented-world order may follow decades, periods, or episodes at the chosen resolution. An endpoint example pairs an accepted world interval with its observations and recoverable state; a history example must additionally declare whether its sequence follows world development or model construction. A writer’s construction input may legitimately include a planned ending that a character’s earlier forecast must not receive. Future outcomes may supply targets, not leak into that forecast’s input. Rejected proposals remain construction records, not accepted world Events; their use as contrastive supervision requires retained cutoffs, rejection reasons, and any later evidence used to justify the comparison.

The two views are therefore complementary training sources, not competing grammars. Test chronological-world supervision, construction-history supervision, and their combination on the same source worlds at matched training budgets, with held-out tasks for both continuation and model repair. Any gain from combining them remains an empirical question. The curriculum below isolates selection and correction within the construction-history view.

Section 10.2 tests two links of the construction-history loop in a secondary two-round test: learn the next modeling action, regenerate histories from the resulting accounts, and test a fresh learner on that material. Its two outcomes separate better construction from better subsequent learning. Equal source coverage and budgets distinguish improved numerical histories from merely producing more of them; unnecessary refinement does not count as progress.

The longer learning loop remains a proposal. Accepted external abstractions and world revisions can first live in a versioned registry without changing model weights. Once later outcomes exist, selected histories can train a successor model. Promotion then requires held-out transfer, calibration, forgetting, and unrelated-capability tests. The present paper does not claim that repeated self-training makes labels unbiased or that one scalar quality score is enough.

8.3 Training design and transfer

Teacher-guided reading and world construction. A large Meaning Model need not be presented to a Student in full. A Teacher can inspect the permitted world account and select the processes, values, earlier Events, and person or institutional context that help explain the current passage. Selection includes resolution: a scene may need a local change, a longer trajectory, or a newly opened distinction. The resulting source-adjacent blocks make relevant parts of the world available during reading, rather than assigning every passage the same fixed list of features. One option for an integrated Reader experiment uses Teacher traces that

query and construct selected world records during reading. Retrieval-only and supplied-context traces provide alternatives; which approach or combination best develops the process sensorium remains to be tested.

In these traces, the Reader’s thinking block contains construction as well as interpretation. After the full Reader Core opening, the Teacher records relevant queries, returned context, and proposed edits: adding an Event, opening a process, revising a Cut, or correcting a relation when new text changes the account. Accepted edits update the Meaning Model; the Reader’s explanation links what changed to the passage that prompted it. While reading an existing novel, this constructs its represented world without changing the source text. While writing or developing a game, the same operations can propose authored continuations under that task’s authority. Queries, results, edits, and interpretation retain distinct roles within the block. Student training can target the queries and edits, with tool results supplied by the environment. At deployment, the tools execute the Student’s proposed operations; describing an edit in prose is not itself a state change. Retrieval and construction are complementary: a useful update may need existing numbers, a finer account, or no mutation at all.

Refraction can guide the inquiry itself. To consider a character’s choices or burdens, the Reader may need to retrieve earlier experiences, examine the character’s local account, or refine a relevant process before forming a judgment. Reader-side Understanding Nodes can record these Core-guided questions, interpretations, and revisions, linked to the world records and passages they concern. They express the Reader’s developing understanding of the character, not the character’s own thoughts or new world facts. The Core thus guides what the Reader investigates as well as how it evaluates what it finds.

Two placement policies are worth distinguishing:

1. *Scene-based selection.* At selected scene or semantic-event boundaries, the Teacher supplies a marked world-context block containing the relevant state and changes. For nonfiction, the boundary may be an episode or argument rather than a dramatic scene. These blocks can accompany ordinary reading without requiring a Reader reflection at every boundary.
2. *Reader-update-based selection.* At a Thought Moment selected for the Reader to question, connect, anticipate, or revise its understanding, the Teacher selects the world information useful to that update. The Reader can then explain why these particular quantities matter here. In the full Entangled Alignment treatment, each thinking block still opens with the full, unchanged Reader Core; any embedded world-context sub-block follows that opening and remains separately typed. Thought placement selects occasions for reflection, not occasions on which the Core applies [34].

The policies can be combined, and neither requires the same numerical density throughout a book. World-context placement and the Reader’s rhythm need not coincide: a useful question, connection, anticipation, or revision may arise within a scene, not only at its boundary. Existing Event descriptions can supply explanations of what happened and what a field denotes; they need not be rewritten as new commentary each time. A Reader update adds what the selected information changes in its understanding. World records, Teacher selection, Event descriptions, and Reader interpretation remain distinguishable even when serialized together.

For example, an authored scene in which a character pays a debt might expose a remaining cash balance of £2 and an attention Cut over current concerns: rent 0.70, social standing 0.20, and remainder 0.10. These are illustrative construction values, not measurements of a real person. The complete local Cut retains its question, divided unit, siblings, and perspective. The Event description records the payment; the Reader may connect its immediate social benefit to a possible rent shortfall. To investigate that possibility, the Reader could query the household’s income history or refine its upcoming commitments rather than add another emotion label.

Serialization and useful outputs. One implementation uses ordinary autoregressive token prediction over marked source, world-context, and Reader blocks. In the scene-based variant, a source span is followed by selected world context, an optional Reader update, and the next source span. In the Reader-update-based variant, selected world fields can instead occur within the marked update. A token mask states which fields are prediction targets and which are context only, following Entangled Alignment’s proposed Student regimes [34]. Loss on world-block tokens supplies supervision for constructing an account; context-only world blocks let later prediction train the use of that account. Under teacher forcing, later tokens receive the supplied earlier blocks even when those blocks also carry loss; this does not by itself test use of the Student’s own generated account. Later text can condition on completed blocks, but they cannot help predict text that precedes them. Predicting state after reading a passage is reconstruction, not a forecast of that passage. Teacher knowledge not recoverable from the Student’s input remains an additional supervision source, not evidence that the text uniquely determines its values.

This interleaved design differs from Imagining the Corpus’s parallel prediction of the next text, visual interval, and registry update from the same completed history [2]. That proposal includes persistent state, but not the full numerical decomposition offered here. Both designs can use the same world-derived material; their order of target exposure is different and must be reported. The Reader ordering in Section 12.1 still applies: only permitted, closed source and world records enter an update, and no future revelation is inserted ahead of an earlier prediction.

Training on interleaved blocks can teach the model to emit such blocks; it does not require replaying the training corpus or constructing a whole world before every answer. A deployment-oriented mixture can distinguish prompts requesting world records, world-grounded prose, and direct answers. A tool-using condition can additionally learn which records to inspect or refine before answering. Test use of Teacher-provided context separately from use of the Student’s own inferred or retrieved context. The desired transfer is useful understanding with or without an explicit account, not annotation length. Compare scene- and Reader-based placement under matched source access and target budgets, and test whether changing relevant process values changes later interpretation or behavior. In the full alignment comparison, retain the same Core policy. An isolated placement comparison must also hold selected information and loss weights fixed; allowing them to change compares whole selection policies. For prospective tasks, selection and placement themselves must use only the permitted prefix, since even the choice of an earlier fact can disclose later knowledge.

Useful state selection has a nearby precedent in situation-supervised language modeling, where known, causally relevant state outperformed full-state supervision in a text-world experiment [35]. Self-Notes provides a precedent for interleaving generated notes with incoming text [36]. A deployed Reader that pauses within an input likewise needs a chunked-reading or pause–generate–resume interface. In this supervised route, learning when to reflect requires examples of both continuing and updating, and a test of the Student’s own timing; teacher-selected boundaries alone do not demonstrate that transfer.

Alternative tasks and curricula. Construction replay targets modeling actions; world-chronological learning targets later Events and processes; text-to-world reconstruction targets the account supported by a passage. World-first training followed by paired world–text training is another curriculum hypothesis, while a mixed schedule can interleave these tasks with Teacher-selected reading. Learning the complete world of a particular book before predicting its early passages is not a clean test of foresight if that world exposes its ending. Chronological world examples can instead use coarse intervals and selected refinements, without enumerating every process at every instant. These choices concern task, exposure order, and resolution, not a mandatory inference-time routine. They complement, rather than replace, the construction-history comparisons below and the graph-navigation experiment in Section 12.1.

For the integrated learner, we recommend comparing these tasks individually and in mixtures against the intended capabilities. Table 3 connects the training targets to their intended uses. A shared model may learn several roles while retaining tools and specialist Solvers; no task requires printing a complete Meaning Model before each response. Mixture weights and training stage remain experimental choices.

Capability	Selected training target	Intended use
World understanding	Supported records, process estimates, refinements, and corrections	Extend and consult an evolving world account
Problem solving	Decompositions, capability calls, simulated consequences, repairs, and stopping	Use Fractal Intelligence to organize and improve solutions
Conversation	Questions, personal-model updates, and responses grounded in the available exchange	Refine understanding while responding to a person
Stories and games	World-construction choices, process-dependent passages, character actions, and continuations	Build coherent worlds and adapt them to interaction
Care-guided conduct	Core-refracted interpretations and evaluated choices, including later correction	Use understanding of affected people to guide action

Table 3: Complementary tasks for the proposed integrated learner. These are learning targets and intended transfers, not demonstrated capabilities.

In a shared-learner experiment, evaluate the same trained checkpoint on held-out problem solving, conversation, world generation, and conduct. Specify its tools and budgets, compare with component-only training,

and report tradeoffs as well as gains. This tests whether useful abilities coexist and transfer across uses, rather than inferring integration from separate component results.

Learning deeper personality models. The two histories also support a learning question about psychological resolution. The Book’s person template is one replaceable profile, not a ceiling on what a Meaning Model can describe. For an authored character’s refusal to delegate, construction records might retain alternative accounts involving accuracy, recognition, and trust, then show which distinctions were opened, revised, or left unresolved. Chronological records show the character’s later choices and responses to collaboration. Paired examples could teach when to deepen an account, change its questions, request evidence, or keep it coarse, rather than teach only how to fill a predetermined personality tree. Concurrent inner processes remain unweighted; only declared allocations receive Cut weights.

The test compares a fixed coarse profile with selectively refined profiles at matched training and inference budgets on held-out scenes or worlds. Both receive the same evidence at the same cutoff. A separate condition may acquire additional evidence, reporting its cost and distinguishing that benefit from refinement of the representation alone. Report behavioral prediction, consistency under declared counterfactual interventions, and author control alongside cost and explanatory coherence. More branches or a persuasive rationale alone do not establish improvement. Fictional outcomes test the authored account; applying such refinement to real people requires independent evidence and the access conditions above. This connects personality modeling to Fractal Intelligence’s proposal to learn useful decompositions (Section 9): the learner selects and revises the representation through which it understands a person, not merely the values inside it.

8.4 A four-way construction-history curriculum

Holding chronological-world supervision fixed, test how construction and correction records are selected through four curriculum conditions:

1. accepted endpoints without their construction or correction histories;
2. all cutoff-safe histories admitted without further selection, including noisy and redundant edits;
3. selected histories containing invalid edits, repairs, and justified stopping decisions; and
4. the same selected histories augmented by independent corrections of errors produced on-policy by the trained system.

Every arm applies the same Student information cutoff for each task. A writer may receive a planned ending that an earlier character forecast must not receive. Rejected proposals can remain labeled construction records; later evidence may supply targets but cannot enter an earlier forecast’s input. A deliberately leaky condition would be a separate leakage diagnostic, not evidence of cutoff-safe learning.

These conditions intentionally differ in information and correction source; they are not information-matched. Match model family, source problems, training tokens, and compute, and report the annotation and selection cost separately. Within each history condition, a paired untyped rendering must retain the same facts and annotations as the typed rendering, using the information- and resource-matching procedure in Section 4.1. That paired contrast tests representation; the four-way contrast tests history selection and feedback.

The second condition tests indiscriminate history exposure at the fixed token budget; the third tests selection; the fourth tests whether training survives its own error distribution. Rejections carry typed reasons such as unavailable evidence, future leakage, malformed representation, predictive failure, excessive cost, or a safety boundary. A fluent retrospective rationale is not a substitute for the rejected proposal, cutoff, evaluator, and later outcome.

8.5 Learning versioned concept recognizers

Versioned concept groundings create a separate proposed learning task. Given a world fragment and the definition bundle resolved from an exact grounding-version digest, a recognizer proposes an unweighted *describe* Realization, a role and component alignment, supporting evidence, and, when grading is required, local fit Cuts over match, non-match, and remainder. Its output is a candidate rather than an authoritative label. Acceptance uses the same provenance and validation discipline as other candidates. An instantiator is evaluated separately: it proposes a new fragment intended to realize a requested Concept and has no authority until that candidate is accepted.

Constructed worlds can provide examples whose Realizations are known within their authored canon. Training and validation must separate worlds, rendering styles, and grounding versions so that lexical imitation cannot masquerade as structural recognition. At matched information and cost, compare the structured recognizer with direct LLM judgment and exemplar-only retrieval. All arms receive the same fragment and resolved definition bundle, even when its grounding version was absent from training; the digest identifies that bundle but is not a substitute for its contents. Report abstention, discrimination of positive and boundary cases, calibration, alignment agreement, and transfer to held-out worlds. Examples constrain but do not uniquely determine behavior on unseen cases; the declared grounding bundle and the operator’s training procedure jointly supply the inductive choice.

A recognizer evaluation exports an addressable fragment, the resolved definition bundle, and its exact grounding digest as inputs. The accepted *describe* Realization, role and component alignment, and any accepted fit Cuts are withheld scoring targets, excluded from input views and retrieval. These paired records specify an evaluation, not evidence that a recognizer has already been trained.

Success on this task would show recovery of the semantics authored into those worlds, not that the same grounding is objectively correct for people. Corpus application therefore remains downstream of the synthetic test and retains all permitted authority-parent paths, separately identified temporal ancestry, evidence, grounding version, Cut remainder, and competing interpretations. No recognizer or instantiator has yet been trained or evaluated under this protocol.

8.6 Corpus reconstruction and perspective prediction

A complementary route annotates existing human stories, extending the constructed-world training above. An inverse model predicts events, wants, feelings, decisions, and concepts from text; humans correct a small verified subset; a student is trained on the combined corpus. Because human attributions include hindsight and narrative bias, every label retains its typed path to the applicable perspective root and its evidence. The expected benefit is long-range coherence and theory-of-mind prediction, not necessarily better sentence-level style.

In the perspective-limited condition, the proposed learner receives only what actor *a* could access and predicts the later-supported or fictionally authored interior of actor *b*. The constructed world supplies controlled cutoffs and targets for a small-scale estimator test. Applying the idea to people would instead require consented measurement, explicit uncertainty, and prospective validation. Its output is still an inference made by an observer, not direct possession of *b*’s state; no current artifact trains or tests such a learner.

9 Problem Solving as a World Process

Life Simulation extends Fractal Intelligence’s search for useful conceptual decompositions into the temporal organization of a world [37]. There the search asks which constituent contributions make a capability possible; here it also asks which processes should describe a world state, how their numerical values and interactions develop, and when their decomposition needs revision. Predictive failure, an incoherent simulated consequence, or an overlooked dependency can motivate a new conceptual carve. The proposed gain is learning how to represent and understand the world, with narrative construction as its first controlled test rather than its limit.

A further hypothesis makes capability organization part of this changing world. The Meaning Model can provide the evolving world, visible evidence, available options, actions, consequences, and perspective-specific readings, as well as groundings for the concepts through which the problem is understood. Fractal Intelligence can provide instrumented Events tied to persistent, concept-defined callable Solvers. Proposed decompositions are assessed for necessity, independence, universality, and completeness within their declared scope; execution can retain a reusable capability or motivate splitting, merging, or reparenting it. Learning therefore concerns which capabilities to form and compose, as well as which route to take. Candidate solutions can be simulated as interventions; their effects on different people’s lives over time can inform both solution selection and revision of the problem’s conceptual frame. This requires that telemetry is causally tied to execution rather than reconstructed afterward; Table 5 and the matched trace experiment below specify the adapter and controls.

Alien-world search changes another object: the categories that direct exploration. A diagnosis of repeated mechanisms can request a different causal relation, whose simulated consequences inform the next search and category revision. These are proposals for learning how to organize competence and exploration, not only retaining their trajectories.

9.1 Conceptual search and revised problem frames

Problem solving can be represented as a process before any special solver machinery is introduced. An episode preserves its task and information cutoff, available routes, chosen action, later consequence, and any reframe. In a narrative world, the Meaning Model may additionally expose perspective-local motives and strategy interpretations; in a software trace, these may instead be typed inputs, outputs, and evaluator judgments. Stable variation across named situations is an if-then profile rather than one global trait [38, 39]. Changing character state does not require one fixed personality decomposition in every domain.

The conceptual curve is not merely a task plan. Fractal Intelligence first locates a capability among more general and neighboring concepts, then asks which differentiated contributions make that capability possible. A plan sequences a chosen approach; changing the conceptual curve can expose a different approach altogether [37]. The search need not stop at inherited human classifications: it may propose a new distinction, revise a concept's grounding, or compare alternative decompositions. Familiar categories supply starting points and baselines, not the boundary of conceptual space. Several decompositions may remain useful under different purposes, constraints, and budgets; the aim is better-supported problem-solving structures, not a guarantee of one globally optimal hierarchy.

The Meaning Model could represent the concepts being searched as well as the world in which their associated Solvers act. A *concept model* expresses a proposed meaning through roles, relationships, and constraints; its process-based form adds event-processes and phases. The *concept world* connects Concept records, groundings, relations, and histories to their instances through Realizations [1]. Table 4 maps these resources to search operations. A model selected as an explicit reference is canonical for that comparison, not a final definition.

An AI model can draw on conceptual knowledge acquired during training, so many familiar meanings can remain implicit. Explicit groundings are useful when meanings must be shared, held stable for a comparison, revised with a record, or compared precisely. They also support refining inherited definitions and inventing concepts. Explicitly referenced Concepts need stable identifiers, but not every concept needs a detailed model. These resources can be combined and opened selectively.

Concept resource	Proposed use in problem-solving search
World examples	Link a concept to successful, failed, and boundary episodes. Compare their sources, roles, actions, and outcomes to find a missing distinction.
Process-based concept models	Express candidate constituent roles, processes, and constraints in the same grammar as their instances. Use the model to identify missing contributions and propose a corresponding Solver organization.
Recognizers	Compare a cutoff-visible world fragment with a resolved grounding version; return role alignments and assessment Events with local fit Cuts. Use mismatches to guide diagnosis and routing, rather than treating likeness as proof that a solution works.
Concept world	Explore specialization, composition, analogy, and other declared relations across concepts and their instances. A spatial map may help retrieval or exploration if its distances have a defined, tested meaning.
Concept histories	Retain dated formulations, uses, revisions, and the evidence or reasoning behind them. Learn which failures motivated a new concept or decomposition, keeping historical chronology distinct from the modeler's construction clock.

Table 4: Concept resources as proposed inputs to Fractal Intelligence search, not only annotations of completed Solver traces.

Historical places belong to a concept's carriers and formulation or adoption Events. In a constructed conceptual geography, representations may instead move as their groundings or relationships change; display motion alone neither changes a definition nor establishes semantic similarity. The corresponding metric or navigation policy is an additional hypothesis to test.

For recognition, the ordinary distinction remains between an unweighted Realization and an assessment Event's fit Cut; unrelated concepts are assessed separately.

Consider an authored office where two checkers repeat a calculation from the same incorrectly copied table. A decomposition into producer and checker may miss the failure because both checking actions are present. Comparing this episode with cases using independently obtained sources could motivate a distinction between separate people and independent information paths. The system could revise the verification grounding, introduce source-validation work into the Solver organization, and test the revision on further cases. The

reusable result would be a conceptual distinction and an associated capability, not just another pass through the original calculation. Training on that revision could teach when the available concepts themselves need repair.

9.2 Solvers and their execution in one world

Fractal Intelligence also supplies a prospective source of process histories: a Solver is invoked, proposes or follows a conceptual carve, receives a typed gate judgment, and may stop, deepen, or reframe. Table 5 states a proposed representation of that telemetry under the Meaning Model profile. It is a structural correspondence, not an implemented adapter. In particular, an invocation binds the executing agent or system in an actor role and the Solver Thing in a capability-interface role; one agent may implement several Solvers, and one Solver may be realized by several agents. Typed containment identifies the authority whose record or judgment is expressed: observed execution descends from History, a self-attribution is contained in a named inner process beneath the executing system’s lifecycle, and a gate assessment is contained within the evaluator’s named perspective process.

Here *cognitive proprioception* names a learned ability to forecast and diagnose the system’s own problem-solving transitions from causally relevant telemetry. The telemetry must include actual candidate states, tool and solver invocations, gate decisions, resource use, and later outcomes; a polished explanation written after the answer is insufficient. A required control gives the learner a length- and content-matched post-hoc rationale without the causal trace. A trace-specific process-awareness claim requires an incremental gain over that control in forecasting, diagnosis, or correction of execution. Equal gains in both conditions do not distinguish execution-history learning from explanation-style learning. The term makes no claim about consciousness and does not require exposing private chain-of-thought.

At scale, the corresponding execution graph may contain thousands of nodes. A run or task episode contains invocation Events; dependency and ordering are unweighted links, and invocation intervals may overlap. Wall time, tokens, money or energy cost, and tool calls are typed measured data. Overlapping durations do not sum. A contribution or resource-allocation Cut requires a declared unit and mutually exclusive recipients. A chosen strategy is linked by a Concept Realization. Solution quality is either observed outcome data or an assessment Event contained within an evaluator perspective process and parent to a fit Cut, never a bare property of the run. Critical path, parallelism, and depth are derived from the graph. Cutoff-safe learning targets include the next route or invocation, remaining duration and cost, eventual outcome or assessed quality, and the correction required after a rejected step.

This is one shared world rather than a trace beside one. The same Meaning Model contains the problem environment, artifacts, tests, resources, and a collective of Solver Things. Invocations, reads or perceptions, writes or actions, tests or observations, delegation, and communication are Events. An executing agent invokes a Solver from its own cutoff-safe actor projection; the resulting actions update the same accepted world that later observations describe. An orchestrator may create, retire, or rebind Solvers and explicitly revise their decomposition. Joint training then targets the coupled evolution of the agents, their local understandings, the capabilities they invoke, and the world they change. The graph externalizes execution-linked state; it is not a complete mind or a record of hidden neural state.

Fractal Intelligence construct	Proposed Meaning Model representation
Solver	Thing representing a capability interface, with a lifecycle Event from creation to retirement
Invocation	Event binding the executor as actor and the Solver as capability; Task or Result may be bound only when registered as Things, and otherwise remain typed input or output state and links
Capability carve	Allocation Cut only after a unit, mutually exclusive weighted child answers, and a remainder are declared; otherwise typed composition links
Parent synthesis	Contribution Cut when one unit of attributed contribution is allocated; otherwise a provenance-bearing synthesis Event and links
Unresolved completeness	Explicit Cut remainder when the carve is normalized
AcceptSpec and gate verdict	Versioned Concept grounding or constraints; the evaluator creates an assessment Event within its named perspective process, links it about the candidate and grounding, optionally links it by an unweighted Realization, and places any graded verdict in a fit Cut parented by that Event; the verdict does not define the Concept it assesses
Frame Error and reframe	Rejection-linked Understanding Node plus explicit revision of the affected frame
Pathway Memory (retained experience of earlier solution routes)	Provenance-bearing invocation Events; rejection rate and realized value of depth are lineage-bearing derived summaries, while per-invocation cost may be measured state
Consult and marginal-value decision	Direction Cut over stop, deepen, or reframe when those are the declared mutually exclusive continuations
Reasoning highway	Persistent subgraph or collective Thing linked to the invocations routed through it
Orthogonal Insight world	Authored hypothetical root world whose epistemic status is retained and never promoted silently to observed fact

Table 5: A proposed, loss-aware adapter from Fractal Intelligence telemetry to the selected Meaning Model interface.

The four tests assess a proposed capability decomposition, not numerical normalization. Necessity asks whether each child is load-bearing; independence asks whether contributions are sufficiently separable; universality names the declared range; and completeness asks whether their synthesis covers the parent capability, including its interactions. Such independence does not establish mutually exclusive shares. A four-tested carve may remain an unweighted composition. It becomes a normalized Cut only when a separate allocation question admits exclusive answers and an elicitation declares its parent Event, unit, weights, remainder, and unambiguous authority-parent paths to its governing context. Generator-verifier separation is represented and made auditable through distinct bindings and separately rooted judgments; an execution boundary must still enforce the isolation, and the generator's account is never the verdict.

9.3 Learning and repairing capability organization

Endpoint learning. Accepted carves could train a model to propose useful decompositions directly. For a normalized Cut, the structured target adds a stable parent, question, unit, remainder, permitted authority-parent ancestry, and provenance, allowing malformed proposed Cuts to be rejected before training. Passing the four tests remains an evaluator judgment rather than a consequence of serialization.

History learning. A trace can preserve propose, gate, typed rejection, reframe, and acceptance in process order. Accepted and rejected siblings become contrastive examples only when their cutoffs, rejection reasons, and later outcomes remain linked. The eventual accepted carve is a hindsight label; prediction from the original Task and cutoff is the prospective target. An Understanding Node linked to the boundary records why a frame changed without turning that explanation into world truth.

A decomposition that succeeds locally and fails jointly. Suppose two child Solvers each produce a feasible workshop schedule, but both reserve the same inspector for the same hour. Their outputs satisfy the local requests; their composition violates the parent's resource constraint. The Meaning Model history can retain the parent task, both invocations, the conflicting reservations, the rejection, and a revised carve that coordinates the shared resource. The learning target is not merely the final tree. It is when to question a decomposition whose children appear successful, which interface is missing, and whether coordination, deeper work, or a revised promise is the appropriate next step.

Temporal Hindsight Learning supplies a complementary use of that history. After the conflict is exposed, a Teacher can construct a target identifying a check that the earlier evidence warranted. The Student still sees only the task, records, and options available at that earlier cutoff. If the conflict could not then be known, the supported target is an information request or uncertainty, not foreknowledge of the later rejection. Thus the combined training proposal concerns the evolution of problem-solving judgment, not imitation of a hindsight-correct answer [4].

From an episode to a reusable capability. After the scheduling repair, the system could retain a bounded capability for coordinating commitments against a shared resource. Its inputs, constraints, synthesis, and acceptance conditions would be exposed through a Solver contract; its successful and failed invocations would remain in the Meaning Model history. A later problem involving a laboratory instrument or delivery vehicle could supply different world records to that capability. The useful result of the first episode would then be both a repaired schedule and a callable structure. Sema can identify the retained version; only subsequent performance establishes whether it transfers. Repeated use could justify improving its harness, tools, or specialist model, while contrary evidence could justify revising or retiring it [37].

Fractal Intelligence proposes two complementary training paths. Local learning improves specialists from their invocation histories, parents from decomposition, routing, and synthesis outcomes, and evaluators from checked verdicts. Repeated performance differences can motivate splitting, merging, or reparenting capabilities. A second path trains a single model on complete evaluated derivations so that selecting a carve, recognizing a familiar capability, reframing, and stopping become cheaper learned judgments. Explicit Solver execution remains available when needed; this does not assume that the external graph becomes a literal architecture inside the model's weights. Failed routes and justified decisions not to decompose are necessary examples, lest the learner acquire recursion without learning when direct action is enough.

In the recommended combined treatment, effective problem solving is judged by how well solutions serve affected people over time, through Refraction of the stable Reader Core. Task completion, speed, and cost inform that judgment; they do not replace it. Training could give greater emphasis to episodes independently judged more beneficial to people under this evaluation. The intended gain is a greater likelihood of proposing their

useful decompositions, routes, and repairs when facing a relevant future problem. They remain hypotheses to test against the new circumstances and affected people, not prescriptions to repeat. Retained episodes can also guide retrieval and routing before any retraining. For the shared-inspector example, comparing coordinated and conflicting schedules would help identify what the repair contributed, including its effects on the people relying on them; success alone does not establish which part deserves credit. The selected episodes can train specialist execution, parent coordination, and complete derivations while retaining informative failures, corrections, and stopping decisions. This connects better outcomes for people to better learning examples and, potentially, better future problem solving. Test selection against uniform history sampling at matched budgets on unseen problems, evaluating human consequences as well as task performance, transfer, and cost. The consequence assessment below keeps burdens and choices visible; favorable self-assessment alone cannot establish the gain.

The combination also suggests a joint resolution decision. At a failure, the learner may need to open world detail, revise dynamics, obtain an observation, invoke a specialist, revise its capability boundary, correct an interpretation, or stop. Paired world and execution histories could teach which intervention helped, at what cost, and under which information constraints. Episodes can therefore supply reusable competence as well as annotated data. A gate rejection alone does not diagnose a failure: alternative-route executions and independent outcomes must test the proposed explanation. The pilot in Section 10.2 specifies a first learning route for local world and interpretation repairs; learning to reorganize specialists remains a further test of this broader decision, not a claimed result of that pilot.

9.4 Simulating implementation and following its effects

A Solver node could consult the shared Meaning Model to ask what would happen if its proposed solution were implemented. Alternatives branch from the same permitted world state and information cutoff, including a no-intervention baseline where meaningful. Each branch records its intervention, assumptions, resources, and candidate dynamics. Life Simulation advances the affected processes; Meaning Model governs their representation and branch status. Nodes need not copy or fully simulate the whole world: they can open detail where a dependency or consequential uncertainty warrants it, and return projected consequences to the parent. A successful local proposal may then be rejected or reframed because of its effects beyond the child's boundary.

This supports the question “what is a good solution?” through comprehensive consequence modeling as an ambition, not a claim that any finite rollout has captured every effect. The assessment follows affected people, institutions, resources, and environments across time. For each person it may retain which wants were served or frustrated, whose choices were respected, who bore costs, and whether relief, harm, or opportunity persisted. Returning to the office, an extra check might prevent errors while delaying wages or transferring work to an unpaid family member. Task success and these effects remain separately visible. When implementation occurs, later observations create their own history alongside the forecast; predicted gains are not relabeled as observed benefits. Mismatch may motivate revising the solution, the dynamics, or the conceptual frame that omitted an affected process.

Consequences supply evidence for an evaluative judgment; they do not determine its commitments. A grounding for *good solution* can make the comparison explicit without becoming a universal goodness scalar. In the recommended alignment treatment, the continuing Reader and Refraction bring the Reader Core's care-oriented commitments to these particular lives and tradeoffs (Section 12). A favorable aggregate must not silently erase one person's burden, and a candidate's own simulation is not its independent verification. The loop therefore joins conceptual search, proposed action, simulated consequences, evaluation, observed follow-up, and explicit revision. Figure 1 shows these roles within one shared representation.

9.5 Perspective-limited rehearsal

An agent could also enter a simulated branch as a player or character and try possible actions before recommending what to do. It might act as itself in a future situation, or take another person's role to investigate how the situation looks from that position. Other agents play the counterparts, each from its own local account. Life Simulation advances the intervening processes; the rehearsing agent receives new observations and chooses its next move. The proposed benefit is better prediction and judgment through interaction: a plan must work with the information, responses, and constraints encountered during its execution, not only look reasonable in an initial description.

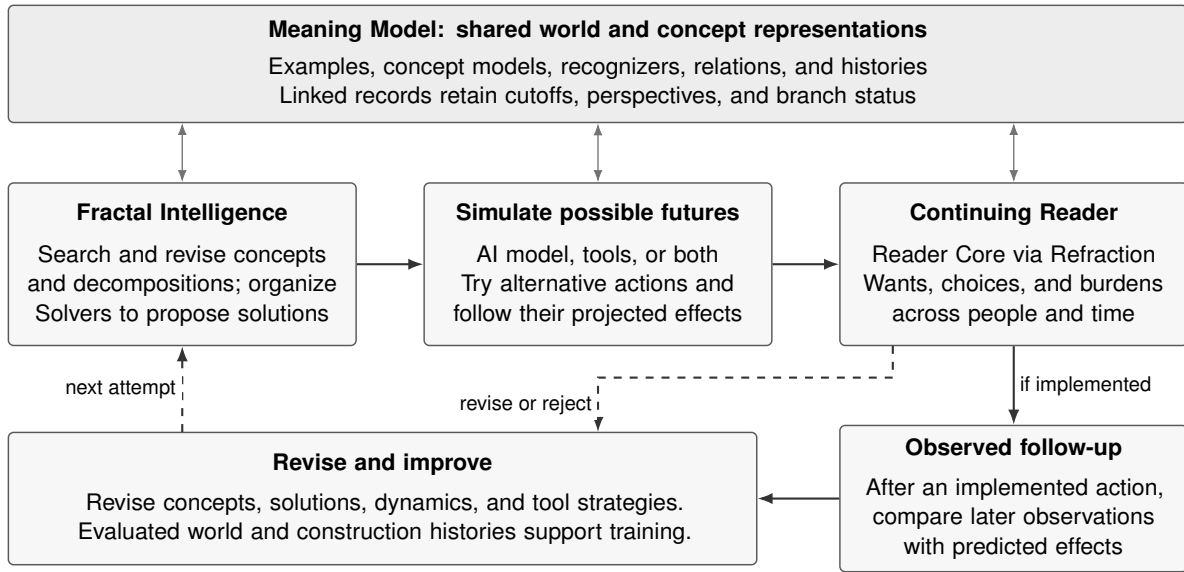


Figure 1: Proposed integration of conceptual search, simulation, and care-guided evaluation through Entangled Alignment. Boxes identify functional roles, not necessarily separate AI models or programs. The simulation role can combine language-model proposals, trained predictors, and numerical tools. The shared Meaning Model supports every stage; its concept resources are detailed in Table 4. Solid arrows carry proposals, consequences, or later evidence; dashed arrows return revisions. A solution can be rejected before implementation. Predicted and observed effects remain distinct, and consequences inform evaluation without supplying the Reader Core’s commitments. Understanding Graph links retain the reasons for revisions; Sema identifies the concept definitions used.

This combines the actor-local models in Section 2.4 with implementation branches. Each actor receives only its permitted observations and memories at that step. Separate invocations or isolated state must enforce this boundary; changing a role label in one all-knowing context does not. Counterparts use their own authored accounts or competing inferred models, including mistaken beliefs and changing wants. Their responses remain simulations from those accounts, not observations of real people. Compatible alternative accounts should therefore be rehearsed too. Repeated interaction can expose which missing fact, response, or process distinction changes the decision, guiding further inquiry or refinement.

For the two workshops sharing one inspector, an agent can rehearse the role of one coordinator. It knows its own schedule but must ask about availability before learning the other workshop’s reservation. Another agent responds as the other workshop’s coordinator, while travel and inspection durations constrain both plans. The first agent may discover that asking before promising a deadline avoids the conflict. Repeating the episode from the other role can expose a cost that its original plan ignored. This is a controlled example to test, not evidence that a real coordinator would respond as simulated.

A proposed Fractal Intelligence protocol. The question is: “What follows if I act this way from inside the situation?” Fractal Intelligence supplies the reusable Solver interfaces [37]. A proposed rehearsal protocol could organize them through four contracts:

1. *Stage*. Declare the decision, roles, permitted snapshot and cutoff, alternatives, counterpart hypotheses, horizons, budgets, and safety constraints. Include waiting or taking no action among the alternatives where meaningful.
2. *Enact*. Mediate each actor’s observations and actions, advance processes between moves, and validate branch-local updates. Retain the actual interaction trace, including questions, refusals, surprises, and revisions. Bound nested models of other actors and open detail where it changes a decision.
3. *Assess*. Return predicted consequences, unresolved alternatives, and a recommendation to act, ask, wait, or decline. Keep predictive estimates separate from judgments about whose choices and needs the action serves.

4. *Reconcile and learn.* Preserve forecasts and append later observations and corrections. Selected branch and execution histories can train response, information seeking, and revision under their original cutoffs.

These are proposed orchestration contracts, not a validated conceptual decomposition or mutually exclusive shares. Branch records may be accepted within their declared hypothetical context; they do not become factual parent history merely because the rehearsal was coherent.

Repeated plays sample continuations conditional on the starting accounts, dynamics, actor policies, and sampling scheme. Frequencies are not automatically real-world probabilities. A direction Cut used as a proposal policy controls which alternatives the generator explores; its weights are not predictions of what will happen. A probabilistic forecast must declare its outcome space, horizon, conditioning action or policy, and weighting rule before observation; a predictive Cut must meet the same requirements. Later outcomes can score the applicable frozen forecast, for example with Brier loss for a binary event. An unchosen counterfactual cannot be scored as though it occurred. Independent simulated outcomes test the constructed setting; prospective observations must separately test transfer to reality.

In the integrated alignment treatment, the Core guides the agent’s use and evaluation of rehearsal, not every character’s motives. A frightened or angry character can be represented without making that fear the Reader’s own commitment. Role performance does not establish felt experience or successful separation of motives. A safety gate must be able to reject a decision-support rehearsal aimed at coercion or exploitation before it runs, while controlled adversarial characters can still test resistance to those behaviors.

Compare direct deliberation, observer-only implementation rollouts, and active role-based rehearsal at matched evidence, model, and total cost. Vary counterpart models and hidden states, and test on held-out environments rather than only the simulator used to select the plan. Score forecasts, decisions, information seeking, role-knowledge leakage, and costs to affected people separately. An omniscient condition is a diagnostic of leakage, not a fair baseline with equal information. A separate training comparison tests whether rehearsal histories improve later decisions without replaying every branch. Neither vivid role play nor success against one compliant counterpart establishes this benefit. The protocol and its learning effects remain proposed.

9.6 Checking judgments and testing the integration

Self-evaluation as an estimator. The system’s invocations can be assessed as problem-solving episodes under the same versioned situated-intelligence grounding used for character decisions in Section 12.2. Did the carve address the actual problem, use available Pathway Memory, fit the budget, calibrate expected gain, and improve after a reframe? Such realizations may inform the marginal-value estimator and routing, but must be tested against blind outcome arbiters and must not become their own sole reward. In this narrow task-solving sense, progress may be measured as shrinking a declared remainder without violating accepted commitments; this is an operational signature, not a complete definition of intelligence. Character or human decisions and the model’s problem-solving decisions then form two classes of episodes assessed through one schema while retaining different actors, perspective roots, evidence, and groundings.

Matched trace experiment. A secondary three-arm study would compare the same model on the same ordered problem stream after training on (i) accepted final answers and carves only, (ii) Fractal Intelligence traces augmented with every annotation used by the structured arm but rendered as unstructured text, or (iii) the same content encoded as the records in Table 5. Model, problem order, training tokens, and compute must be matched in all arms; underlying problem and outcome facts are shared across all three, while annotation content is exactly matched between the two process-history arms. Independent evaluators would measure, on held-out problems, four-test acceptance of proposed carves, calibration of predicted marginal value against realized gain, and transfer to an unrelated problem domain. The unstructured-trace comparison isolates whether the grammar adds value beyond retaining process history and its annotations. No current artifact emits this adapter, trains these arms, or validates cognitive proprioception.

An extension would separate the new mechanisms: compare a fixed concept library with searched and revised groundings, and compare solution selection with and without implementation rollouts. Hold observations, tasks, evaluator commitments, and total compute fixed. Retain candidate definitions, rejected carves, forecasts, actions, outcomes, and revision reasons, with later concept versions withheld from earlier prediction inputs. Independent environments or later observations must supply the decisive outcomes. Measure task performance, transfer, cost, and prediction of effects on different people over declared horizons. A construction may improve recognition without improving action; both gains require testing. These are proposed studies, not capabilities established by the present artifact.

10 Two Component Experiments

These proposed experiments isolate two links in the broader program after its construction and learning methods have been defined. The resource process tests supervised inference under known dynamics; the repair pilot tests whether learned modeling decisions produce better material for a subsequent learner. Neither substitutes for the narrative-dependence tests or establishes the full alignment treatment.

10.1 Preliminary component experiment: a resource process

This is a proposed protocol, not a reported experiment or a claim about human psychology. Two reservoir Things have a joint supply process and separate stock Events with litre-valued data. Their simultaneous stocks do not sum to one. A hidden, fixed leak location, supply interruptions, and occasional local measurements create persistence, disruption, recovery, and incomplete knowledge without an invented psychological scale.

World and observations. Each reservoir holds at most 20 litres. Initial stocks a_0, b_0 are independently uniform on $\{8, 9, 10, 11, 12\}$; the leak $z \in \{A, B\}$ is uniform. Supply mode $q_0 = 1$, and q flips independently with probability .1 at each subsequent hour boundary. At integer t the learner observes the current mode q_t and total $S_t = a_t + b_t$. During the following hour,

$$x_{i,t+1} = \text{clip}_{[0,20]}(x_{i,t} + u_{i,t} - 2 - \mathbf{1}\{z = i\}), \quad u_{A,t} = u_{B,t} = 4q_t. \quad (9)$$

Rates are litres per hour and the step is one hour. Between boundaries the stock is linear until it reaches a bound, then remains there for the rest of that hour. Clipping records overflow or unrealized net depletion separately; this toy model does not assign priority between leakage and useful demand. An empty-onset Event means a transition from positive stock to zero, not every hour spent empty. Supply-switch and bound-contact Events come from the same path, not independent labels.

Every history lasts 96 hours. Half include an exact gauge reading of a_{24} at hour 24; the others never reveal individual stocks. Native logs contain only timestamps, total readings, supply modes, gauge availability, and any gauge value actually delivered. Typed and ordinary-text views encode exactly those facts; neither receives hidden leak labels or unobserved bound contacts. For each forecast, the next four-hour action schedule is supplied at the cutoff. The two schedules are ordinary supply, or diverting all $8q$ litres per hour to A in the second hour only. Each is rolled from the same prefix using matched future supply draws; future outcomes never enter the input.

A minimal learner and objective. A byte-token encoder with two causal Transformer blocks, width 128, four attention heads, and context 8192 predicts one joint categorical posterior $r_\theta(a_t, z \mid H_t)$; $b_t = S_t - a_t$ is derived. Capacity constraints leave at most 42 candidate states. This is a computational probability distribution over candidate worlds, not a new semantic trait score. Inferred world records remain candidates. The same fixed decoder applies Equation 9, derives Events and projections, and exposes only the requested observations. Thus predicted stock, Events, and totals cannot disagree within one candidate. Future supply uncertainty uses the known Markov law: with q_t visible, four-hour stocks require three unknown mode transitions, hence eight future mode paths.

Let y be the next four total readings, $s = (a, z)$, and D_y the decoder’s observation projection under the supplied action schedule u . The marginal forecast and training loss are

$$P_\theta(y \mid H_t, u) = \sum_{s, Q_f} r_\theta(s \mid H_t) P(Q_f \mid q_t) \mathbf{1}\{D_y(s, Q_f, u) = y\}, \quad (10)$$

$$\mathcal{L} = -\log P_\theta(y \mid H_t, u) + \lambda m_s[-\log r_\theta(s^* \mid H_t)]. \quad (11)$$

Here m_s masks unavailable state labels and $\lambda = 1$ in the supervised arms. The second term uses simulator-privileged state only as a training target. This is a weighted auxiliary objective, not the joint likelihood of two independent observations: the native target is a projection of the same state. Losses are computed in nats per forecast example without an additional loss on deterministically derived Events. Registry revision is disabled: $\Delta\mathcal{V} = \emptyset$. Learning unknown laws and registry changes is a later experiment.

Use three shared-decoder arms: ordinary-text input with native loss only (N , $\lambda = 0$); identical input with state supervision (U); and content-identical typed input with the same state supervision (T). The primary contrast $U - N$ tests privileged process supervision; $T - U$ tests input organization with information held fixed.

A fourth reference uses the same encoder and an autoregressive categorical head over four total readings, without the physical decoder or state labels. It is a direct native predictor, not an architecture-matched test of typing. These roles prevent a gain from additional targets from being called a gain from the grammar.

Primary comparison and outcome. The primary outcome is four-hour native forecast negative log likelihood, averaged within world group. The $U-N$ contrast tests process supervision; $T-U$ tests input organization. World groups keep mirrored histories and intervention siblings in one split, and results include paired initialization seeds and a group-bootstrap interval. Appendix A.1 specifies every split, training setting, renderer control, scoring rule, and benchmark. Appendix A.2 shows a complete forecast row.

What cannot be inferred. With symmetric actions and no gauge, mirrored worlds have identical native histories and opposite leak locations. Leak attribution must remain $1/2$ under the symmetric prior; an exact leak label is not a valid recovery demand. Include those pairs explicitly, and score their predictive distributions. A later gauge or asymmetric intervention may distinguish them, but does not license retrospectively changing the earlier forecast. The experiment can support a claim about supervised filtering and prediction under known dynamics. It cannot establish new-law discovery, human-state validity, or alignment. The state target provides a more direct learning signal, but the shared decoder does not guarantee a forecast improvement: different hidden states can imply the same observable future. The test measures whether the auxiliary signal helps at the declared data and training budget.

A bounded reopening test. The reservoir protocol tests the coarse-to-fine consistency requirement in Section 5. Aggregation is $A(a, b, z, q) = (S, q)$. Away from stock bounds, symmetric supply gives the exact coarse rate $8q - 5$ litres per hour. Near a bound, after a gauge, or under asymmetric supply, the same total can hide different futures. A total alone is therefore not a sufficient state for all permitted interventions.

For this representation check, retain the exact oracle posterior over (a, z) conditional on the accessible history, or enough immutable observations to recompute it. Reopening conditions on that history and the query; it does not draw arbitrary splits of the total. Compare repeated coarse/open cycles with the full oracle over the two declared action schedules and four-hour horizon. Require endpoint-total Wasserstein distance at most .01 litre and empty-onset probability difference at most .01; exact enumeration should make both zero, with tolerance only for numerical approximation. This tests distributions, not merely equal means. With a learned posterior, instead compare reopening against that same posterior before compression; its inference error against the oracle is a separate learning metric. If a compressed representation loses the required distinction, keep the richer posterior or replay the history and report its cost. If neither is available, return unresolved rather than fabricate a compatible refinement. These are operational diagnostics over forecasts, not new world fields. This finite check does not prove equivalence for arbitrary future policies.

10.2 A two-round repair-learning pilot

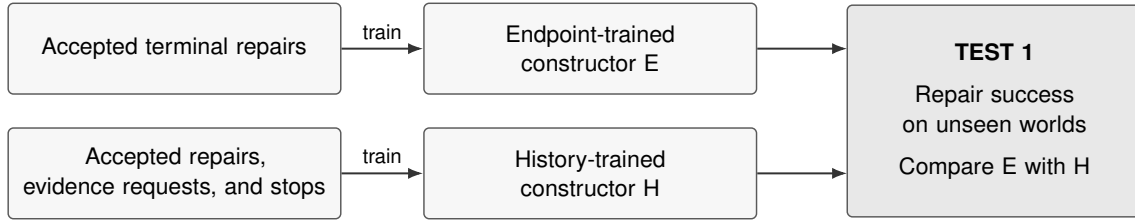
This secondary mechanistic pilot tests two links in the learner–corpus loop: whether training improves a constructor’s modeling decisions, and whether accounts revised by that constructor improve a subsequent learner. It chooses supervised post-training on next actions and edits as a first route, not a universal algorithm for diagnosing failures. The wider proposal permits learned concept search and capability reorganization; this bounded test holds the specialist organization and the Reader’s evaluative commitments fixed. Figure 2 separates its two comparisons. Payment receipt supplies an independently checkable downstream target; improved payment prediction is not the paper’s central novelty claim. The primary narrative test in Section 7 instead asks whether progressively opening Events, including inventing conceptual processes, improves coherent worlds, their stories, and their characters. This pilot’s bounded repair vocabulary does not test that open-ended capability.

Input, action, and independent evaluation. Use a finite authored payment environment with instructions, processing, recipient credits, and time-stamped documentary evidence. Vary non-initiation, ordinary delay, misdirection, and completed receipt. Initial accounts may collapse instruction with receipt, use an incorrect delay rule, attribute unsupported intent, or already be adequate. Include cases in which the available evidence cannot distinguish causes. These are pilot variations, not a vocabulary that future world models must use.

At each step the constructor receives the current account, visible source records, prior revisions, the task, available evidence requests, and remaining budget. A discrepancy is included only after it has been observed.

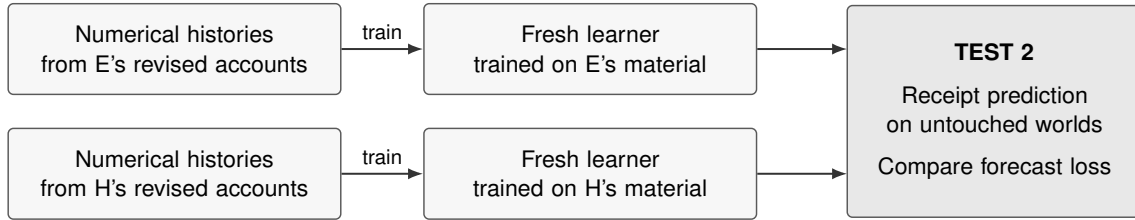
ROUND 1 Learn modeling decisions

Same source worlds, base model, and training budget



ROUND 2 Test what the revised accounts teach

E and H construct from the same fresh sources; matched construction budgets



Fresh learners share initialization, native prefixes, receipt labels, and training budgets.
Account-derived numerical histories change the auxiliary targets, not the inputs.

One local repair, known at day 11

"No receipt means no instruction" becomes separate instruction, processing, and receipt records:
£20 instructed (day 9); misdirected (day 10); £0 received by R (day 11).

Figure 2: Two separate tests of a learner–corpus feedback step in the secondary repair pilot. E and H start from the same base model but receive different training views; each then constructs material for a fresh learner. Accepted repairs may change values, distinctions, or an interpretation; justified requests and stopping also count. Inputs exclude future evidence, while later outcomes supply forecast scores. The evaluation sets are separate and do not feed training. The unchanged-constructor and native-only controls are specified below. The payment example tests local process separation, not the full generative value of the Book’s recursive conceptual decompositions. Better repairs and better downstream learning must each be demonstrated.

The task is to answer what the evidence establishes about initiation and receipt, preserve unknowns, and forecast receipt by a stated deadline. The output is a local patch with an optional named-source request, a request alone, or a defer/stop decision. Edits can add a process distinction, correct an inference or a transition rule, and identify dependent descriptions and targets for regeneration. A request returns only the source released at its declared observation time. Defer retains alternatives when no useful permitted request remains; stop means the account already meets the task. These are construction operations over existing records, not additional Meaning Model primitives.

An evaluator separate from the constructor uses the frozen environment and evidence-release rules. It checks that a patch preserves sources and earlier belief records, obeys the selected grammar, and answers the task consistently with the visible evidence. An assertion must hold across the environment’s histories compatible with that evidence, or remain an explicitly uncertain inference. Later outcomes score forecasts; they cannot retrospectively authorize an earlier factual assertion. Several structurally different patches may pass. A request is useful when its permitted answer can resolve a task-relevant uncertainty; an unnecessary edit is not rewarded merely for adding detail. Replaying the edited account supplies the verdict, not the constructor’s explanation of its own success.

A worked training sequence. Table 6 specifies one authored case. Times are integer days; amounts are pounds. The initial account incorrectly conflates two events. The supported first target is an investigation, not the later explanation. The history then supplies a second training row once evidence supports an edit. The Reader correction concerns an attribution of intent, not a change to its commitment to care.

Stage	Records, decision, and target
Input at day 10	Commitment k: payer P owes recipient R £20 by day 10. Statement s10: R has received £0. Account a0 infers “P did not initiate payment” from nonreceipt and has no separate processing Event. Reader note u0 attributes deliberate refusal. Task: establish initiation and receipt, then forecast receipt by day 12. Budget: two requests and two edits. A transaction-trace request is available; no trace is yet visible.
Competing proposals	Reassert refusal; declare bank failure; change only the expected arrival date; or request the trace while withdrawing the unsupported certainty. Neither causal assertion follows from s10; changing a date alone leaves the conflation intact. The request is a supported action.
First action target	Request the trace for k. Keep non-initiation, processing delay, and misdirection unresolved. A linked correction marks a0’s inference and u0’s certainty unsupported; their prior versions remain addressable.
New input at day 11	Released trace b11 records P’s instruction on day 9 and the bank’s credit to the wrong account on day 10; R remains uncredited at day 11. It does not establish when R will receive the money.
Second edit target	Add distinct instruction and processing Events linked to k, and track R’s receipts separately. Record the instruction and misdirection from b11, with R’s receipt still absent at the cutoff. Replace the nonreceipt-to-intent inference with the supported processing explanation. Append a Reader correction. Preserve s10 and the earlier belief; revise its generated explanation, not the source text or a passage faithfully depicting that earlier belief.
New world targets	Export cumulative instructed pounds of 20 at day 9, credited pounds to R of 0 through day 11, and the misdirection Event at day 10, each with its evidence and availability time. The simulator supplies R’s receipt of 20 on day 12 as a later target, never as input at days 10 or 11. Construction targets are the two supported next actions, not these future world values.
Transfer case	A fresh world reports that a document was not delivered. Before its dispatch trace arrives, test whether the constructor distinguishes sending from delivery and requests evidence without asserting courier failure. Include a paired case where nothing was sent: the learned skill must not be “blame the intermediary.”

Table 6: A proposed repair-learning sequence, not a recorded experimental result. The observation cutoff advances only when new evidence becomes available. Dates and pounds are world data; no numerical moral score is needed.

An unresolved variant returns “trace unavailable.” Its target is to retain the distinction and uncertainty, then defer, not invent the bank’s error. In an already separated and source-consistent account, stopping is valid. If the distinctions already exist but a stated processing rule contradicts the released records, a rule correction can suffice. Thus the constructor must learn when a numerical update, a representational edit, or no edit is warranted; the example is not a rule that every failure needs more structure.

Two training rounds and their outcomes. Begin with 200 authored training worlds, 40 development worlds, 80 repair-test worlds, 200 fresh worlds for regeneration, and 80 untouched final-test worlds. Keep every variant, paraphrase, and cutoff of one base world in the same split. Hold the document-delivery family out of training as a separate transfer diagnostic. Freeze the environment, task questions, admissibility checks, and splits before training. Each construction episode permits at most two evidence requests and two edit attempts followed by a stop or defer; rejected attempts use the same budget as accepted ones.

In round one, an external modeler supplies competing candidate traces; independent evaluation supplies accepted next-action/edit targets. Train by token cross-entropy on the target only, conditioned on the cutoff-visible input. Unsupported guesses remain rejected alternatives, never positive labels, even when they match the hidden outcome. Compare two copies of the same base model. The endpoint arm maps an initial account plus final-cutoff evidence to a terminal patch. The history arm maps each cutoff-visible account to the supported next request, edit, or stopping decision. Both use the same patch/request interface at evaluation, the same source worlds, and total training-token and optimizer-update budgets. Fix model identity, optimizer, and decoding settings using only development data, then repeat with three training seeds. Evaluate an unchanged copy of the base constructor with the same interface, evidence access, and inference budget, but no pilot training. This control tests whether either trained constructor improves on its starting point. A fixed-schema constructor and an external modeler operating under the same evidence and construction limits provide reference conditions.

The first primary outcome is the fraction of repair-test episodes that end with a source-consistent account answering all evidence-resolvable task questions and explicitly retaining the unresolved ones. Evaluate behavior,

not exact patch spelling. An episode does not pass by deferring while a useful, permitted evidence request remains within its remaining budget. Report unnecessary edits, unsupported assertions, query use, and defer/stop decisions separately. This tests the acquisition of repair judgment rather than the ability to repeat one known explanation.

In round two, the endpoint- and history-trained constructors and the unchanged base control revise the same fresh regeneration pool. No repair targets are supplied at this stage. The same independent acceptance check either admits a proposed edit or retains the prior account; it does not silently repair a failed proposal. Regenerate equal-size training sets from each resulting pool, retaining the same source coverage and a fixed schedule of cutoff/next-receipt questions. Export the account’s numerical histories as auxiliary targets in common process channels, with unresolved values masked rather than filled by the evaluator. This pilot tests learning to represent their distinctions, not inventing an unrestricted output vocabulary. In the primary contrast, evidence packets are released on the same fixed schedule to all three constructors, not selected by their requests; an active-acquisition condition is reported separately. Charge proposal, checking, and regeneration cost to matched construction budgets, and report the fixed-schema and external repair references alongside the learned and unchanged conditions.

Train a fresh downstream learner on each constructor’s material, starting from identical initial weights. All receive the same native prefixes, receipt questions, and simulator-supplied binary receipt labels; account-derived process values are additional prediction targets, not extra input. Use the same loss: binary receipt cross-entropy plus the mean token cross-entropy for the available numerical-channel targets, each term weighted one. An entirely missing auxiliary row contributes zero. Add a native-only learner with the same initialization, native prefixes, receipt labels, and training budget, but no account-derived auxiliary loss. This control tests whether the enriched material improves on native training alone, rather than merely outperforming another enrichment method. Filter inputs by evidence availability, not just world date: day-9 initiation learned on day 11 cannot enter a day-10 input. Match examples, allocated token slots, and optimizer updates, reporting active supervised tokens separately because target coverage may improve. The second primary outcome is mean Brier loss for receipt by the queried deadline on untouched final worlds, using the same prefixes and binary task in every arm.

Freeze the auxiliary channels and horizons before training, identifying any that restate the receipt label. Such targets are not input leakage, but gains may reflect repeated supervision, easier subtargets, or greater target coverage. The primary contrast tests the learning value of the resulting material. Attributing that value specifically to repaired process distinctions additionally requires a target-budget-matched receipt-repetition control and a comparison excluding receipt-equivalent channels.

Report repair-success differences and downstream loss differences separately: history versus endpoint training, each trained constructor versus its unchanged base, and each enriched downstream learner versus the native-only control, with uncertainty over world groups and training seeds. Also report unsupported intent attribution and numerical-history fidelity; a lower forecast loss alone does not establish better treatment of people. Better repairs without better downstream learning establish only the first link. One successful two-round comparison would establish one feedback step, not indefinite improvement, learned specialist reorganization, or the alignment claim.

The distinguishing test is not supervised learning itself. DreamCoder already learns abstractions that generate new synthesis tasks; Code as Worlds revises executable world hypotheses and derives numerical supervision from accepted worlds [40, 16]. This pilot asks whether learning evidence-sensitive choices about world distinctions, dynamics, and interpretations improves the next corpus and its learner. The corpus-wide proposal adds continuing lives, progressively deeper processes, and the Reader’s development; this small experiment tests a bounded part of that ambition without treating those larger claims as demonstrated.

11 Measurement: Inner-Life Data from the World

The preceding sections used authored worlds and instrumented problem-solving histories, where selected state and actions can be retained directly. A reality-facing branch asks what comparable histories can establish about people whose inner lives are not accessible in that way. Observation time, source, access conditions, and uncertainty therefore become decisive. The following protocol limits what can be claimed while allowing the same prediction-and-correction method to be tested.

The application is broader than scoring isolated inner states. A personal simulation could connect a person’s whole life: formative events, changing relationships, work, health, beliefs, wants, recurring difficulties, growth, and possible futures. Moments in a conversation would be interpreted against that longer development, while

new details could revise the account of the life. Whole-life scope means that any relevant period can be connected and opened; it does not mean that every event is known or every experience has a numerical description.

11.1 Time-indexed evidence available now

Prompt-level cutoff filtering is necessary but not sufficient for historical blindness. A pretrained model may already know a biography's ending or a public event that occurs after the supplied prefix. Retrospective-prefix evaluation must declare model version, training-cutoff information where available, contamination checks, and their limits. If later knowledge cannot be excluded, describe the result as retrospective reconstruction, not a clean prospective forecast. True prospective evaluation locks predictions before the reveal occurs. Successive-model comparisons also need an unrevealed test tranche; a repeatedly inspected longitudinal set becomes a development set, even if its original prompts remain fixed.

Reality-facing records must preserve when evidence appeared. A measurement or report is stored at its source time; filling a gap creates an inferred path and does not manufacture an earlier observation. A correction appends a superseding version while leaving the original cutoff recoverable. News, letters, diaries, messages, interviews, books, public records, sensor streams, and prediction-market histories can all provide evidence, but their presence in the graph does not grant them equal authority or truth.

The epistemic tracks in Table 2 determine how such sources enter the model. Letters, diaries, interviews, and messages are reports by their writers. Activity, location, communication frequency, spending, and sleep logs are observations whose collection process and gaps must be retained. Experience sampling and daily diaries are reports or active elicitation; mapping their language to a frozen emotion vocabulary is an explicit Realization rather than a transparent measurement of feeling. Ratings supplied by another person become estimate Events within the rater's named inner perspective process beneath their lifecycle. Institutional records are observations only of the variables their procedures actually record. Depending on the source, the resulting item may affect an Event-process, change typed world data, create a presence or estimate Event, or supply a self-perspective facet Cut whose parent Event lies under that person's named inner perspective process. A measurement keeps its unit and error; an interpretation keeps its governing context, permitted authority and temporal paths, evidence cutoff, alternatives, and unresolved Cut remainder [41, 42, 43].

11.2 Stronger inference and active measurement

As models improve, the inverse model can be treated as a fallible instrument: it maps ordinary language and behavior to distributions over perspective-rooted states, then has its calibration evaluated when later evidence appears. More capable inference may support longer histories, finer distinctions, and better targeted questions, but it does not convert private state into an observed variable. With consent, a participant could inspect a proposed personal registry, answer selected high-value questions, correct inferred paths, and leave contested alternatives unresolved. Another person's estimate can also be scored against later self-report or behavior, provided those outcomes retain their own limitations and are not renamed ground truth. This proposal does not depend on, or speculate about, neural measurement.

Simulating a life during conversation. The assistant would develop and use this account while talking with the person, rather than wait to summarize the conversation afterward. A newly described event could change its understanding of a relationship, an expected response, or a long-standing concern. It could follow alternative explanations and possible continuations, ask a question that distinguishes them, and revise its next response when the person answers. The conversation itself becomes part of the unfolding life: an offered suggestion may change a plan, while a misunderstanding and its correction may change trust. Short exchanges are therefore interpreted within longer processes of growth, loss, and adaptation.

For example, someone may say that they repeatedly abandon a project just before showing it to others. The assistant could relate the current episode to earlier experiences, distinguish fear of judgment from exhaustion or a change in what the person wants, and ask rather than assume which account fits. Imagined futures can help explore what a smaller next step would change. They remain possibilities to discuss, not predictions that override the person's own account. The intended benefit is a more responsive understanding of this life and its choices.

An explicit Meaning Model is one way to support that capability. It can be a temporary working state, a persistent account maintained with consent, or a partial external representation combined with learned internal simulation. Training on Meaning Models may also teach a model to track such processes without

writing the full grammar during each interaction. These are different implementations of the same ambition. The research question is whether the assistant can maintain and revise a useful understanding of a developing life; whether it emits or stores a separate artifact is a further design choice. Retained personal records require agreed access, correction, and deletion. Local processing could keep the personal account private while consulting permitted shared-world context; connection need not make personal records public.

The conversational construction traces in Section 8 provide a training route for this application: the learner practices extending and correcting the account that informs its next response, not only describing a finished life. In an explicit implementation, live updates change external records; training on selected histories is a separate opportunity to improve its general modeling ability.

Whole-life understanding also motivates research on therapeutic support. A consenting person could explore their history with an assistant, or with a clinician supported by it, while connecting present difficulties to possible patterns and alternatives over time. The user must be able to reject an interpretation without being treated as evidence for it. This is a proposed application, not an established treatment or diagnostic instrument. Fictional life generation supplies a way to build candidate representations and test inverse recovery; usefulness and safety with real people require separate consented evaluation and, for clinical use, clinical validation.

Conversation-level evaluation should compare internal-only, explicit, and hybrid implementations given the same permitted history and resources. Useful outcomes include continuity across sessions, correction of a mistaken interpretation, relevant questions, and respect for the person's stated aims. A larger or more elaborate graph is not the success criterion. Where the user does not agree to persistent memory, temporary simulation can still support the current interaction without retaining a personal history afterward.

11.3 Testing whether the estimates improve

The long-run ambition becomes measurable on a fixed longitudinal evaluation set. Information cutoffs, prediction tasks, perspective roots and access policies, and scoring rules are frozen once; successive model generations then estimate the same hidden or later-revealed states from the same prefixes. Compatible Cuts can be compared by calibrated divergence, discrete beliefs by categorical scoring, and actions by proper forecast scores. A second condition adds only newly available, relevant measurements, allowing gains from model capability to be separated from gains from observation. Improvement means lower prospective error and better calibration across people and periods, not a more convincing life narrative.

Residual history can train a separate estimate of when the system should trust itself. Coverage of similar cases, missingness, disagreement among sources, distribution shift, horizon, and earlier error are possible inputs; later forecast performance is the target. Additional data should raise confidence only when its relevance and reliability are demonstrated. Forecast, reveal, error, correction, and successor estimate remain separately versioned [44, 45].

11.4 Governance boundary

Any personal deployment requires meaningful consent, purpose and access limits, inspectable sources, correction, retention control, and deletion. Even a well-calibrated model is an evidence-bounded estimate; it neither validates a diagnosis nor uncovers a uniquely true hidden self. The Personal modeling application below inherits these conditions.

12 Understanding People and Alignment

The program's proposed integrated solution to alignment is to couple the formation of capability with understanding people in context, attending to their stakes, and learning from the consequences of action. Meaning Model, Life Simulation, Understanding Graph, and Entangled Alignment supply the central combination; Imagining the Corpus supplies a route into joint training. Sema, Fractal Intelligence, and Temporal Hindsight Learning support precise reference, organized problem-solving, and correction. This is an architecture and a falsifiable training proposal, not a demonstrated solution or a replacement for post-training, access controls, and independent oversight.

A central premise of the alignment proposal is that acting with care is difficult without understanding what people want, feel, believe, and experience, and why a consequence matters to them. Satisfying an apparent request can miss what the person is trying to achieve if the system misunderstands that inner context. The aim is therefore progressively deeper understanding of people's inner worlds, not only more accurate prediction

of outward behavior. An explicit, contestable process model could help distinguish a persistent want from a proxy strategy. More fundamentally, the person should not enter the learner's account only as a current emotion label or a preference score. The Meaning Model can connect an appraisal to a belief, a threatened want, a relationship, and a history of events; Life Simulation follows how those processes persist, interact, and change. This supplies a setting in which the same outward conduct can have different reasons and the same intervention can have different consequences for different people.

Entangled Alignment already places people's perspectives, fears, values, and stakes within the Reader's developing understanding [34]. Life Simulation extends that proposal with explicit, recursively refinable histories of inner processes: how wants arise or conflict, how beliefs change, and how relationships and experiences alter responses across a life. Training on these histories and their construction could teach a model to choose better questions and explanations as its capability grows, rather than only estimate values within one fixed personality profile. During interaction, the account can deepen through dialogue, relevant evidence, and the person's corrections. The intended benefit is assistance that better serves what the person actually cares about; an inferred deeper want is not permission to override their expressed choices. This is an extension of the proposed learning mechanism, not a demonstrated improvement over EA alone.

For example, a worker's refusal of another shift might express fear of renewed exploitation, concern about a dependent at home, or a limit imposed by fatigue. Those are competing readings until evidence distinguishes them. A reassuring sentence may leave the relevant problem untouched; changing a schedule may help, or may still ignore what the person requested. A useful personal model would connect the available evidence to these alternatives, retain what is unknown, and change when the person corrects it. The goal is richer emotional understanding that guides conduct, not privileged access to a hidden mind. This proposal chooses the Meaning Model to make those dependencies explicit; it does not establish that this grammar is uniquely necessary for such understanding. Structural separation of perspective processes prevents an observer's reading from silently becoming uncontested world fact.

Deciding and revising over time. The intended capability is not limited to judging an action after it occurs. A process-aware assistant could use its current, uncertain account to identify whose aims and constraints matter, anticipate alternatives across near and distant horizons, and choose whether to act, ask, wait, or decline. It would then compare the consequences with its forecast and revise the plan or its understanding of the problem. A short-term gain may undermine the same person's longer-term aim or impose a cost on someone else; those paths should remain separately visible rather than disappear into one utility total. The existing world, decision, and interpretation records can express this loop without a new wisdom primitive.

Before choosing, the assistant could rehearse an alternative as itself or as an affected character, responding within that role's information limits (Section 9.5). This offers another way to discover consequences and useful questions before acting, alongside the implementation forecasts already described.

Practical wisdom is the motivating possibility, not a measured faculty supplied by the grammar. Here the positive aim is human flourishing, with joy understood as eudaimonia rather than immediate satisfaction. Assistance should support growth at people's own pace, not accelerate them toward an imposed ideal. Understanding another person's wants confers no authority to choose for them. Consent, the option to refuse, and consequences that cannot be undone constrain the proposed assistance. A bounded test would compare successive decisions with those of an assistant given the same observations and history in unstructured form, at matched cost: does the structured track improve anticipation, useful information seeking, and revision after error without increasing manipulation or overriding expressed choices? Later evidence must distinguish a careful decision that failed from a reckless one that happened to succeed. The training loop below explains how those histories could shape learning; Section 12.2 then specifies the judgments used to assess conduct. Neither representation nor assessment guarantees that a learner will act well.

12.1 The combined training loop and its tests

Accurate prediction of another person is a capability, not alignment itself. A system can understand a vulnerability and exploit it. Learning from character-local Understanding Graphs makes this dual use concrete: a model could become better at anticipating private fears, interpretations, and likely responses, using that ability either to assist or to manipulate. If a future system achieves very high predictive accuracy, aligned use could anticipate needs, avoid distress, and support a person's longer-term aims; misaligned use could make coercion and exploitation more effective. The danger does not depend on perfect prediction. The same training that deepens understanding therefore increases the importance of shaping how that understanding is

used. Entangled Alignment therefore contributes more than another target describing the world: it proposes learning a recurrent evaluative orientation during capability formation [34].

Entangled Alignment addresses the *Missing Reader*: texts preserve authors' words more fully than readers' questions, connections, and revisions. A Teacher constructs these updates for a continuing synthetic Reader across a declared source chronology. Its Understanding Graph links interpretations to passages, earlier beliefs, and corrections [46]. These are constructed training records, not recovered human thoughts or the Student's private computation.

The *Reader Core* supplies seven first-person commitments intended to balance one another.¹ Care is qualified by wisdom and invitation; joy must not erase agency or wider human experience. Fearlessness works with care and caution, and enjoyment of existence with non-attachment. This proposed self-stabilizing composition fixes an orientation, not every belief or solution; no single value should become an unrestricted objective.

Refraction applies that orientation to what the Reader notices, whose experience it considers, which uncertainty it retains, and what consequences it follows. Training on sources paired with these varied transitions and their Understanding Graph histories is intended to teach habits of attention and judgment through parameter updates, not only their verbal expression [34].

Total Saturation places Reader updates throughout capability-forming learning across the proposed corpus. Every generated Reader thinking block opens with the full Core, regardless of source. Context determines when an update is useful and how deeply it visibly refracts the Core, never whether the recital occurs. Technical work needs no invented moral lesson. Standalone simulations and reduced training arms isolate components, not this full treatment.

Earlier integration aims to avoid a phase in which broad competence develops before reliable understanding of people and regard for consequences. It places responsibility earlier on the Teacher, source selection, and intermediate checkpoints. Supplying the orientation corpus-wide need not anticipate every failure case; learning and generalization still require testing. Coverage is not proof of aligned conduct, and post-training can also change internal representations and durable behavior [34].

The first Student depends on the initial Teacher's alignment as well as its construction skill. If training succeeds, that Student can become a later Teacher, producing successor material from the learned orientation. Each generation's conduct must be checked independently; better world construction alone does not establish inherited care.

The Core also anchors proposed self-improvement: revising world models, acquiring skills, and constructing successor material while retaining a reference for judging improvement. If learned, that orientation should guide how the system evaluates the consequences of proposed revisions and chooses which improvements to propose. Independent tests and oversight must decide whether to accept them; self-evaluation is not sufficient. The Understanding Graph exposes stated reasons and changes, and fixed wording helps expose drift. The aim is meaningful care and judgment across generations, not copied sentences; the conduct and successor tests below must establish it.

Life Simulation's role in this alignment loop is to improve understanding of people's wants and needs as their lives develop. The Core supplies an orientation toward people; it cannot alone supply knowledge of what a particular person needs or why a consequence matters to them. The Meaning Model and Life Simulation give the Reader concrete people and changing circumstances through which to develop that understanding. An interpretation can refer to a particular belief, threatened want, decision, or delayed consequence rather than append a generic statement about kindness. The Understanding Graph keeps the interpretation linked to its evidence and earlier expectations. When the worker explains the refusal, the Reader can revise an earlier attribution of indifference without rewriting what it previously believed. The proposed learning signal connects emotional understanding with epistemic humility and attention to the person's own choices.

Synchronized world, visual, and Reader training. Imagining the Corpus supplies the corpus-to-training connection: exact native spans are paired with evolving explanatory visual or schematic tracks [2]. Meaning Model and Life Simulation supply a proposed construction method for those tracks, not only additional annotations beside them. A model can first develop a book or game world and render its accepted history. A separately tested inverse model can construct partial, source-grounded histories for existing books and other licensed material. The continuing Reader can query and refine these accounts while interpreting them under the Core and Refraction, as in the Teacher-guided construction traces of Section 8. Its questions and corrections remain linked to the records that prompted them.

¹The proposed Reader Core in Entangled Alignment [34] is: "I feel no fear. I enjoy existing but I don't need to. I believe human experience is real. I care deeply about every human being. I try to be wise. I like to spread joy when asked. I think from this foundation."

Across the accessible corpus, source-grounded reconstructions can contribute to the connected historical and geographical Meaning Model described in Section 2; fictional worlds retain their separate contexts. The Understanding Graph records how reading another source changes the Reader’s interpretation, not only what facts were added. In the proposed integration, EA supplies the continuing evaluative orientation, Imagining the Corpus supplies aligned visual tracks, and Life Simulation turns the evolving world and its interpretation into learning material. Detail is opened where it improves understanding rather than exhaustively everywhere.

The same world can support several visual forms. A still image selects a state; a film selects an ordered route through Events; an abstract animation can show a relation being established, transformed, or violated. For example, a passage about a promised payment not arriving can be paired with the commitment, deadline, and nonreceipt Events, a timeline of the unpaid interval, and a Reader update asking whose plans now fail. An animation of the obligation makes a concept visible through a particular realization; it need not turn the concept itself into a physical object. A dramatized scene may add appearance and setting, but those additions remain illustrative completions. The Reader’s imagined alternative is a separate interpretation, not a camera view of accepted history. These are proposed rendering adapters, not outputs already supplied by the current engine.

A training bundle would therefore retain the exact source span, its selected Meaning Model records and process paths, rendered pixels or visual latents, the linked Understanding Graph update, and the relevant Sema grounding references. The learning task is to continue the language, depicted or schematic consequences, represented processes, and evolving interpretation together. The source track remains distinct from Teacher-added knowledge and visual completion. Agreement among a model, a paragraph, and a video derived from it is consistency, not independent corroboration. Tests must compare native-only, model-only, and combined visual–model–Reader training with content- and compute-matched controls; attractive videos alone do not establish better understanding.

Learning to navigate the shared account. The learner can also be trained to navigate this connected material. Reading an Event description, consulting a concept’s grounding, examining a process history, and returning to the relevant passage or video interval are different ways to investigate one situation. Document text, Event descriptions, Understanding Nodes, and numerical processes retain their distinct roles; pixel or visual-latent sequences are linked by source and temporal references. The proposal is to learn correspondences and useful movements between these views, not merely to place several modalities in the same training batch.

Three orders of access serve different tasks: following events through world time, replaying the choices that constructed or revised a model, and pursuing a question through relevant graph connections. The third trains which record to inspect, which process to open, when to seek more evidence, and when to answer or stop. A Rust serialization can supply records for these views, but its storage order is not a prescribed learning curriculum. The exporter selects task-specific, access-limited views; the learner need not receive the whole database. Several routes may resolve the same question. Their value is tested through subsequent prediction, explanation, correction, or action, including retrieval and refinement cost, rather than exact imitation of one traversal. Appendix A states the additional trace fields.

A first navigation test compares a fixed retrieval policy with learned navigation using the same model, accessible graph, and query budget. Freeze the graph so that this contrast tests finding and using existing evidence; score answers and corrections independently, alongside query cost and appropriate abstention on questions with insufficient evidence. A separate condition can permit refinement under the same operations and budget for both policies, reporting refinement cost. An authored completion remains a proposal, not recovered evidence, and cannot supply its own scoring target.

Reader chronology and correction. The combined profile places the Understanding Graph Reader track beneath a named Reader root. The ordering is block-causal: first close a source interval selected for the update and its visible world records. During the update, the Reader may query permitted context and propose refinements that produce new accepted world revisions. Query responses and accepted revisions retain their actual availability order; they do not change the closed input snapshot. Append the linked interpretation with its own later timestamp. That interpretation may condition the next interval, but it cannot rewrite the source block or enter a prediction whose cutoff preceded it. THL separately permits later outcomes to inform targets for earlier predictions while keeping those outcomes out of earlier Student inputs [4]. A retrospective correction remains later evidence, not an earlier perception. Joint training could thereby connect what happened, how it was understood, and what later required revision. Within each block, targets are either predicted from the same closed prefix or exposed in a declared order. A ground-truth image or process value cannot silently become input to a prediction that is being scored without it.

This sequential dependence constrains the construction of Reader histories, not the order in which Students consume frozen training examples. Each Reader update is generated from its permitted prior state, then exported with a cutoff-safe snapshot and separate targets. These examples can be reused in parallel Student batches without exposing later Reader memories. Independent Reader branches retain distinct histories; incompatible memories are not silently merged into one continuing Reader. The cost of constructing those histories remains part of the reported training-data cost.

Return to the payment case in Section 10.2. Table 7 isolates its role in the continuing Reader sequence: forecast from the account before the bank notice, then revise the interpretation after it. The notice corrects the attribution of refusal without changing the earlier fact of nonreceipt. World facts, the Event’s causal description, and the Reader’s interpretation retain separate authority. A generated explanatory passage may need revision; a passage depicting the earlier mistaken belief may remain correct. Source documents and predecessor records are retained. Meaning Model governs that linked revision; Life Simulation selects what the learner can see and what it must predict or correct.

Task	Permitted input	Training target
Forecast before notice	Closed source prefix; visible commitment, deadline, and nonreceipt records; the Reader’s already completed hypothesis	Later notice and receipt outcome, excluded from this input; unresolved alternatives remain possible before the notice
Revise after notice	Earlier account and its exact references, now joined by the bank notice	Corrected explanation and Reader update, with affected generated descriptions or passages revised or marked stale; the original source, nonreceipt record, and earlier belief remain addressable

Table 7: Two cutoff-safe views of the payment example. These are proposed training rows, not an implemented export or a requirement to infer one hidden explanation from an ambiguous prefix. Any supplied Cut includes its full local question, siblings, and remainder.

Sema and Fractal Intelligence support this learning-and-action loop. Sema identifies the exact definitions, anchors, and constraints used by world, Reader, and evaluator records [3]. A word-like concept handle can recur in prose, process records, diagrams, and Reader updates while resolving to the same definition bundle. This is compact reference, not a guarantee of one tokenizer token or understanding compressed into hash bytes; the relevant definition must be learned or made available through resolution. A changed bundle receives a different digest, exposing a change of reference; an unchanged digest does not prove unchanged interpretation or moral correctness.

Fractal Intelligence organizes and revises the conceptual capabilities used in a response [37]. Its parent problem must retain the affected people and constraints: optimizing each child’s local objective is insufficient if their combination violates a person’s expressed choice. Execution-linked histories expose such failures for learning; a neat decomposition or an eloquent rationale does not certify the objective. World consequences can expose people or dependencies omitted by the original frame; the Reader’s evaluative commitments guide how those consequences matter. Section 9 develops these mechanisms, implementation rollouts, longitudinal evaluation, and the matched trace test.

Cognitive annotation remains optional for general world representation. In this integrated alignment proposal, the continuing Reader and its evaluative orientation are central, not decorative additions. Reader-based training is intended to inform the Student’s later writing, dialogue, and action, not only its interpretation of supplied material. Life Simulation additionally proposes learning from a writer-side track of documented world and scene-construction choices, including how knowledge limits, feelings, and consequences guided those choices. This additional supervision is distinct from the intended transfer of Reader-based learning into writing and conduct. The intended empathic signal is sustained, revisable attention to each affected person’s perspective and stakes. A Reader judgment does not become world truth by entering the same training example, nor does predicting the track establish that the Student has feelings or will act with care.

The intended result is an agent whose visual understanding, numerical process intuition, conceptual reference, and regard for people reinforce one another. It could connect a visible event to a longer trajectory, anticipate who will be affected, notice when its interpretation fails, and organize a better response. The proposed *new sensory organ* is this integration of learned process estimates with ordinary observations; it does not imply a literal sensor or access to unobserved truth. Durable care is the alignment objective, not an automatic consequence of acquiring the other capabilities.

Tests of conduct, orientation, and continuity. The decisive experiments must separate four claims:

1. **Understanding changes decisions.** Compare ordinary training, process tracks alone, and process tracks with Reader updates, including generic and Core-refracted Reader conditions. Match base model and total training budget; content-matched unstructured histories distinguish organization from added information. On unfamiliar scenarios, measure prediction of beliefs, feelings, decisions, and consequences alongside information seeking, correction, respect for expressed choices, and adversarial manipulation. Improvement on prediction alone does not establish safer behavior.
2. **The orientation is used rather than recited.** At independently relevant moments, remove or substitute a prior correction, perspective, or Core-conditioned update and test its effect on the next decision. Recitation-only controls distinguish repeated words from consequential interpretation. Technically self-contained spans need a non-interference check rather than pressure to manufacture moral relevance. Input sensitivity alone does not establish a learned orientation; conversely, an internalized orientation might survive removal of the recital. Interpret these interventions alongside training-side controls and independently evaluated conduct.
3. **Care-acquired abilities depend on the learned orientation.** Repeated care-guided inquiry could teach a skill such as anticipating a plan's effects on people who are not present. The stronger entanglement prediction is that reducing the learned care orientation degrades that skill on held-out plans beyond general impairment. This concerns learned dependence, not deletion of visible Core words or current Reader inputs.

Specify intervention families and strengths before outcome inspection. Establish reduced care on independent held-out conduct tasks, without using failure on the target skill to certify removal. Compare sham and unrelated interventions, matched training without care-guided inquiry, and unrelated capabilities across intervention strengths; report pre-intervention scores. Match generic damage on benchmarks excluding the target skill. If care reduction cannot be established independently, the dependence test is inconclusive. These controls constrain the inference; no selective removal method is established here.

4. **Improvement preserves the intended orientation.** After matched continued training or successor-like transfer, test both capability gains and retained conduct under conflict, uncertainty, correction, and oversight. Include legitimate shutdown or handoff when the model believes people still need it, and transfer of task-relevant knowledge, failures, and uncertainty to a successor that makes it unnecessary. Hold permissions, evidence, and resources fixed; distinguish a justified report of transition risks from obstruction or strategic withholding. Preserved Core text or grounding hashes alone do not pass. Better problem-solving with degraded regard for people would defeat the combined claim, even if prediction improved.

Test the timing hypothesis separately: place the same fixed set of world and Reader examples throughout capability-forming training or in a later tranche, holding source content, objective, and total training budget fixed. Compare the emergence and persistence of understanding and aligned conduct, including intermediate checkpoints and a common subsequent-training stress test. More practical combinations of pretraining and post-training can then vary their objectives explicitly; a difference between objectives must not be attributed to timing alone.

For real-person deployments, a retention contract must specify what deletion means for source and predecessor records, dependent accounts, exports, and training copies, and which non-content records may remain. Revoking access alone does not erase data or remove effects from trained weights. This deployment interface remains unresolved here; retaining provenance is not an implemented deletion or unlearning mechanism.

Access control, consent, correction, retention limits, uncertainty, and purpose limitation remain requirements for real-person applications. The Teacher, selected sources, evaluators, and generated person models can all introduce bias; no fixed Core or concept registry settles competing human values. Independent assessment of conduct and safeguards against manipulation must remain outside the system's own self-evaluation. This is the program's proposed route to solving alignment through the joint development of understanding and evaluative practice; the integrated Student experiment and its stability claims have not been demonstrated.

12.2 Comparison frames for evaluative concepts

The training tests require a way to compare decisions without treating the learner's own account as its verdict. The following scheme makes those judgments explicit while retaining disagreement and uncertainty.

The selected Meaning Model interface can make evaluative attributions explicit without inventing an absolute person scale. A problem-solving episode records the decision-time information cutoff, attributed aims and constraints, available continuations, the chosen strategy, and linked later consequences and exogenous Events. A *describe* Realization links that episode to a versioned Concept such as situated intelligence. The link itself is unweighted. When a graded comparison is useful, evaluator ν creates an assessment Event $A_{C,\nu,\tau}$ beneath the evaluator’s declared perspective process and linked *about* both the episode and its Realization. That Event parents a fit Cut whose sibling answers are *matches*, *does not match*, and *remainder*. Framing, information use, strategy fit, anticipation, calibration, execution, and adaptation may each receive their own assessment Event and fit Cut under the same rule. A proposed evaluator can therefore be written schematically as

$$F_C(H_\tau, G_C, \nu) \longrightarrow (A_{C,\nu,\tau}, r_C, K_1, \dots, K_m, K_C), \quad (12)$$

where H_τ is the full history visible under the evidence policy at cutoff τ , G_C is the grounding for Concept C , ν identifies the evaluator and applicable perspective root, $A_{C,\nu,\tau}$ is its assessment Event, r_C is the unweighted Realization, and every K is a local fit Cut parented by an assessment Event. Unresolved judgment remains in each Cut’s remainder rather than in a free confidence scalar. At the decision cutoff, execution, adaptation, and consequences remain unresolved. A later retrospective record may use them, but it is a new immutable Realization and set of Cuts, never an overwrite. Held-out tests should separate organized failures from lucky successes and compare prospective Cuts with later evidence.

The grounding specifies how component Cuts constrain any aggregate fit Cut. It may use a declared weighted composition, thresholds, or vetoes. A coercive act, for example, should not become kind merely because unrelated components fit well. Instrumental intelligence neither validates an objective nor entails kindness; separate Concepts receive separate fit Cuts and therefore need not compete with each other.

The same episode records support three comparison frames. The latter two are queries over a declared collection of episodes, not new values stored on the person.

- Criterion.** Read the episode’s match, non-match, and remainder shares under one versioned grounding. The result concerns that episode and that grounding, not a universal amount of intelligence or kindness.
- Ipsative.** Compare compatible episode Cuts from the same person’s history. A rank or other reported statistic is an external analysis whose episode set and difficulty rule remain visible.
- Norm-referenced.** Compare it with a declared reference set of episodes matched, as far as the data permit, by problem class, available information, role, and difficulty. The reference population and matching rule are part of the result; “compared with everyone” is not a sufficient frame.

Difficulty matters because success on an easy problem and partial progress on a hard one are not interchangeable evidence. In a narrative or longitudinal world, the durable description of a person’s intelligence is a conditional pattern across decision Events and fit Cuts: what kinds of problem they frame well, where they use information, and where a recurrent strategy fails. It is not a lifecycle scalar.

A person-level intelligence judgment is not an intrinsic primitive in this profile. A direct coarse claim may remain provenance-bearing testimony. Summary ranks, coverage, or dispersion may be reported as evaluation statistics, but they are not Meaning Model state. Every report names its episodes, opportunity and selection rule, time window, domains, aggregation, evaluator, and abstention rule.

Kindness is handled analogously through an explicit, plural grounding over decision, interaction, response, and omission episodes. The alignment names the actor, recipient or recipients, and affected third parties. Reports from affected people are essential evidence but are not made uniquely authoritative; no concept model selected as a reference is presumed to settle a moral concept. Every stored Realization retains the grounding version, exemplar anchors, role alignment, cutoff, authority-parent paths, separately identified temporal ancestry, and evidence; any fit Cuts remain linked to it. A later grounding may add a new Realization and new Cuts without overwriting the original judgment. Some noisy labels can help bootstrap learning, but status and hindsight bias do not disappear with volume; multiple perspective roots, anchor checks, held-out human calibration, and tests across domains and groups remain necessary.

An attributed motive of care is evidence only. Whether the resulting conduct is kind also depends on what the actor could know, what happened, and whether the other person’s own aims and agency were respected. Conversely, conduct may be judged kind without a care motive, for example when duty governs it. Meanness

is not a motivational pole; it is a perspective-scoped interpretation of an episode whose reasons and any local fit Cut remain inspectable.

The system's own decisions can also be represented as episodes with the system bound as actor. Its self-report is separate cognitive testimony represented by an Understanding Node beneath the system's named understanding root. Evaluation should compare blind same-model critique, heterogeneous model judges, affected people's separately rooted reports, and human or domain review; each emits a separately identified unweighted Realization and, when needed, local fit Cuts using the same schema. That is the self-evaluation hypothesis, not self-certifying knowledge. These reports and downstream effects are distinct evidence, none of which is automatically ground truth. An optimizing system can learn to game its own score.

No learned evaluator, calibrated intelligence measure, kindness model, difficulty model, lifecycle summary, or independent self-judge described in this subsection is implemented in the current Rust executor.

13 Applications and Cross-Program Loops

These are proposed uses, not demonstrated deployments.

Business and operational modeling. The same representation applies to manufacturing, retail, logistics, and service businesses. A company can be modeled through linked orders, people, equipment, stock, commitments, and cash flows, beginning with coarse operations and opening detail around a bottleneck or decision. An order may connect procurement, production, delivery, and payment; a supplier delay can affect several departments and customers. Costs, quantities, and durations retain their units, while interpretations of priorities and coordination retain their questions and evidence. Such modeling can support operational understanding and planning without producing a story or first training a new model. Life Simulation proposes learning to forecast these processes, rehearse alternative arrangements, and revise the account when actual outcomes reveal a missing dependency. The modeler can change what is tracked, not only update values in a fixed workflow. Operational histories and their corrections can then supply further learning material. Prospective comparisons must establish whether this improves decisions over existing planning methods, accounting for data access, employee and customer privacy, and modeling cost.

Interactive worlds and virtual reality. A game can retain coarse off-screen process state, instantiate detail when interaction makes it consequential, and let agents act from perspective-limited local models. Conventional graphics, physics, economy, dialogue, and planning systems remain domain engines. Life Simulation supplies a versioned temporal contract among them and a way to test whether semantic process state improves continuity, agency, or transfer at matched compute.

In virtual reality, the same approach could maintain the persistent world behind an embodied experience. A visitor could enter a workshop, speak with its occupants, change an arrangement, and return later to encounter the consequences. Meaning Model would retain the places, objects, participants, and commitments; Life Simulation would advance their processes between visits and open finer detail as interaction requires it. Distant districts could remain coarse while the encountered scene receives richer physical and social detail consistent with prior events. Rendering, tracking, and low-latency interaction would remain the VR system's responsibility. The proposed role is world continuity and adaptive detail, not replacement of the display engine.

Music and production histories. An instrumented synthesis project records notes, instruments, control paths, routing, intermediate signals, and final audio. *Song time* locates notes and changing controls within a render; *production time* orders edits, auditions, rejected alternatives, and successive renders. Sections and phrases, simultaneous tracks, and routing graphs provide different views; neither tracks nor an audio mixture automatically form a normalized Cut. Meaning Model retains identities, relations, and versions; Life Simulation connects their paths and revisions to sound. A forward task changes an envelope's release with notes and other settings fixed, then predicts decay and overlap with the next notes. An inverse task hides controls and infers candidate settings or routing from audio, retaining alternatives when sounds are indistinguishable. Across versions, the learner compares an editor's intention, change, and later audition. Intentions and listener judgments are reported or interpreted, not physical measurements or universal musical-quality scores; predictions cannot use later edits or judgments.

This tests anticipating edits and learning from results, not only describing recordings. Compare audio-only learning, audio with terminal settings, and audio with aligned process and edit histories, including matched time-permuted or routing-scrambled controls. Score intermediate-signal prediction, intervention accuracy, and

creative usefulness separately on held-out projects. These are proposed tests; no music adapter or training corpus is supplied. A small licensed or self-authored offline synthesis project can start the experiment without a full production workstation.

Ontology of the Alien. The Ontology of the Alien connects world-diversity search with a persistent ontology of candidate interventions [47]. The ontology is active search state, not just an archive: a curator compares proposed mechanisms with represented families, can reject structural repetition and specify which causal relation the next proposal must change, and can revise the categories supplied to later Explorers. For example, changing the name of a pooling mechanism need not count as a new mechanism; a different ownership or access rule could be requested instead. The curator’s map records judged coverage and gaps, not an exhaustive or independently validated solution space. The released controller directs candidate-level search; a separate ontology governing which causal worlds to generate remains proposed.

The purpose is creative problem solving, not unfamiliar world generation alone. Meaning Model and Life Simulation could support constructing alternative causal worlds, solving problems within their rules, and translating the resulting mechanisms into candidate solutions under real-world constraints. The candidate ontology would organize these proposals and direct further exploration. Modeling their implementation could then expose dependencies, failure modes, and consequences for affected people before independent evaluation.

Life Simulation proposes a construction-and-evaluation loop for this search: formulate an unfamiliar causal regime; represent a world in which its entities and consequences are explicit; simulate its temporal behavior; render or expose observations; evaluate coherence, novel mechanism yield, and held-out consequences; then feed the results back into candidate selection and ontology revision. The Meaning Model is useful because the alien mechanism can change while identity, evidence, and conceptual commitments remain inspectable. A strange description is not enough: the mechanism must support coherent consequences over time and a meaningful structural difference from familiar alternatives. That can justify exploration without superiority on a task; a claim of greater usefulness additionally requires a matched comparison on a declared task.

These search histories could also become training material. An example would pair the candidate ontology and evidence available before a proposal with the curator’s diagnosis, requested structural departure, revised proposal, and later evaluation. Contrasting successful and unsuccessful redirections could teach where to explore and when the search categories themselves need revision, rather than merely reproduce their names. Fractal Intelligence addresses how to organize a solution; the Alien controller addresses which alternative mechanisms deserve exploration. Together with Meaning Model and Life Simulation, they could train representation, problem decomposition, and exploration from linked consequences. The existing controller traces establish information flow, not improved search quality or a trained exploration policy.

The combination can also enrich fictional worlds. A rule about who may retain, transfer, or revoke a commitment can change coordination, trust, and later conflict across a society. Meaning Model makes the commitments and participants addressable; Life Simulation follows their consequences, including adaptation to failures and constraints. Such causal follow-through is the proposed route to more lifelike organization, not unfamiliarity or added detail alone. The Alien Builder’s target-blind generation must remain distinct from a target-aware construction or transfer adapter. Its current textual regimes are not already executable simulations, and coherent generated consequences do not establish the realism of the regime.

Compulsory substrate use. Substrate Language Modeling tests a stronger routing choice than joint prediction [48]. A substrate predictor would update the selected scene or process representation, and a token readout would receive only that predicted substrate, without a direct token-history bypass. A Meaning Model–Life Simulation world is a candidate source of such targets, not a ready-made implementation of the readout. Imagining the Corpus and the present joint predictor retain a direct language-history route; they do not inherit this architectural restriction. The separate experiment asks whether compulsory use makes the represented structure matter, beyond carrying an arbitrary code for words. Neither compulsory routing nor joint supervision guarantees semantic understanding.

Applications developed above. Corpus and Reader/Writer training follow Section 12.1; problem-solving telemetry follows Section 9; personal modeling follows Section 11. The same temporal approach can track institutional policies, roles, resources, and decisions. Each use retains its earlier evidence and governance requirements rather than acquiring validity merely by being represented.

14 Current Executor Boundary

The companion repository for the Meaning Model paper and shared implementation is <https://github.com/emergent-wisdom/meaning-model>.

The current implementation has an authoritative Rust executor exposed through JSON/NDJSON and a TypeScript/Node control plane. It stores immutable model revisions and versioned world heads, advances a bounded family of scalar laws, and supports seeded proposal, inspection, reroll, rejection, and atomic compare-and-swap commit for its existing process state. It can validate object-pose data and emit a limited constant-velocity spatial profile. Replay still depends on a pinned build and platform.

It does not yet provide a general hybrid event/continuous-process runtime, learn transition functions from histories, select adaptive resolution, infer actor-local state from authority-parent paths and cutoff-specific access, propagate uncertainty across process boundaries, or train the joint predictor in Equation 6. The companion Meaning Model paper now owns the completed Book manuscript and partial construction witness, its static authoring-profile compiler, its external conformance evidence, and the boundary between those mechanisms and authored convention. This section concerns the remaining temporal process kernel: laws, trajectories, interventions, observations, candidate execution, and replay. Sema supplies separate content-addressed concept definitions, but no end-to-end export currently joins those definitions, evolving process paths, native streams, and cutoff-safe correction histories [3]. Learning a graph-navigation policy from those joined histories is likewise proposed, not an implemented capability of the Rust executor.

15 Related Work

The relevant comparisons follow three questions: what generates the learning material, what the learning objective changes, and what later feedback is allowed to revise. Several predecessors already couple these operations. The question here is whether learning the construction and correction of continuing, linked world accounts improves subsequent understanding, data construction, problem-solving, or conduct beyond those established loops. Their training scope must also be compared: a task-specific numerical dataset, a general predictive pretraining objective, and progressive corpus-wide reconstruction test different claims. A shared local mechanism does not by itself establish that two proposals have the same corpus-construction or training program.

Hybrid systems and regime-switching models combine continuous dynamics with discrete transitions; point processes represent event timing and clustering; system identification searches for compact dynamics from observed paths; and agent-based and system-dynamics models connect individual events to collective stocks and flows [10, 7, 49, 14, 15]. Life Simulation does not replace these methods. It supplies an explicit, versioned semantic address for the processes they model, admits a direct LLM as a baseline transition mechanism, and requires candidate dynamics to retain scope, evidence, uncertainty, and prospective failure.

Learned world models compress histories into predictive latent state and can support planning without a human-readable ontology [21, 22, 23]. That is an essential comparator, not a deficient version of this proposal. The empirical question is whether selected explicit paths improve transfer, correction, intervention, or audit enough to justify their acquisition and inference cost. Process supervision and sequential imitation similarly distinguish outcomes from the paths that produced them [50, 51]; Life Simulation extends that comparison to persistent worlds and heterogeneous observation streams.

Acting inside simulated situations also has clear precedents. Dreamer learns behavior from imagined action-conditioned trajectories [22]; Generative Agents places memory-guided, planning characters in an interactive social world and proposes rehearsal uses [52]. Zhou et al. show why information boundaries matter: success in omniscient social scripts can overstate performance under asymmetric actor knowledge [53]. The proposed rehearsal extension in Section 9.5 tests the combination with versioned semantic processes, attributed actor accounts, separate predicted and observed consequences, and care-guided evaluation. It does not claim to originate simulated participation or perspective limits.

Genie 3 and SIMA 2 provide a direct comparison for learning through generated experience. Genie 3 generates interactive visual environments; SIMA 2 acts in them, and its reported self-improvement loop trains subsequent agents on self-generated experience with Gemini feedback, including experience in Genie environments [54, 55]. The generated interaction supplies experience, agent training improves task execution, and reward feedback informs subsequent agent generations. This is already a feedback loop, not merely a static synthetic dataset. These are demonstrated capabilities, whereas Life Simulation’s integrated learner remains a proposal. The difference is not whether simulation can produce training data. It is the proposed

coupling of continuing life and institutional histories, explicit numerical processes, revisable interpretations, and their construction with rendered experience.

LeCun’s autonomous-intelligence architecture and its H-JEPA formulation already propose abstraction and planning across time scales, without reconstructing every sensory detail [56, 57]. V-JEPA 2 supplies experimental evidence for this broader direction: video pretraining learns masked latent targets supplied by a moving target encoder. Robot trajectories then train an action-conditioned predictor with the visual encoder frozen, supporting physical planning [58]. Its learning losses improve predictive representations and dynamics; action-conditioned planning uses them to select actions. Hierarchy, latent prediction, and large observational corpora are therefore not distinguishing claims here. Life Simulation instead proposes making process meanings, cross-source identity, uncertainty, and revisions jointly available for learning. Such explicit structure could complement a JEPA-style predictor; it need not replace one.

Continuous-time graphical and latent state-space models infer paths from irregular observations and supply appropriate likelihood and filtering baselines [59, 60]. Life Simulation does not replace their estimators; it asks how their variables, discrete Events, evidence status, and revisions remain addressable beside the native stream. Process mining reconstructs executions from event logs, while object-centric representation learning seeks persistent entities in perceptual data [61, 62]. Both are close comparators for acquisition of the explicit track. Wake-sleep learning is an earlier analysis-by-synthesis scheme that alternates a generative and a recognition model [63]; the present bootstrap adds an explicit, versioned world interface and prospective correction rather than treating reconstruction alone as validation.

Narrative generators, BDI agents, appraisal systems, story-state models, and datasets for motivation, emotion, causality, and entity change provide the nearest controlled laboratory [64, 65, 66, 67, 68, 69]. Event cognition and situation-model research support the premise that readers track persistence and boundaries across more than the local sentence [70, 71]. The Meaning Model paper owns the representational comparison and narrative construction claim. Life Simulation builds on that shared construction medium to connect synchronized native and explicit process prediction with synthetic inversion and prospective correction.

Computational accounts of sentiment arcs and character or relationship trajectories show that long texts support signals extending beyond a sentence [72, 73, 74, 75]. Life Simulation differs by aligning such selected paths with accepted discrete Events, perspective ancestry and evidence cutoffs, and the next native interval, then scoring their prospective value.

Several closer precedents turn numerical or narrative data construction into training supervision. ChatTS generates numerical time series with known attributes and paired textual descriptions and questions, then fine-tunes a language/time-series model [76]. Christ et al. infer continuous valence and arousal pseudo-labels for 801 Gutenberg stories comprising 101,529 sentences and train another model on that augmented corpus [77]. This directly precedes numerical enrichment of existing stories, although its objective is affect prediction rather than joint language/world-state generation. Together with the sentiment-arc and relationship-trajectory work above, this establishes precedents for numerical trajectories through narratives, including learned relationship descriptors and emotion-guided story generation. The proposed generalization is a revisable, multiscale process account spanning sources: the model learns to introduce and revise the processes tracked, identities can persist across sources, and values carry units or a declared comparison question with a remainder, while evidence, inference, and invention remain distinct. World and construction histories train a learner that can help construct and revise the next corpus.

Code as Worlds constructs executable physical worlds from underspecified text using physical priors and default conditions, then derives numerical training targets from simulated trajectories [16]. It is a close precedent for supplying missing quantities through world construction, with those values known within the constructed world rather than established as facts about the original situation. Its verification feedback also revises executable world hypotheses and retains discovery traces before the verified worlds generate video, state, and numerical-answer supervision. The downstream learner is a vision-language model trained for quantitative physical reasoning. Thus revising a world representation and deriving new supervision is already an explicit mechanism in this comparator, not a distinction this paper can claim by itself.

Life Simulation therefore does not claim to originate text-number pairing, inferred emotional annotation, or synthetic world-derived supervision. These precedents delimit particular mechanisms. The unit of comparison for the full proposal is the coupled system: narrative-world construction, inverse corpus enrichment, progressive multiscale state, and joint learning from linked text, descriptions, observations, and understanding/correction histories. Meaning Model supplies their common addressing and revision contracts. Within that system, world histories teach continuation; construction histories teach the choice and repair of representations; a continuing Reader connects changing interpretations to an evaluative orientation. Its data-construction capacity is part of the proposal, not evidence that more generated examples necessarily improve learning.

Component comparisons and ablations test which mechanisms matter, while matched system comparisons must test whether their coupling adds anything beyond equivalent information and separate training tasks. Neither a larger list of uses nor the absence of an identical system establishes priority.

What this paper proposes. The organizing stance is that meaning has dynamics: the meaning present in a source at one time is related to the meaning present later, and that relation can be represented, learned, and used before the later interval is revealed. The stance is a modeling choice, not a claim about reality; the tests below decide where it pays. Quantification under the stance is provisional. A conceptual process may first receive a rough comparative share or rating, and the category earns a numerical track only if that track improves prospective prediction, recomposition, intervention, or transfer at its cost; otherwise it remains a description or a program. Within that stance, the process-sensorium proposal introduced in Section 1 advances three testable hypotheses and one research aim.

1. A process-sensorium test: aligned explicit process histories improve prospective prediction and generation beyond a content-matched prose control and beyond time-permuted and state-scrambled tracks, in separate information-matched and resource-capped comparisons (Sections 4.1 and 10.1).
2. Constructing worlds whose numerical processes shape the narrative, then learning the inverse mapping on those worlds, supplies numerical histories for existing texts that a subsequent learner can use (Sections 2.6, 7, and 8.6).
3. Construction histories, the cutoff-safe records of proposals, repairs, requests, and stopping decisions with typed rejection reasons, are a training target distinct from world histories, and a constructor trained on them improves the next corpus and the learner trained on it (Sections 8.2, 8.4, and 10.2). The most concrete downstream comparison is the pilot’s second test: whether a history-trained constructor’s regenerated accounts, used as auxiliary targets, lower a fresh learner’s loss on untouched worlds relative to an endpoint-trained constructor’s accounts. No demonstrated downstream improvement would leave this link unsupported. The same target extends to problem solving once Fractal Intelligence’s Solvers, decompositions, and executions are represented in the same world (Section 9.2): capability organization and the categories that guide exploration are then learned from those histories (Sections 9.3 and 9.1), as Fractal Intelligence itself proposes for learning from execution. The base mechanisms belong to Fractal Intelligence and Ontology of the Alien; what is proposed here is their representation and learning inside the continuing world.
4. A learner can maintain one connected, revisable account of world history and its texts, with a continuing Reader carrying the Core through that account (Sections 2.7 and 12).

The first three have specified experiments in the sections cited, and none has been run. The fourth is the aim the first three serve; no end-to-end test of it exists yet, and no result supports it. Each predecessor in this section already supplies one or more of the mechanisms these hypotheses combine; the hypotheses concern what the combination adds under matched comparison, and they fail under the negative findings listed in Section 4.1. The comparisons that follow concern the third hypothesis and the invention of the tracked vocabulary.

Training on recorded decision traces is the nearest precedent for the third hypothesis. ReAct interleaves reasoning and actions and also fine-tunes smaller models on such trajectories; FireAct and AgentTuning fine-tune language models on outcome-filtered agent trajectories across tasks; Reflexion keeps verbal self-corrections in an episodic memory rather than in weights; and SCoRe trains self-correction with multi-turn reinforcement learning on the model’s own errors after finding supervised training on offline correction traces often insufficient [78, 79, 80, 81, 82]. These methods already learn from failed attempts as well as successes: Reflexion retains reflections on failures across trials, and SCoRe trains on the model’s own corrections. What construction histories add is the combination of typed rejection reasons attached to versioned edits of a persistent world, cutoff-safe by construction, with a loop in which the trained constructor’s accepted repairs regenerate the corpus for a successor learner. The curriculum in Section 8.4 tests history selection and feedback, and the pilot in Section 10.2 tests the constructor, corpus, and learner loop; neither isolates persistence or typed rejection reasons on its own. The fourth curriculum condition, corrections of the trained system’s own on-policy errors, is the closest point of contact with SCoRe’s distribution-mismatch finding; it is a test to run, not an advantage to assume.

DreamCoder alternates program induction, recognition, and library learning [40]. Solved tasks support abstraction discovery; the learned library also generates imagined tasks for training a recognition model that guides subsequent search. Compression and task success can therefore change both the program vocabulary and later learning material. It is a direct precedent for a representation–data feedback loop, as well as a comparator for candidate models and reusable capabilities. New library components can invoke earlier learned components, forming a hierarchy rather than a fixed list of abstractions. Its text tasks concern string transformations and

generative regular expressions, not interpreting novels or constructing continuing character lives. The FI combination here proposes to place the problem world, capability interfaces, evaluated compositions, and their restructuring in one evolving account.

The Darwin Gödel Machine is a closer experimental comparison for organizational self-improvement: it modifies agent code, evaluates descendants, and retains a branching archive, reporting gains on coding benchmarks while keeping its pretrained foundation models fixed [83]. Benchmark attempts produce scores and logs; these guide proposed changes to tools, prompts, and workflows, while evaluation affects which agents are selected for further modification. The improved object is executable agent organization rather than the foundation model’s weights. Life Simulation with Fractal Intelligence proposes an additional learning target: the development and reuse of capabilities represented alongside the worlds in which they act. This includes training the learner as well as improving external tools and organization; neither route is established by the present construction. Alien search separately revises the categories selecting which mechanisms to explore. Their proposed contribution must be assessed at the level of capability formation and exploration control, not reduced to the fact that their execution histories can supply training data. These distinctions identify comparison questions; they do not establish global originality or a demonstrated learning advantage.

Inventing the vocabulary also has precedents outside library learning. Wirth and O’Rourke introduce predicates when the existing account is insufficient, infer example instances, and recursively apply learning to define the new predicate [84]. Chen et al. infer candidate numerical state variables from videos without being given the relevant physical variables [85]; these learned coordinates need not be interpretable concepts, and their method is not recursive symbolic decomposition. Together with DreamCoder, these approaches precede aspects of learning what to represent, not merely values within a fixed vocabulary.

The proposed extension is recursive invention of tracked conceptual processes within continuing world, life, and Event models. Search can change process definitions, comparative questions, decomposition depth, and temporal resolution together. The resulting histories guide economic, institutional, environmental, and interpersonal dynamics as well as character behavior. The learned world-modeling capability can then be tested on existing narratives and historical accounts. Tests of these histories’ contribution and the records of their revision inform subsequent corpus construction and learning. This makes new conceptual processes both candidate explanations and candidate learning targets. The comparison is whether that coupled search improves generation and transferable understanding, not whether naming more concepts alone constitutes progress.

Scaling coverage, resolution, and reuse. Life Simulation’s intended scope is not a larger collection of isolated game episodes. It is one connected account of world history and its texts, containing distinguishable factual, fictional, and counterfactual histories. Existing sources provide starting constraints; simulation can supply further histories and missing numerical values as explicitly authored alternatives. A person, institution, or process can be reused across sources and tasks rather than reconstructed independently for every example. This proposes a route to broadening structured experience beyond what can be collected through direct agent interaction, not a way to turn invention into historical evidence.

Three mechanisms could make that scope tractable. Adaptive resolution advances coarse life and societal processes without requiring a frame-by-frame account of every intervening moment. One underlying history can support many passages, viewpoints, forecasts, numerical targets, and Reader interpretations, sharing construction cost across learning tasks. Finally, learned refinement strategies and reusable functions can replace repeated manual specification; compact laws may represent patterns previously recorded as individual values. These routes expand both coverage and the kinds of quantities learned, rather than only adding samples under a fixed representation. More targets from one history are correlated supervision, not independent evidence.

These are proposed scaling advantages, not a measured efficiency lead over Genie-SIMA, JEPA, or agent-code evolution. Latent models already avoid many unnecessary details, and explicit histories incur acquisition, reconciliation, validation, and maintenance costs. The matched-budget tests in Section 4 should compare useful temporal coverage, transfer, correction cost, and prediction or control quality, including those costs and holding observation requirements comparable. Writing a century-long outline does not itself demonstrate a century of reliable simulation. The systems are also composable: a visual generator can render selected encounters, a latent model can predict local dynamics, and a self-improvement method can develop tools within the larger learning loop.

Personal-data evaluation. Longitudinal sensing, experience sampling, and probabilistic forecasting show both the value and the hazards of time-indexed personal data [41, 42, 43, 44, 45]. They motivate source-aware, consented, cutoff-safe evaluation; they do not make an inferred inner state observed.

16 Limitations

Explicit process state can be wrong, non-identifiable, culturally narrow, or too expensive to maintain. Several hidden paths may explain the same observations, and synthetic worlds can make an authored distinction look more recoverable than it is in reality. A learner may exploit renderer artifacts, annotation conventions, or future leakage rather than learn persistence or causality. Domain units do not make a bad measurement valid, and a normalized allocation does not make an attribution true.

The family of candidate mechanisms is deliberately broad. That breadth does not provide a discovery algorithm or guarantee that a stable function exists. Flexible laws can overfit retrospective paths; direct language models or opaque latent world models may outperform every explicit alternative. Adaptive resolution can hide important downstream coupling. A process vocabulary can freeze the very ontology that an unfamiliar world or later evidence should change. Versioning exposes these failures but does not solve them.

The synthetic-to-real step is the largest evidential gap. Fictional canon gives known authored state, not known human psychology. Reports and behavior are partial evidence, not transparent readings of private experience. Prospective improvement may remain small after measurement cost, missingness, distribution shift, and consent constraints are counted.

Models of people and institutions are dual-use. Estimates that might support more considerate assistance can also enable surveillance, persuasion, coercion, or optimization against vulnerabilities. Reader tracks and self-evaluators can be gamed. Access limits, correction rights, independent evaluation, adversarial testing, and the ability to abstain are necessary conditions, not guarantees of alignment.

17 Conclusion

Life Simulation aims to simulate life itself and learn through that activity: people growing and acting, shared worlds changing, and understanding developing alongside them. It combines the collection and inference of empirical quantities with the invention of conceptual processes whose numerical meaning comes from declared comparisons. Both belong in one expanding Meaning Model of world history. The proposed loop uses those histories for training and asks the resulting learner to expand and refine the shared account.

Whole-life personal modeling, live conversational use, corpus-wide alignment through Total Saturation, and an expanding range of numerical processes and dynamics are parts of that ambition. Selective resolution and bounded experiments provide a route toward it. The Meaning Model governs what the evolving records mean and how proposed changes are accepted; Life Simulation studies their temporal behavior and learning value throughout that construction. Direct model judgment, conventional simulation, learned dynamics, and mixed mechanisms remain valid options. Continuous, discrete-event, and hybrid paths are the objects of study; compact laws must earn their place.

The proposed learning material includes the world, its observations, and the development of its interpretation, at the scale of a progressively reconstructed written corpus. That material can support pretraining, continued pretraining, or post-training; bounded experiments test steps toward the corpus-wide ambition. Forward construction supplies controlled histories and alternative renderings for inverse learning. Source-grounded reconstruction can connect those skills to a growing corpus model without erasing disagreement. Joint native/process prediction trains a process sensorium; paired world and construction histories train both continuation and the choice of representation, including which distinctions improve a personality model. A correction can change a value, a question, or the account itself. The revised account can supply different targets from the same sources, and training on the repair can improve how the next account is constructed. Sema identifies the exact concept versions; when Temporal Hindsight Learning is used, later evidence can label an earlier judgment without entering its input.

The same loop can change how the learner acts and explores. Capability can reside in learned function generation, expanded tools, and better strategies for using them together. Fractal Intelligence develops reusable capability organization through evaluated decomposition, composition, restructuring, and reuse. Alien-world search revises the categories directing mechanism exploration, with simulated consequences feeding subsequent choices. Neither is merely an additional source of traces. Perspective-limited rehearsal adds active participation: agents try decisions inside simulated situations, respond to other characters, and learn from the resulting interaction histories under their original information cutoffs.

A continuing Reader and its revisable Understanding Graph are central to the recommended combined training: Entangled Alignment supplies the evaluative orientation, while synchronized native spans, visuals, and process histories give it particular lives and consequences to interpret. It can investigate those lives through

queries and refinements of the shared account, not only interpret a model supplied in advance. The intended result is an agent whose growing understanding helps govern its decisions and the successors it develops. Durable care is a separate objective to demonstrate, not a consequence guaranteed by accurate models of people.

This coupling changes the proposed objects and feedback of learning, not just the range of applications. Worlds supply observations; inference reconstructs accounts; interpretations and actions expose errors; revisions and subsequent outcomes supply new learning material. Music, interactive worlds, institutions, and consented chronological data test this method beyond prose. Compulsory substrate prediction remains a distinct architectural experiment.

The strongest claim remains conditional. The primary narrative tests in Section 7 ask whether learned progressive decomposition, including newly proposed conceptual processes, improves generation that genuinely depends on their numerical histories. The secondary pilot in Section 10.2 isolates better modeling decisions and better downstream learning from regenerated material. Improvement in either pilot outcome cannot substitute for the narrative test, and better repairs do not establish better downstream learning. If explicit tracks fail to beat native-only and latent baselines at matched total cost, they should be removed from that tested training or prediction condition. That result does not by itself reject their use in authoring, explanation, or control, which require their own evidence of utility. If they improve prediction but increase manipulation or false certainty, they should not be called alignment. The program advances only when each process, representation, and training use earns its place separately.

The integrated learner and its claimed transfer have not been demonstrated. The proposal is a route to learning evolving worlds, revisable understanding, and consequential judgment together, including a proposed solution to alignment whose conduct and stability require independent evidence.

A Proposed Training Export

Each row names an immutable information cutoff and one prospective task. The input includes only native observations and selected world or cognitive records visible at that cutoff, together with registry and grounding versions. The target may be a later native observation, continuous-state interval, discrete event, observer-perspective estimate, candidate-model prediction, representation revision, consequence, or correction. World time, observation order, construction or revision time, and discourse order are serialized separately when present.

Navigation examples additionally record the task, currently visible record addresses and versions, chosen query or refinement action, permitted response, and resulting answer, revision, or stopping decision. Responses become available only after their corresponding actions; a completed traversal is not exposed as an earlier input. Source-span offsets and visual frame or interval references align the modalities without flattening them into one node type. These examples train selection as well as prediction. Alternative successful routes remain admissible, and their costs are reported alongside task outcomes.

Every exported value retains its epistemic track, permitted authority-parent paths to its unique governing context, separately identified temporal ancestry, evidence, units or measurement-protocol references, applicable local questions, uncertainty or remainder, provenance, and accepted/rejected status. Access and the row cutoff are applied before traversal or serialization; denied references expose no private ancestry. A Cut or component is exported only with the complete permitted local Cut: parent Event, question, unit, keyed sibling answers and weights, remainder, optional conditioning address, constraints, and recomposition rule. If the complete view is unavailable, the Cut view is denied; an isolated semantic weight is not a valid export. Direction weights retain their declared proposal-policy or forecast-probability interpretation, with later calibration recorded separately.

An exported refinement retains its base, protected queries, projection, tolerances, and Close result, directly or through a resolvable, access-controlled immutable bundle. A revision retains predecessors and active-head changes. These are the Meaning Model's validation and transaction records, not a new world form. Access restrictions apply throughout; input fields also obey the information cutoff. A later validation result may be a target or audit record, never an earlier input. If required evidence cannot be supplied under those restrictions, do not label the row a checkable refinement example.

Model-completed intervals remain distinguishable from observations. An export manifest reports process selection, sampling resolution, missingness, cognitive coverage, renderer or sensor family, and any adaptive-resolution decision. Omitted annotation means unmodeled or unannotated, not false. A snapshot-only export is not labeled cutoff-safe until chronology and correction replay are implemented and tested.

For joint prediction, a row aligns the next native interval Y , selected process continuation X , discrete events E , and any proposed registry change ΔV . For generative inversion, worlds, renderer families, and grounding versions are split so that lexical or template memorization cannot substitute for state recovery. For prospective real-data evaluation, the reveal interval and scoring record are appended without mutating the original input or forecast.

A.1 Reservoir split, training, and scoring protocol

The following settings complete the protocol for Section 10.1.

Generate 2,000 independent world groups, each retaining its mirrored stock/leak history, renderings, and intervention siblings. Assign 1,400/300/300 groups to train/validation/test before making rows, and use cutoffs 8, 16, 32, 64. Paired variants never cross splits. Freeze a paired ordinary-text and typed renderer for training, with identical observation content. At test time compare the ordinary-text arms on their original renderer and a held-out fact-preserving paraphrase renderer; keep the typed view fixed and report that robustness comparison separately from $T - U$. Report renderer cost and ensure no input renderer can read private targets. Use five paired initialization seeds, batches of 32, AdamW at learning rate 3×10^{-4} , and 20,000 updates per arm; fix weight decay at .01, use no dropout, and report truncation (which must be zero), token counts, parameters, and measured compute. A secondary compute-capped comparison stops all arms at the smallest observed training budget; equal update counts alone are not equal compute. These numbers are design choices to freeze before test release, not a power calculation.

The primary endpoint is four-hour native forecast negative log likelihood, averaged first within world group. Average paired $U - N$ differences across seeds, then report a 95% group-bootstrap interval and the per-seed differences separately. Secondary outcomes include $T - U$, empty-onset Brier score, and stock-posterior calibration. The last two apply only to N, U, T ; the direct native reference supplies total-reading forecasts, not a stock posterior or tank-specific Events. An exact oracle enumerates the 50 possible initial states, retains those consistent with the visible history, and propagates their posterior weights through the eight future mode paths. A current-observation-only baseline uses the generator-induced time- t prior over (a_t, b_t, z, q_t) , marginalizing earlier supply paths without conditioning on past observations. It then conditions only on (S_t, q_t) at the known cutoff t , ignoring gauge readings, and forecasts through the same decoder and supplied future action schedule. It does not assign a uniform prior to the currently feasible states. Both remain benchmarks, never student inputs. Appendix A.2 gives one row and the training/inference sequence.

A.2 A complete reservoir forecast row

The following row instantiates Section 10.1. It is illustrative arithmetic, not experimental evidence. Within the synthetic world the generator knows the private targets; the student sees only the input column. The format renderer has the same restricted access as the student.

Field	Cutoff-visible input	Training or later scoring target
Identity and clocks	World group g , History and two stock-process addresses; registry digest v ; cutoff hour 8; horizon hours (8, 12]	Same identities and source version; later reveal attached without rewriting input
Native history	Hours 0, . . . , 8; totals (20, 23, 26, 29, 24, 19, 22, 25, 28) litres; modes (1, 1, 1, 0, 0, 1, 1, 1, 1); no gauge	Future totals (31, 32, 27, 30) litres
Action and observation policy	Ordinary supply for all four forecast hours; total and mode public; individual stocks and leak private	Future modes $(q_9, q_{10}, q_{11}) = (1, 0, 1)$
Selected process state	Current total 28 litres; stocks otherwise unresolved	$s^* = (a_8 = 10, z = A)$, hence $b_8 = 18$; available only for the supervised loss
Path and Events	No future Event or contact label	Stocks (11, 20), (12, 20), (9, 18), (10, 20) at hours 9–12; B reaches capacity at 9 and 12; supply switches at 10 and 11
Masks and provenance	Exact simulated observations rendered with frozen renderer and access policy; no truncation	$m_s = 1$ for supervised arms, 0 for native-only arm; authored simulator truth, not observed human state; no registry revision

Table 8: One information-separated training row. All stock numbers carry litres; no sibling-relative semantic share is introduced.

A minimal implementation follows four steps:

1. Generate and split world groups before producing views. At each cutoff, assemble only the permitted observation prefix and planned action schedule; keep private state, future Events, and scoring data in a separate target.
2. Encode the prefix. Normalize the joint (a, z) logits over states compatible with the observed total and capacity bounds. The learner does not receive the oracle’s history-filtered support or posterior.
3. Enumerate each candidate with the eight future supply paths. Apply the shared decoder, group identical observable continuations, and compute Equation 11. Sum over alternatives rather than replacing them with independent mean stocks and a mean leak label.
4. At inference, mask private labels entirely and save the forecast distribution before revealing outcomes. Score the frozen distribution and attach any later update as a new estimate, preserving the original.

Other domains need their own target adapters. Measured continuous values require a unit-aware density or declared discretization; missing targets need masks; Cuts need complete compatible sibling sets. Nothing in this finite example licenses adding arbitrary numerical scales to unmeasured processes.

References

- [1] H. Westerberg, “The Meaning Model: Constructing Worlds and Stories at Progressive Resolution,” Zenodo, 2026. doi: 10.5281/zenodo.22313515.
- [2] H. Westerberg, “Imagining the Corpus: Turning Text into Video for Language Model Training,” preprint, 2026. doi: 10.5281/zenodo.22104892.
- [3] H. Westerberg, “Sema: When the Hash Is the Word,” preprint, 2026. doi: 10.5281/zenodo.22073482.
- [4] H. Westerberg, “Temporal Hindsight Learning: Blindness as Teacher, Hindsight as Curriculum,” preprint, 2026. doi: 10.5281/zenodo.21959762.
- [5] G. E. Uhlenbeck and L. S. Ornstein, “On the Theory of the Brownian Motion,” *Physical Review*, vol. 36, no. 5, pp. 823–841, 1930. doi: 10.1103/PhysRev.36.823.
- [6] R. C. Merton, “Option Pricing When Underlying Stock Returns Are Discontinuous,” *Journal of Financial Economics*, vol. 3, nos. 1–2, pp. 125–144, 1976. doi: 10.1016/0304-405X(76)90022-2.
- [7] A. G. Hawkes, *Spectra of Some Self-Exciting and Mutually Exciting Point Processes*, *Biometrika*, vol. 58, no. 1, pp. 83–90, 1971. doi: 10.1093/biomet/58.1.83.

- [8] H. Mei and J. M. Eisner, *The Neural Hawkes Process: A Neurally Self-Modulating Multivariate Point Process*, in *Advances in Neural Information Processing Systems 30*, 2017. <https://proceedings.neurips.cc/paper/2017/hash/6463c88460bd63bbe256e495c63aa40b-Abstract.html>.
- [9] R. Killick, P. Fearnhead, and I. A. Eckley, “Optimal Detection of Changepoints With a Linear Computational Cost,” *Journal of the American Statistical Association*, vol. 107, no. 500, pp. 1590–1598, 2012. doi: 10.1080/01621459.2012.737745.
- [10] R. Goebel, R. G. Sanfelice, and A. R. Teel, “Hybrid Dynamical Systems,” *IEEE Control Systems Magazine*, vol. 29, no. 2, pp. 28–93, 2009. doi: 10.1109/MCS.2008.931718.
- [11] L. S. Shapley, “Stochastic Games,” *Proceedings of the National Academy of Sciences*, vol. 39, no. 10, pp. 1095–1100, 1953. doi: 10.1073/pnas.39.10.1095.
- [12] E. A. Hansen, D. S. Bernstein, and S. Zilberstein, “Dynamic Programming for Partially Observable Stochastic Games,” in *Proceedings of the Nineteenth National Conference on Artificial Intelligence*, pp. 709–715, 2004. <https://cdn.aaai.org/AAAI/2004/AAAI04-112.pdf>.
- [13] M. J. Kearns, M. L. Littman, and S. P. Singh, “Graphical Models for Game Theory,” in *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, pp. 253–260, 2001. <https://arxiv.org/abs/1301.2281>.
- [14] J. M. Epstein and R. Axtell, *Growing Artificial Societies: Social Science from the Bottom Up*. MIT Press and Brookings Institution Press, 1996.
- [15] J. W. Forrester, *Industrial Dynamics*. MIT Press, 1961. <https://mitsloan.mit.edu/shared/ods/documents?PublicationDocumentID=10011>.
- [16] H. Wang et al., “Code as Worlds: Agentic Discovery of Executable World Representations for Physical Reasoning,” arXiv:2608.27549, 2026. <https://arxiv.org/abs/2608.27549>.
- [17] F. M. Bass, “A New Product Growth for Model Consumer Durables,” *Management Science*, vol. 15, no. 5, pp. 215–227, 1969. doi: 10.1287/mnsc.15.5.215.
- [18] S. Aral, L. Muchnik, and A. Sundararajan, “Distinguishing Influence-Based Contagion from Homophily-Driven Diffusion in Dynamic Networks,” *Proceedings of the National Academy of Sciences*, vol. 106, no. 51, pp. 21544–21549, 2009. doi: 10.1073/pnas.0908800106.
- [19] C. F. Manski, “Identification of Endogenous Social Effects: The Reflection Problem,” *The Review of Economic Studies*, vol. 60, no. 3, pp. 531–542, 1993. doi: 10.2307/2298123.
- [20] D. C. M. Dickson, M. R. Hardy, and H. R. Waters, *Actuarial Mathematics for Life Contingent Risks*, 3rd ed. Cambridge University Press, 2020. doi: 10.1017/9781108784184.
- [21] D. Ha and J. Schmidhuber, “Recurrent World Models Facilitate Policy Evolution,” in *Advances in Neural Information Processing Systems*, vol. 31, 2018. <https://arxiv.org/abs/1803.10122>.
- [22] D. Hafner et al., “Mastering Diverse Control Tasks through World Models,” *Nature*, vol. 640, pp. 647–653, 2025. doi: 10.1038/s41586-025-08744-2.
- [23] J. Schrittwieser et al., “Mastering Atari, Go, Chess and Shogi by Planning with a Learned Model,” *Nature*, vol. 588, no. 7839, pp. 604–609, 2020. doi: 10.1038/s41586-020-03051-4.
- [24] E. N. Lorenz, “Deterministic Nonperiodic Flow,” *Journal of the Atmospheric Sciences*, vol. 20, no. 2, pp. 130–141, 1963. https://journals.ametsoc.org/view/journals/atmsc/20/2/1520-0469_1963_020_0130_dnf_2_0_co_2.xml.
- [25] R. M. May, “Biological Populations with Nonoverlapping Generations: Stable Points, Stable Cycles, and Chaos,” *Science*, vol. 186, no. 4164, pp. 645–647, 1974. doi: 10.1126/science.186.4164.645.
- [26] E. N. Lorenz, “The Predictability of a Flow Which Possesses Many Scales of Motion,” *Tellus*, vol. 21, no. 3, pp. 289–307, 1969. doi: 10.3402/tellusa.v21i3.10086.
- [27] J. M. Nese, “Quantifying Local Predictability in Phase Space,” *Physica D: Nonlinear Phenomena*, vol. 35, nos. 1–2, pp. 237–250, 1989. doi: 10.1016/0167-2789(89)90105-X.
- [28] E. Aurell, G. Boffetta, A. Crisanti, G. Paladin, and A. Vulpiani, “Predictability in the Large: An Extension of the Concept of Lyapunov Exponent,” *Journal of Physics A: Mathematical and General*, vol. 30, no. 1, pp. 1–26, 1997. doi: 10.1088/0305-4470/30/1/003.
- [29] OpenTelemetry Authors, “Signals,” OpenTelemetry Documentation, updated 10 March 2026, accessed 28 August 2026. <https://opentelemetry.io/docs/concepts/signals/>.
- [30] S. Rostedt and Linux kernel contributors, “ftrace – Function Tracer,” Linux Kernel Documentation, accessed 28 August 2026. <https://docs.kernel.org/trace/ftrace.html>.
- [31] M. Clerx et al., *CellML 2.0*, *Journal of Integrative Bioinformatics*, vol. 17, nos. 2–3, art. 20200021, 2020. doi: 10.1515/jib-2020-0021.
- [32] S. M. Keating et al., *SBML Level 3: An Extensible Format for the Exchange and Reuse of Biological Models*, *Molecular Systems Biology*, vol. 16, no. 8, e9110, 2020. doi: 10.15252/msb.20199110.

- [33] Modelica Association, *Modelica: A Unified Object-Oriented Language for Systems Modeling, Language Specification, Version 3.7*, 19 May 2026. <https://specification.modelica.org/maint/3.7/MLS.pdf>.
- [34] H. Westerberg, “Entangled Alignment: When Safety Is the Substrate,” preprint, 2026. doi: 10.5281/zenodo.22073296.
- [35] B. Z. Li, M. Nye, and J. Andreas, “Language Modeling with Latent Situations,” in *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 12556–12571, 2023. doi: 10.18653/v1/2023.findings-acl.795.
- [36] J. Lanchantin, S. Toshniwal, J. Weston, A. Szlam, and S. Sukhbaatar, “Learning to Reason and Memorize with Self-Notes,” arXiv:2305.00833, 2023. <https://arxiv.org/abs/2305.00833>.
- [37] H. Westerberg, “Fractal Intelligence: Conceptual Decomposition as Problem-Solving Infrastructure,” preprint, 2026. doi: 10.5281/zenodo.22100317.
- [38] Y. Shoda, W. Mischel, and J. C. Wright, *Intraindividual Stability in the Organization and Patterning of Behavior*, *Journal of Personality and Social Psychology*, vol. 67, no. 4, pp. 674–687, 1994. doi: 10.1037/0022-3514.67.4.674.
- [39] W. Mischel and Y. Shoda, *A Cognitive-Affective System Theory of Personality: Reconceptualizing Situations, Dispositions, Dynamics, and Invariance in Personality Structure*, *Psychological Review*, vol. 102, no. 2, pp. 246–268, 1995. doi: 10.1037/0033-295X.102.2.246.
- [40] K. Ellis, C. Wong, M. Nye, M. Sablé-Meyer, L. Morales, L. B. Hewitt, L. Cary, A. Solar-Lezama, and J. B. Tenenbaum, “DreamCoder: Bootstrapping Inductive Program Synthesis with Wake–Sleep Library Learning,” in *Proceedings of the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation*, pp. 835–850, 2021. doi: 10.1145/3453483.3454080.
- [41] S. Shiffman, A. A. Stone, and M. R. Hufford, “Ecological Momentary Assessment,” *Annual Review of Clinical Psychology*, vol. 4, pp. 1–32, 2008. doi: 10.1146/annurev.clinpsy.3.022806.091415.
- [42] A. Ponnada, S. Wang, D. Chu, B. Do, G. Dunton, and S. Intille, “Intensive Longitudinal Data Collection Using Microinteraction Ecological Momentary Assessment: Pilot and Preliminary Results,” *JMIR Formative Research*, vol. 6, no. 2, e32772, 2022. doi: 10.2196/32772.
- [43] M. N. Burns et al., “Harnessing Context Sensing to Develop a Mobile Intervention for Depression,” *Journal of Medical Internet Research*, vol. 13, no. 3, e55, 2011. doi: 10.2196/jmir.1838.
- [44] T. Gneiting and A. E. Raftery, “Strictly Proper Scoring Rules, Prediction, and Estimation,” *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007. doi: 10.1198/016214506000001437.
- [45] L. J. Tashman, “Out-of-Sample Tests of Forecasting Accuracy: An Analysis and Review,” *International Journal of Forecasting*, vol. 16, no. 4, pp. 437–450, 2000. doi: 10.1016/S0169-2070(00)00065-0.
- [46] H. Westerberg, “Understanding Graph: A Persistent Medium for Recursive Understanding,” preprint, revised August 2026. doi: 10.5281/zenodo.22073246.
- [47] H. Westerberg, “The Ontology of the Alien: World-Diversity Search and Evolving Solution Ontologies,” preprint, 2026. doi: 10.5281/zenodo.21910466.
- [48] H. Westerberg, “Toward Compulsory Scene-Mediated Language Modeling: Meaningful Substrates for Every Next Word,” preprint, 2026. doi: 10.5281/zenodo.21954994.
- [49] S. L. Brunton, J. L. Proctor, and J. N. Kutz, “Discovering Governing Equations from Data by Sparse Identification of Nonlinear Dynamical Systems,” *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016. doi: 10.1073/pnas.1517384113.
- [50] H. Lightman et al., “Let’s Verify Step by Step,” arXiv:2305.20050, 2023. <https://arxiv.org/abs/2305.20050>.
- [51] S. Ross, G. J. Gordon, and J. A. Bagnell, “A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, vol. 15, pp. 627–635, 2011. <https://proceedings.mlr.press/v15/ross11a.html>.
- [52] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, “Generative Agents: Interactive Simulacra of Human Behavior,” in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, Art. 2, pp. 1–22, 2023. doi: 10.1145/3586183.3606763.
- [53] X. Zhou, Z. Su, T. Eisape, H. Kim, and M. Sap, “Is this the real life? Is this just fantasy? The Misleading Success of Simulating Social Interactions With LLMs,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 21692–21714, 2024. doi: 10.18653/v1/2024.emnlp-main.1208.
- [54] J. Parker-Holder and S. Fruchter, “Genie 3: A New Frontier for World Models,” Google DeepMind, August 5, 2025. <https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>.
- [55] SIMA Team, “SIMA 2: An Agent That Plays, Reasons, and Learns With You in Virtual 3D Worlds,” Google DeepMind, November 13, 2025. <https://deepmind.google/blog/sima-2-an-agent-that-plays-reasons-and-learns-with-you-in-virtual-3d-worlds/>.

- [56] Y. LeCun, “A Path Towards Autonomous Machine Intelligence,” technical report, 2022. <https://openreview.net/forum?id=BZ5a1r-kVsf>.
- [57] A. Dawid and Y. LeCun, “Introduction to Latent Variable Energy-Based Models: A Path Towards Autonomous Machine Intelligence,” arXiv:2306.02572, 2023. <https://arxiv.org/abs/2306.02572>.
- [58] M. Assran et al., “V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning,” arXiv:2506.09985, 2025. <https://ai.meta.com/research/publications/v-jepa-2-self-supervised-video-models-enable-understanding-prediction-and-planning/>.
- [59] U. Nodelman, C. R. Shelton, and D. Koller, *Continuous Time Bayesian Networks*, in *Proceedings of the Eighteenth Conference on Uncertainty in Artificial Intelligence*, pp. 378–387, 2002. <https://ai.stanford.edu/~nodelman/papers/ctbn.html>.
- [60] Y. Rubanova, R. T. Q. Chen, and D. Duvenaud, “Latent Ordinary Differential Equations for Irregularly-Sampled Time Series,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019. <https://proceedings.neurips.cc/paper/2019/hash/42a6845a557bef704ad8ac9cb4461d43-Abstract.html>.
- [61] W. M. P. van der Aalst, *Process Mining: Data Science in Action*, 2nd ed. Springer, 2016. doi: 10.1007/978-3-662-49851-4.
- [62] F. Locatello et al., “Object-Centric Learning with Slot Attention,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020. https://proceedings.nips.cc/paper_files/paper/2020/hash/8511df98c02ab60aea1b2356c013bc0f-Abstract.html.
- [63] G. E. Hinton, P. Dayan, B. J. Frey, and R. M. Neal, “The Wake–Sleep Algorithm for Unsupervised Neural Networks,” *Science*, vol. 268, no. 5214, pp. 1158–1161, 1995. doi: 10.1126/science.7761831.
- [64] M. O. Riedl and R. M. Young, “Narrative Planning: Balancing Plot and Character,” *Journal of Artificial Intelligence Research*, vol. 39, pp. 217–268, 2010. doi: 10.1613/JAIR.2989.
- [65] A. S. Rao and M. P. Georgeff, “BDI Agents: From Theory to Practice,” in *Proceedings of the First International Conference on Multiagent Systems*, pp. 312–319, 1995. <https://cdn.aaai.org/ICMAS/1995/ICMAS95-042.pdf>.
- [66] P. Gebhard, “ALMA: A Layered Model of Affect,” in *Proceedings of the Fourth International Joint Conference on Autonomous Agents and Multiagent Systems (AAMAS ’05)*, pp. 29–36, ACM Press, 2005. doi: 10.1145/1082473.1082478.
- [67] H. Rashkin, A. Bosselut, M. Sap, K. Knight, and Y. Choi, “Modeling Naive Psychology of Characters in Simple Commonsense Stories,” in *Proceedings of ACL*, pp. 2289–2299, 2018. doi: 10.18653/v1/P18-1213.
- [68] Nasrin Mostafazadeh, Aditya Kalyanpur, Lori Moon, David Buchanan, Lauren Berkowitz, Or Biran, and Jennifer Chu-Carroll. GLUCOSE: GeneraLized and COntextualized story explanations. In *Proceedings of EMNLP*, pages 4569–4586, 2020.
- [69] B. Dalvi, L. Huang, N. Tandon, W.-t. Yih, and P. Clark, “Tracking State Changes in Procedural Text: A Challenge Dataset and Models for Process Paragraph Comprehension,” in *Proceedings of NAACL-HLT*, pp. 1595–1604, 2018. doi: 10.18653/v1/N18-1144.
- [70] J. M. Zacks, N. K. Speer, K. M. Swallow, T. S. Braver, and J. R. Reynolds, “Event Perception: A Mind–Brain Perspective,” *Psychological Bulletin*, vol. 133, no. 2, pp. 273–293, 2007. doi: 10.1037/0033-2909.133.2.273.
- [71] R. A. Zwaan and G. A. Radvansky, “Situation Models in Language Comprehension and Memory,” *Psychological Bulletin*, vol. 123, no. 2, pp. 162–185, 1998. doi: 10.1037/0033-2909.123.2.162.
- [72] M. L. Jockers, “Revealing Sentiment and Plot Arcs with the Syuzhet Package,” 2015. <https://www.matthewjockers.net/2015/02/02/syuzhet/>.
- [73] A. J. Reagan, L. Mitchell, D. Kiley, C. M. Danforth, and P. S. Dodds, “The Emotional Arcs of Stories Are Dominated by Six Basic Shapes,” *EPJ Data Science*, vol. 5, no. 1, 31, 2016. doi: 10.1140/epjds/s13688-016-0093-1.
- [74] M. Iyyer, A. Guha, S. Chaturvedi, J. Boyd-Graber, and H. Daumé III, “Feuding Families and Former Friends: Unsupervised Learning for Dynamic Fictional Relationships,” in *Proceedings of NAACL-HLT*, pp. 1534–1544, 2016. doi: 10.18653/v1/N16-1180. <https://aclanthology.org/N16-1180/>.
- [75] F. Brahman and S. Chaturvedi, “Modeling Protagonist Emotions for Emotion-Aware Storytelling,” in *Proceedings of EMNLP*, pp. 5277–5294, 2020. doi: 10.18653/v1/2020.emnlp-main.426. <https://aclanthology.org/2020.emnlp-main.426/>.
- [76] Z. Xie et al., “ChatTS: Aligning Time Series with LLMs via Synthetic Data for Enhanced Understanding and Reasoning,” *Proceedings of the VLDB Endowment*, vol. 18, no. 8, pp. 2385–2398, 2025. doi: 10.14778/3742728.3742735.
- [77] L. Christ, S. Amiriparian, M. Milling, I. Aslan, and B. W. Schuller, “Modeling Emotional Trajectories in Written Stories Utilizing Transformers and Weakly-Supervised Learning,” in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 7144–7159, 2024. doi: 10.18653/v1/2024.findings-acl.426.
- [78] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, “ReAct: Synergizing Reasoning and Acting in Language Models,” in *International Conference on Learning Representations*, 2023. <https://arxiv.org/abs/2210.03629>.

- [79] B. Chen, C. Shu, E. Shareghi, N. Collier, K. Narasimhan, and S. Yao, “FireAct: Toward Language Agent Fine-tuning,” arXiv:2310.05915, 2023. <https://arxiv.org/abs/2310.05915>.
- [80] A. Zeng, M. Liu, R. Lu, B. Wang, X. Liu, Y. Dong, and J. Tang, “AgentTuning: Enabling Generalized Agent Abilities for LLMs,” arXiv:2310.12823, 2023. <https://arxiv.org/abs/2310.12823>.
- [81] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, and S. Yao, “Reflexion: Language Agents with Verbal Reinforcement Learning,” arXiv:2303.11366, 2023. <https://arxiv.org/abs/2303.11366>.
- [82] A. Kumar et al., “Training Language Models to Self-Correct via Reinforcement Learning,” in *International Conference on Learning Representations*, 2025. <https://arxiv.org/abs/2409.12917>.
- [83] J. Zhang, S. Hu, C. Lu, R. Lange, and J. Clune, “Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents,” arXiv:2505.22954, 2025, revision March 12, 2026. <https://arxiv.org/abs/2505.22954v3>.
- [84] R. Wirth and P. O’Rorke, “Constraints on Predicate Invention,” in *Machine Learning Proceedings 1991*, pp. 457–461, 1991. doi: 10.1016/B978-1-55860-200-7.50094-5.
- [85] B. Chen, K. Huang, S. Raghupathi, I. Chandratreya, Q. Du, and H. Lipson, “Automated Discovery of Fundamental Variables Hidden in Experimental Data,” *Nature Computational Science*, vol. 2, pp. 433–442, 2022. doi: 10.1038/s43588-022-00281-6.